

Phil 404 The Philosophy of Science

Numbered Transcript

PHIL 404 • Dr. James Orr

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Lecture 1: Science as Philosophy

At a glance — Orr introduces the philosophy of science and what the field is about.

- 1.1 Welcome to this eight-lecture series in Philosophy of Science. What you may ask is: what's the philosophy of science all about? Well, I'm going to come on to that in a moment. But what I want to start with is to note the extraordinary dominance that science has in our culture, in our popular culture, in our political culture, across our universities. It's become really the benchmark of what counts as real, what counts as true.
- 1.2 In fact, shortly before he died, the great Cambridge cosmologist and physicist, Sir Stephen Hawking, wrote a book, co-wrote a book, called "The Grand Design" in 2010. And he wrote this. "Philosophy is dead. Philosophy has not kept up with modern developments in science, particularly physics. Scientists have become the bearers of the torch of discovery in our quest for knowledge." Very striking image, that. Stephen Hawking says that the torch of discovery has gone from philosophy, where it was with the philosophers for the best part of 2,000-3,000 years, to the scientists. Philosophers just can't keep up.
- 1.3 What's so striking is that, for thousands of years, the deepest questions that human beings could ask were always really in the domain, understood to be in the domain, of philosophy, not of science. Science was understood to be a branch of philosophy. When Isaac Newton, another great, great Cambridge physicist and philosopher, wrote his famous "Principia Mathematica," Principles of Mathematics, in 1687, he titled it, the full title he gave it was "Philosophiae Naturalis Principia Mathematica," the mathematical principles of the philosophy, philosophiae, naturalis, of nature, of natural philosophy.
- 1.4 Science was one branch of philosophy. It was taken for granted that the most fundamental questions that we can ask were not simply to be settled within that branch, natural philosophy. Why does anything exist? Why anything? Why this? Why me? What happens to us when we die? Is biological death death? What is the nature of reality? How did the universe begin? What are the fundamental ingredients of nature? What is consciousness? What is life? What is mind? How do we understand beauty? All of these questions were understood to be philosophical questions.
- 1.5 But such has been the success of science, the natural sciences, or natural philosophy, as it used to be called over the last 250 years, a success that is undeniable, that there's been colonizing of other parts of knowledge by science. This is what's motivating the great Stephen Hawking's claim. What he's saying is that all of those questions now belong to physics. They belong to cosmology, to neuroscience—to science, not philosophy.

- 1.6 So if we're interested in the nature of ultimate reality, that's a question for quantum mechanics, wave functions, superpositions, et cetera. If we're interested in questions about the beginning of the universe, where the universe came from, that's a question for the cosmologists to settle. If we're interested in the fundamental nature of reality, that's a question for the particle physicists, the particle scientists, people finding bosons and quarks and neutrinos and so on. Neuroscience explains the mind. It explains consciousness, or you might say, unkindly, it explains consciousness away.
- 1.7 So this is philosophy's role on this quite-dominant popular view. That book, "The Grand Design," in 2010, was a popular book. It was a book that you would find in ordinary bookshops. It wasn't an academic tome. It was Hawking's reflections on the end of his long and extraordinarily successful life as a cosmologist about the nature of philosophy. Philosophy was now, really, for him—and this is a view that I think is broadly representative—the philosopher is a backseat driver, occasionally being asked for some advice here and there, but normally just being told to shut up.
- 1.8 The idea that philosophy can contribute substantively to knowledge is now, I think, quite a controversial one. Philosophers are more like sports commentators. They might know a lot about the sport; they might have some interesting and engaging things to say, but they don't know how to play the game. This, as I said, is a complete reversal of the historical relationship, as it's something that had been the case for 2,000 to 3,000 years, something very, very new. Whether we think it's a good thing or a bad thing, a true thing or a false thing, it is something that is striking and, historically speaking, without precedent.
- 1.9 Just back to this question of the dominance of science in our popular culture, and the grip that it has on us, and the strength and authority that it has on us. Think of all the debates over the COVID-19 pandemic. "Follow the science. Trust the science." It's not a suggestion. It's a command. It's a moral imperative: "follow the science." In one's mind—capital S, T, M. Climate change: "the science is settled. If you don't think it is, you're a denier." Again, that's a heavily freighted, morally freighted claim, that. This isn't wrongthink, it's evilthink if you dissent from the science.
- 1.10 Evidence-based practices: "What does the data show?" "Studies demonstrate." All of these implicitly invoke the authority of the natural sciences or often the social sciences—economics, sociology, anthropology. Even there, the sciences that involve human beings, again, you can see the colonizing force of science. Why should our economic behavior, or why should the behavior of society, be understood as something that can be studied as a science? Doesn't that just imply we're nothing more than just scientific objects of inquiry, as human beings? Well, maybe that's true.
- 1.11 But if it's true, it's not something that science can settle. That's a philosophical question. Whether we are souls, for example. Maybe we're not. Maybe we are. What's the scientist going to tell us about the question of whether or not we have souls? Is he going to put us through some clever scan and say at the end, "We've done the most

comprehensive scan that any neuroscientist or physiologist or biologist knows about, and we couldn't find a soul anywhere"? This is not an impressive philosophical conclusion. Again, it may be the case that we don't have souls, but it is not a question for the scientist to settle that, as a scientist. These are, and have always been understood to be, philosophical questions.

- 1.12** We think also of the authority of science in courts of law, the so-called Daubert standard, D-A-U-B-E-R-T, which says that expert testimony must be based on scientific knowledge. It must be testable, it must be peer-reviewed, generally accepted in the scientific community. As we see in all of these matters, science has the final say. The conclusions of the scientists are dispositive. I think, in many ways, this is something that is damaging to science, because science, as we'll see in the course of these lectures, walks along this very delicate tightrope.
- 1.13** On the one hand, it operates by invoking a certain kind of authority. Now, this is what our best experimenters and investigators in the natural world have determined to be the character of the natural world. There are institutions, there are Nobel Prizes, there are certain conventions that afford authority to our best scientists, and that's a good thing. It's a good thing. On the other hand, though, science always operates through a kind of skepticism. Scientists are always testing. "Well, why? Could we test that again? This theory doesn't quite—it explains a lot, but it doesn't explain everything. This theory actually explains a lot more."
- 1.14** And so, the tightrope is between, on the one hand, a kind of dogmatism, a commitment to the authority of science, and on the other, the awareness that science always proceeds through testing, through skepticism, through a healthy skepticism. If Einstein hadn't been skeptical or hadn't been willing to be skeptical about Newton's classical framework for physics, he would never have come up with his general theory of relativity. And so, it's this delicate balance between authority and skepticism, and that's how science moves forward.
- 1.15** If I discovered that my daughter was being taught physics from a textbook dated 1895, I'd be very concerned. If my daughter was being taught maths from a textbook dated 1895, I'd be thrilled. I'd be over the moon. Why? Because science moves forward. It's constantly moving forward. We expect it to move forward. Maths? Well, we crack theorems, I suppose, but broadly speaking, if it's true that $2 + 2 = 4$, it's always true that $2 + 2 = 4$. That's not going to change much. Newton's law of universal gravitation—that two objects attract each other in a relationship that is inversely proportional to the [square of the] distance between them—it's not exactly false, but it's been relativized, as it were; it's been shown that it's true only under certain conditions.
- 1.16** So again, this is what's so fascinating about the philosophy of science. How do we think about truth if it's a truth that dominates society, dominates our courts and our labs and our policy and our media, but it's a truth that keeps changing? You think we've got it, and then it slips through our fingers, and we call it progress. So these are some of the

questions that we're going to be exploring over the next eight lectures. Where does science get its authority from? How do we justify it? How do we demarcate what counts as science from what counts as not science?

- 1.17 Newton probably spent a lot more time writing about alchemy than he did about physics or the cosmos. He was fascinated with alchemy. Was alchemy a science for him? He didn't operate. It was all natural philosophy. It was all philosophizing about the natural world. We'll be looking at those questions, theoretical questions; how scientific theories change, what justifies the change; the problem of induction, which is the problem of how we spot patterns in the world that can sustain future predictions about the behavior of the world.
- 1.18 Can we say that, because we've seen that the sun comes up every morning over the last 13.8 billion years, that it must come up tomorrow morning? What is the nature of that must? Is it as strong as it—that it must be the case that $2 + 2 = 4$? Or might it be the case that, actually, as the stock market people like to tell us, that past performance is not necessarily an indicator of future performance?
- 1.19 What does it mean to explain something scientifically? I can explain why you're watching this by appealing to personal explanations. You wanted to learn a bit more about the philosophy of science. Is that a scientific explanation? Well, not really. The scientist could probably give me a scientific explanation of what's going on in your brain as you're thinking that, and what's happening in your body as you're turning on the lectures to watch them. Is that an explanation that competes with the personal explanation? Is it an explanation that is actually—is it the complete explanation of what's really going on? And the personal explanation, your reasons for acting, are just illusions? I have to think about those questions as well.
- 1.20 Can you reduce all sciences to physics? Does biology collapse into chemistry? Does chemistry collapse into physics? Does physics collapse into quantum physics and cosmology? That'll be the second half of the course, Lectures 5 through to 8. But in the first half of the course, I want to do something quite unusual, something that isn't typically done in most, I think, really, in any philosophy of science courses across the West.
- 1.21 I'm going to put in—the first half, I'm going to think about the history of science, or rather the history of the philosophy of science, how science or natural philosophy begins to emerge, how reflecting on the natural world begins to emerge, because I think that's going to give you a historical context that will make sense of what comes later.
- 1.22 Philosophy of science as a subject today tends to be dominated by what C. S. Lewis once called "chronological snobbery." This is the idea that anything that has come before is to be treated with some skepticism, and the further away it is in history, the more skeptically it should be treated. This is the idea that, as time passes, we just get

wiser and cleverer and more insightful. This is what Lewis calls a kind of snobbery, a kind of condescension, and I want to show how science emerges.

- 1.23** It's often the story is quite a self-congratulatory one. Once upon a time, it was gloom and darkness and despair. Then in 1687, God said, as Alexander Pope puts it, "God said, 'Let Newton be!' And there was light." We call that, that whole period is now called the Enlightenment, as if everything that had come before—Plato and Aristotle and Anaxagoras and Aquinas and Augustine and Anselm, Al-Farabi, Al-Ghazali—all of these were just sort of in a kind of darkness and confusion and gloom, and as I hope to show, that that is not at all the case. If anything, the Enlightenment represents the culmination, the peaking, of a process that was already well underway by the turn of the first millennium. And arguably, as I'll suggest later in this lecture, it goes right back to before Socrates and Plato themselves.
- 1.24** But again, the questions we're going to be asking about induction, about explanation, about reduction, these are philosophical questions. To think that you can simply answer all philosophical questions by reference to the sciences is to be in the same situation as the man who's had a bit too much to drink one night, and you find him on the pavement—sidewalk in American—looking for his keys under the streetlight. You say, "Well, you can't see, the keys aren't here, but have you tried looking beyond the street light?" And he says, "No, I'm not going to do that." You say, "Well, why not?" "Well, because this is where the light is shining."
- 1.25** And the person who thinks that all philosophical questions have to be reduced to scientific ones are a little bit like that man, say, "Well, look, this is where the light of science is shining. I'm not going to look beyond that. I'm not going to consider that perhaps there's a non-scientific answer to some question." But it seems to be obvious to me, at any rate, that there just are truths that are not scientific truths.
- 1.26** $1 + 1 = 2$. Is that a scientific truth? I don't think so. It's just a mathematical truth. In fact, it's stronger than any scientific truth could be, because it's true no matter what. No Einstein is going to come along and say, "Actually, you guys thought it was, you thought that $1 + 1 = 2$, but actually in certain circumstances, it could be 3." That's not going to happen. You could say that, not only are mathematical truths not scientific truths, they're stronger than scientific truths ever could be, as I was saying earlier about the comparison of those hypothetical 1895 textbooks for my daughter.
- 1.27** If it's true that $1 + 1 = 2$ is a mathematical truth, not a scientific truth, there are infinitely many non-scientific truths. Because if $1 + 1 = 2$ is a non-scientific mathematical truth, so is $1 + 2 = 3$, and so on and so forth, and we assume that goes on for infinity, at least it would be very odd, it would cry out for explanation, if, at 1,003,000,000,406, it ceased to be the case that you could add 1 and get to the next number. That would cry out for explanation. That would be finitude. Infinity in philosophy is much simpler than finity, than finitude. Because whenever a line is drawn,

you can ask the child's question, "Why, Daddy, why? Why there, and not just a little bit further?"

- 1.28** Then I think there are also actually many other truths that are not reducible to scientific truths, that are probably a little bit more contentious—questions about who we are, our own private thoughts and imaginings, our sense of subjectivity, our sense of self, or the feelings that we might have towards other people. It may be the case that there is something going on in our bodies that corresponds or correlates with those feelings or those thoughts, but it's very contentious to say that they are exactly the same thing.
- 1.29** Just because there's traffic between, say, our brain states and our mental states, our conscious states, doesn't mean that we know which way the traffic is going, and it certainly doesn't mean that we can identify all of our mental states with all of our brain states, and so on and so forth. If philosophy is not dead, if it has an essential role in examining the foundations of science, then Hawking is wrong, and science's claimed total authority is itself a philosophical claim.
- 1.30** It can't be a scientific claim, otherwise he'd be arguing in a circle, right? "Science says that science is the most authoritative way of understanding the universe." That's just arguing in a circle. You have to step outside of the streetlight and be able to give some account of what the discipline is.
- 1.31** So this lecture series is going to be a lecture series in the philosophy of science. We're going to be examining science philosophically. We're not doing science; we're thinking about science— its methods, its foundations, its scope, and its authority. To do that is not to be anti-science; it's to be respectful of science. It's to understand it and to categorize it as an avenue of human inquiry with its own particular history, its own heroes, its own villains—just as any other avenue of human inquiry, whether it's mathematics or literary theory or history or, indeed, philosophy itself.
- 1.32** To question what science is is not to be skeptical about its practical successes. It's not to be skeptical about the technology that makes it possible for you to be watching this lecture now, or about medicine or engineering. All of these, of course, flow from scientific understanding. But those successes don't necessarily justify the authority that is sometimes attached to science. It doesn't necessarily mean that every question is a question that science can ultimately settle.
- 1.33** You might think, "Well, look, I'm not a philosopher. I'm not a scientist. Why should I care about these abstract questions? What's the relevance? What relevance does this course have for me?" Well, to start with, as I suggested at the beginning, we are all of us living in a culture where we are constantly being asked to trust the science, to follow the science, in domains that affect your life and that affect mine—health decisions, vaccines, medications, diets, policy questions, climate, economy, education, and of course, existential questions about free will and consciousness and meaning.

- 1.34 Again, if we just think that those are scientific, these are all reducible to scientific questions, that it's the neuroscientist who can help us get through a bad day, then that's a contentious claim and one that needs to be investigated. Because scientists disagree. As I said, that's how science progresses. Studies conflict. How do we adjudicate if we're not experts? I hope that, over the course of this series, we'll be exploring the glories of science and its insights, but also understanding its methods and its limitations.
- 1.35 As we start to understand the limits between what counts as science and what counts as not science, we will educate ourselves in grasping what the claims are for science in all of these different aspects of our lives, how we evaluate them, how we weigh them up. We live in a culture, of course, where "scientific" is almost the word, the adjective "scientific" is almost the same as true. And "unscientific" is almost the same as, is almost a synonym for false. And this affects what views are acceptable in our public debates, in our political debates.
- 1.36 And so, understanding what counts as something that's scientific versus something that's probably scientific overreach is fundamental to our discourse in the public square and for intellectual freedom. So the roadmap for the first half of the course is more of a historical one. We'll go right back to the beginnings of Western philosophy, to the dawn of thought, even beyond, even behind Socrates. And we'll come up all the way through, in Lectures 1, 2, 3, and 4, into the Middle Ages, and then into the so-called Enlightenment. That will set us up for the second half of the course, where we'll be looking at more conceptual and systematic questions.
- 1.37 So, to begin at the beginning. Around about 600 BC, shall we say, just as the Iliad and the Odyssey, or the oral traditions of those great Homeric poems, just as they're beginning to be crystallized in their final form, natural phenomena were explained, essentially mythologically. There was an earthquake— that's Poseidon. The seasons change when Persephone goes down to Hades. Thunder—oh, Zeus is having a bad day. These were seen as totally normal ways of explaining natural phenomena, very poetically powerful ways of understanding natural phenomena.
- 1.38 But more than that, they were understood to be literally literal explanations. Natural events are explained by supernatural agents with humanlike psychology, not always very well-adjusted human beings, humanlike psychology. The point here is that nature is fundamentally unintelligible, at least through rational investigation. I mean, it's intelligible. The reason there's thunder is because Zeus is having a bad day. That's a kind of explanation, but it's completely unfalsifiable. There's nothing to test it. These are appeals to personal or superpersonal explanations, anthropomorphic explanations, not natural ones.
- 1.39 There are no laws to discover. There are no patterns that can tell us whether Zeus is going to have a bad day tomorrow or not. We just don't know. He's Zeus. We're not going to tell him what to do. If he's having a bad day, he's having a bad day. So, that's

where we are at, round about 600 BC. In a place called Ionia, which is roughly the west coast of modern-day Turkey, there emerged two or three city states, one in particular Miletus, where thinkers began to emerge who we might call, not implausibly, the first natural scientists.

- 1.40** This, by the way, is a corner of the Mediterranean where trade is really taking off, trade between different cultures—Babylonian cultures, early Israelite culture, Egyptian culture, Phoenician culture, and yes, Greek culture. The great thing about trade is that it's not just goods that get exchanged, but ideas, technologies. 600 BC, this is round about the time that the first coins are being struck; coins are being invented by the Lydians. It's round about this time, actually a little bit earlier, that the alphabet, as we know it, begins to emerge, almost certainly as a technology for inventories and to take notes and to keep records and so on and so forth.
- 1.41** A few thinkers began to emerge in the sixth century BC, 600 to 500 BC, who started to come up with explanations that we might laugh at today, but which we can identify as attempts to be natural explanations—no appeals to Zeus, no appeals to Poseidon or to Persephone. I think you can make a good case for saying that this is the birth of something like natural philosophy, a philosophy of nature, a way to understand nature, a kind of science.
- 1.42** So Thales, famously, the pre-Socratic philosopher, Thales, T H-A-L-E-S, claimed that all of reality is ultimately water. All things come from water. It's reputed, if I recall correctly, that he died by falling into a well because he was looking up at the stars, so he met a watery end, appropriately. Now, what's going on? He also, by the way, we know that he predicted an eclipse in 585 BC, correctly. That gives us a nice date that locks him in, into the first half of the 6th century BC, his dates.
- 1.43** By the way, we've got very little that remains of these guys. We've got fragments that we have to try and piece together; and they're very poetic as you sort of read them, but they were attempting something like philosophical or scientific treatises. So, what are we to make of this claim? All things come from water. Note what isn't there. It's not, all things come from Zeus. Everything that happens at sea is to be explained by Poseidon. There's a principle, one principle, that explains everything, all diversity: water.
- 1.44** In many ways, that is what scientists are trying to do. They're trying to come up with a theory that most unifies what we can see and experience. The reason that Einstein displaces Newton is that his theory was a more unifying theory than Newton's, which did work, but only under particular conditions. Einstein's worked under those conditions and more conditions, that traced the very edges of space-time. It was unifying. Again, we can laugh at Thales, thinking that these are all coming from water, but he's trying to come up with this sense that everything comes from one thing. Got to have some sense of unity.

- 1.45 Secondly, it was a naturalistic explanation. He's appealing to some feature of nature, water. You can see it. It's natural, it's observable, it's not supernatural. It's testable. You can investigate this claim empirically, rationally. As we'll see later, Anaximander comes along and says, "No, it's not all water, and here's why." So, Thales [was] the first human to really propose a natural principle as a fundamental explanation.
- 1.46 Anaximander, who I've just mentioned, comes a little bit later and says, "No, it can't be water. It can't be water because you can ask the question, 'What explains water? What explains water?'" Water is determinant, it's fixed, it needs some explanation, just as air would need it. Anaximander says there is a fundamental principle, but whatever it is, it's indefinite, it's boundless, you can't catch it. Because if you could, you could ask the question, "Why? Why that? Why water, and not air, or rock?" So, Anaximander says it's something that is infinite, indefinite, indeterminate.
- 1.47 Now, what's interesting about this? Modern science today is constantly making theoretical posits. It's constantly saying we must assume that there is this unobservable entity that explains everything else, or explains a lot of other things. The more science we're doing now, in fact, most of the science we're doing now, we can't test, because our mathematics and our theorizing of our best scientists is outstripping their technological capacity to come up with the labs that can test a lot of these claims.
- 1.48 So, famously, the English physicist, Peter Higgs, round about the mid-1960s, posited something called the God particle, the Higgs boson particle, just with a pencil and a piece of paper, at his desk, just doing a lot of maths. Some people thought this was a plausible hypothesis. Other people said, "No, it's not going to work." It took, I think, another 25 years for the Large Hadron Collider at CERN in Geneva to establish that Higgs was right, that there is indeed a Higgs boson particle. He posited something theoretical that he couldn't see, it wasn't observable, but it made so much sense of what we could observe that we were justified in positing it. In some way, this is what Anaximander is doing in response to Thales. It's got to be something that we can't explain in which the definite things can be rooted, can be grounded.
- 1.49 A couple of other thinkers, very quickly: Heraclitus, a wonderful poet and thinker, for whom nature was a state of constant flux and dynamic flow. He famously says, "You can't step into the same river twice." You try and step into the Mississippi once, try stepping into it again—it's a completely different river, because it's a completely different constellation of H₂O, or water particles, around your foot on the second occasion. Actually, there was a philosopher who came later, who said, "You can't even step into the same river once, because you put your foot in, and there's a whole load of different particles."
- 1.50 So, they're thinking in these rational ways. Heraclitus says the world is informed by logos—it's a Greek word for rational principle or intelligibility. And nature is—they don't use the language of law, because law is just something that's human; it's something just for the city-state. So the Greek mind can't cope with a law of nature in

the way that Abrahamic monotheists, Jews, Christians, and Muslims, who believed in a creationist framework, immediately could understand the notion of a law of nature. But the Greeks could think of this notion of there being patterns and regularities in nature that you don't need to appeal to the gods to explain. They're predictable and regular; there's a rational ordering principle to nature.

- 1.51 And so, I think we see some big leaps forward with the pre-Socratics, that nature is intelligible, that it's orderly, that explanation should be unified. Non-observable theoretical entities can be legitimate. Observation and reason, married together, help us get closer to the truth. Mathematics can be involved in the physical world, as Pythagoras thought and as Plato believed. So these are all very distinctively scientific ideas that we are happy associating with science, even today—actually, not just associating with them, but seeing them actually as central.
- 1.52 Before we leave the Greeks, a few words about the great Aristotle. He's born in 384 BC, dies in 322 BC. He's a hillbilly from Macedonia, right up in northern Greece, and he's hardly even seen as Greek. The Macedonians were seen as—they had terrible accents, and you couldn't really understand them, and they shouldn't really be allowed at the Olympic Games, until Alexander came along, and then they all bowed down before, had to bow down before Alexander, who was, famously, Aristotle's pupil.
- 1.53 Aristotle really is the first figure in human history, we can say, who is thinking in extraordinarily powerful scientific ways. Before Aristotle, there's no such thing as physics, "ta physika," the things of nature, as he described it. Before Aristotle, there's no such thing as psychology. There's no such thing as biology. Aristotle is the first person to lay out all the principles of scientific explanation, the importance of the senses, the importance of applying the mind to what we receive in sense experience, the importance of observation, and so on and so forth. The importance of common sense.
- 1.54 His biological works were vast. He dissected animals. He observed embryology. He's the first to really catalog species, sometimes in a remarkably accurate way, certainly in ways that would dominate what we would call the sciences, or scientific thinking, for many, many hundreds of years. In fact, Aristotle's belief in, his cosmological belief, that the universe has existed infinitely far back into the past, the universe did not have a beginning, was the standard view, the so-called steady-state theory, of the origins of the universe, up until the 1960s.
- 1.55 Famously, Fred Hoyle, who was a fellow of St. John's College, Cambridge, said to the young Stephen Hawking, when he'd come up with this theory that actually, no, the universe began, Aristotle was wrong, replied, "So, what are you going to call this theory then, the Big Bang?" As it was meant as a joke, as a satirical, as a cruel put-down, but that is indeed what it came to be called. What was so extraordinary about it was that it wasn't just a Big Bang in cosmological terms; it was a Big Bang in terms of the explosion

of a theory that Aristotle had set down in the fourth century BC, 2,450-odd years before.

- 1.56** So he's an extraordinarily powerful mind, an obsessive investigator and catalogue of the world. But it wasn't just science for him. He comes up with, he invents logic; that is to say, he's the first person to try to set out the laws of thought, how it is that we think, how it is that we form arguments and move from premises to conclusions. He's thinking about the role of mathematics and Euclidean geometry, and how we go about demonstrating scientific conclusions, so an extraordinary figure.
- 1.57** The fact that a lot of Aristotelian science came to be discredited or attacked much, much later—we're talking now in the 17th century, really, is when it gets underway. It's beginning in the 14th century. But really, with Descartes and Newton in the 17th century, that's when he really starts to be challenged. For all that, what's been challenged in his thought is not their unscientific nature, but the fact that it's the wrong kind of scientific insight. And so, Aristotle is not to be dismissed. He's somebody we need to be thinking about as an important figure in the development of the philosophy of science.
- 1.58** Some of the things he didn't quite get, for example, is the importance of how mathematics connects to physics, for example. He tended to think in terms of the qualities of things, the essences of things, rather than in quantifying terms. We've got to measure things, and we've got to look not at the mysterious essence of things, but really at how things behave in the world.
- 1.59** And although he was brilliant at observing things, he hadn't developed the so-called hypothetico-deductive method; that is, he hadn't developed this idea that we should be trying to manipulate the world in particular ways, trying to come up with experiments, controlled experiments, shift different variables. All of that comes later.
- 1.60** Some of his claims turned out to be empirically false. Some of them you would have thought would have been seen to be self-evidently false when he was alive. His claim that women have fewer teeth, for example. He could have just checked his wife's teeth. He was just wrong. So sometimes he was too busy looking at the ground. That heavy objects fall faster, he believed. Galileo shows this is false. That the heart is the seat of thought. False, but in what sense is the brain the seat of thought? It's not obvious, but we do know that thinking and feeling is much more tightly connected to the brain than it is to the blood pump.
- 1.61** Aristotle's view that natural objects have purposes, functions, that the oak tree was always designed, was always going to be the endpoint of the acorn, that acorns are directed towards becoming an oak tree. This claim is one that modern science gets very uneasy about, can't involve future states in present explanations; but actually, as it turns out, if you look at philosophy of biology today, they're constantly talking about, or if you just talk to biologists, "What's the flipper for? What's its purpose? What's its

function?" "Well, it's so the thing can swim." It's a totally normal way of talking. "What's the heart for?" "Its function is to pump blood."

- 1.62** That seems like, for Aristotle, that was a totally normal explanation. But for scientists today, from the 17th century onwards, this was seen as spooky. There's a politically incorrect quote attributed to the Cambridge biologist, J. B. S. Haldane, 1920s-1930s, who said that "function is to the biologist what a mistress is to the cad." He won't be seen with her, but he can't live without her.
- 1.63** Biologists won't be associated with talk of function and purpose, but you can't talk about the natural world if you're not invoking function terms, purpose terms. That's obviously what the acorn is for. It's not designed to be, it's not aiming towards being a willow tree. It's obviously directed towards a future state of being an oak tree, which it may or may not attain, but that's got to be part of, says Aristotle, of what the explanation, what is a complete explanation of what an acorn is, should involve.
- 1.64** But still, it's fair to say that we're still missing, in this period, this idea of manipulating nature to try to test it, "putting nature on the rack," as Francis Bacon would say later. We're still not talking in terms of quantitative predictions, as modern physics now does. And crucially, we're not seeing at this point science as distinct from philosophy; it's still a kind of philosophy. And on top of that, one might add, there's none of the institutional framework that was the engine room for the great scientific discoveries from the 17th century onwards.
- 1.65** What do I mean by that? You didn't have the universities. You didn't have the Royal College of Science. You didn't have the Nobel Prize Committee in Sweden. You didn't have the great journals, the great system of peer review, of guilds of scientists testing each other's work and scrutinizing each other's work. This comes much later. But what I want to say is just in this first lecture that the foundations are laid here; that nature is rational, that we can test it, we can investigate it, we can explain it to varying degrees of success, we can argue about it. And we don't need to be, at any of those points, we don't need to be invoking supernatural or nonnatural explanations.
- 1.66** I think we can say that the pre-Socratic philosophers, the philosophers who came before Socrates, 470 to 399 BC, and Aristotle, are laying the foundations for natural philosophy. But it's not until 16th-, 17th-, 18th-century Europe, 2,000 years later, that we can really say that the scientific revolution happens.
- 1.67** One of the questions I want to look at in the next lecture is: why Europe? Why us? Why not China? Which, after all, was a much more ancient civilization, vastly superior to people living in tribes, living in mud huts on the rainy northwestern rim of Eurasia, with terrible food and terrible teeth. Why not China?
- 1.68** Why not the Islamic world? In the golden age between 750 AD and 1000 AD, when Islam preserved much of the best of Greek thought, including Aristotle? Why not India? Incredibly advanced civilization in mathematics and in philosophy and in logic. No, it

was Europe that incubated the scientific revolution. I want to look at that question in more detail in the next lecture. This is the so-called Needham question, N-E-E-D-H-A-M, named after Joseph Needham, the founder of modern sinology, the founder of Chinese studies. Also a Cambridge man, I'm glad to say.

- 1.69** And this was his question: why, if you were a proverbial alien and you landed on Earth in, say, 1500, and you had to predict which corner of the planet would incubate the greatest scientific revolution that human history had ever seen or possibly would ever see, would the proverbial alien pick Europe? Almost certainly not. Almost certainly not. We'll look at that in the next lecture as a way, as a springboard into thinking more deeply about that crucial transition, or I should say, culmination of these scientific insights, in the scientific revolution of the 17th and 18th century.
- 1.70** To conclude this first lecture, remember that opening question of Stephen Hawking's, or that opening claim, "Philosophy is over." Science has put it to rest. Science is here now. It's science that's taken on the torch of discovery. So our question is: is Hawking right?
- 1.71** My own instinct is that philosophy always buries its undertakers. What I want to try and show in the course of this series is that it's quite clear that philosophy also buries its undertakers, even if he is the greatest cosmologist since Newton and Einstein.
- 1.72** What I've tried to suggest in this lecture is that science is a branch of philosophy. It's part of philosophy. It's a way of inquiring into the world as it presents itself to us in a particular aspect, just as mathematics is a way that reality presents itself to us in a particular guise, or the processes of history is a way that reality can be understood in a particular mode. And so, too, with the sciences. It's just another branch, a very important branch, of human inquiry.
- 1.73** And so, if I'm right, then philosophy is not dead; and this course is not a waste of time, because science needs philosophy. And dare I even say these words? Stephen Hawking was wrong. Thank you for listening.
- 1.74** When you were talking about the acorn and the oak tree, and I was thinking, could the purpose of an acorn be to feed squirrels?
- 1.75** My quick answer to that would be that, certainly for Aristotle and people in the Aristotelian tradition, they would say it is a necessary condition of being an acorn that it has being an oak tree in its future, potentially. It is not a necessary condition of acorns that, one day, it has a future state of being eaten by squirrels or pigs or whatever.
- 1.76** That is to say, Aristotle would say that, when you're answering the question, "What is X?", you can give an answer in terms of a shopping list of certain necessary conditions that are intrinsic or characteristic of that thing X. Just as you might say, "It is characteristic of the X, being a human being, that a human being has 32 teeth." It doesn't mean that when you lose a tooth, you cease to be a human being.

- 1.77 What it means is that it is a normal condition, a normal descriptive property, of being a human being, that all other things being equal, you will have, even Aristotle's wife would be born with 32 teeth. It's normal or normative for human beings to have 32 teeth. If you lose a tooth, then you are defective as a human being, with respect to having a full complement of teeth, a normal complement of teeth.
- 1.78 So, similarly with the acorn, you could say it is a normative condition; it is an essential characteristic or property of the acorn that it's going to become an oak tree, not a willow tree. If it then gets eaten by a squirrel, and its purpose in life has been cruelly frustrated by the squirrel, that doesn't mean it ceases to be an acorn or it ceases to be the kind of thing that ought to have become an oak tree. But it was never part of its destiny; it was never part of its destiny, to be eaten by a squirrel, whereas it is part of its destiny, so to speak, to become an oak tree.
- 1.79 I was thinking about just the process of thought, and how these thinkers were positing the purpose of an oak tree is to produce acorns for squirrels to eat.
- 1.80 They didn't tend to think in terms of balance within ecosystems, if that's what you mean. There's that wonderful clip that goes around online sometimes of what happens in Yellowstone National Park after the introduction of wolves. I don't know if you've seen that film. They introduced wolves, and over the course of 10-20 years, it completely transformed the landscape of Yellowstone because of, what seems to be, you might think of it as being a disruptive, negative introduction, or disruption of the Yellowstone ecosystem. But through all sorts of complicated—you should look up the video—it's extraordinary how things transform.
- 1.81 And so the Greeks, no, they weren't thinking quite in those terms of interconnectedness and the interrelationship of complex food chains and ecosystems and so on. They were more interested in what is it that's essential to this particular thing, more than they were thinking in holistic terms about, how do these different things interact with each other?
- 1.82 But even if they were thinking in those terms, they wouldn't think that, say, wolves have some complex but indirect relationship as part of their identity to—I think, it's otters. There was some complicated connection between wolves and otters, which changed the whole structure of the rivers around Yellowstone, and I can't remember the exact sequence of the links. They would say it's still a happy accident that there is this propitious way in which these different things interact in the world.
- 1.83 Now, once religion comes along, or once the creationist religions come along, there you could start to have a more plausible theological explanation about saying, well, the extraordinary interaction of all of these different animals or elements or objects in the world does have a theological explanation. This is God's creative blueprint for this flourishing ecosystem. It's not just an accident that all these things get along together so well. It's not just an accident that some random rock in a faraway galaxy had, within

itself, the enabling conditions for carbon-based life, when the chances of that happening, the chances of all of these different forces intersecting so exquisitely, so perfectly, is astronomically improbable.

- 1.84 But that's not the way the Greek mind operates, because it's not operating within a—there are rumors of a creation doctrine. You see it in Plato's *Timaeus*, one of his dialogues; there's a creation narrative there. But it's still basically foreign to Greek and Roman thought, that idea of one God who's the source of all reality, with enormous powers and supreme rationality, who orders and renders all of nature intelligible and legible, who writes one book of nature, just one author. That's not really a thought that the Greek mind can cope with.

Lecture 2: The Scientific Revolution

At a glance — The Scientific Revolution and its intellectual upheaval.

- 2.1** Welcome to Lecture 2 in this eight-part lecture series on the Philosophy of Science. In this lecture, I want to dive into a question that's a bit of a hobbyhorse of mine, to be honest. It's quite niche. You're not going to find it on many, or indeed, any courses on the philosophy of science in most Western faculties. But for me, it's a question that's fascinated me since I was a teenager.
- 2.2** I think that, in answering it, we can actually start to understand how it is that the whole world of science and the scientific revolution emerged. All of the different conditions that happened to intersect at just the right time, in just the right place, that, as it were, provided together, provided the engine room, for the scientific revolution, and therefore the engine room for the modern world. What would the modern world be, after all, without the scientific revolution?
- 2.3** This question, as I referred to it in the previous lecture, is the so-called Needham question, which is in effect the question, when it comes to science, the scientific revolution, "Why Europe?" Why Europe, when it might so obviously have been, much more likely to have been, China, Babylon, the Mayans, the Incas, the Aztecs, Islam, India, and so on?
- 2.4** Because what we didn't really explain in the first lecture is how we're to understand why it is that science suddenly really takes off in 1500-1600. Why that 2,000-year gap between roughly the death of Aristotle, the first emergence of the first natural philosophers, you might say, and things really taking off in the 1500s—and in that part of the world, in Europe, not in Greece, specifically, but really, in mainly northern Europe and, to some extent, in Italy?
- 2.5** Aristotelian science had spread all around the Mediterranean. It was successfully preserved in places like Baghdad and Cairo, under the Abbasid caliphate, from 750 AD onwards, up to, well, in the case of Baghdad, up until the horrific sacking of Baghdad by the Mongols in 1258 AD, one of the greatest civilizational tragedies in human history. It was said that the Euphrates ran black and red—black from the ink of the books the Mongols had thrown into the river, those great works now lost to us, thousands of them that had slowly accumulated in the libraries of Baghdad over the previous 500 years or so, and red with the blood of the librarians who tried to protect their collections.
- 2.6** So Aristotelian science was pervasive, from Córdoba in Spain, to Baghdad in modern-day Iraq, from Athens to Rome to Carthage in North Africa. And yet, that's not where the scientific revolution happened. A little bit about Needham, Joseph Needham, 1900 to 1995. He was a biochemist, actually, a British biochemist. Ends up at Cambridge, Gonville and Caius College, Cambridge, and becomes a historian, improbably, of

Chinese science, Chinese science and Chinese technology. He really set his mind to the study of China as a discipline, now called sinology, the study of China. But nobody had really done it in this systematic way before, at least in the West.

2.7 He wrote this extraordinary work, "Science and Civilization in China," which he started in 1954, and I think he was probably really still working on it 40 years later in 1995. 27 volumes. 27 volumes. "Science and Civilization in China." What his work revealed pretty early on was that China was, technologically at least, way ahead of Europe for most of history, certainly from, let's say, 200 BC to 1400-1500 AD. Chinese inventions that transformed Europe—the compass, paper, the printing press, gunpowder—these were Chinese inventions that simply reached Europe many centuries later.

2.8 And so, this gave rise to the so-called paradox, the Needham paradox, or the Needham question. Why did modern science, the mathematization of hypotheses about nature, why, with all of the implications for advanced technology, why is it that it took off in the West, round about the time of Galileo?

2.9 This isn't just my hobbyhorse; this is not just Needham's hobbyhorse. It's not just an interesting historical puzzle: Europe versus China. It's philosophically crucial for our inquiry in this series. It's fundamental to answering the question: why science? What is science? Those philosophical questions about science.

2.10 Because if modern science was just the result of a bit of human curiosity and observation and intelligence, it would have emerged with Aristotle. The scientific revolution would have happened 2,000 years before it did. Aristotle was nothing if not curious. He was nothing if not obsessive about observing and cataloging the world. He was nothing if not intelligent. As Dante calls him, "il maestro di color che sanno." The master, he's the master of all those who know anything. Thomas Aquinas calls Aristotle, simply, "philosophus," the philosopher.

2.11 So, this suggests that modern science requires a very specific set of conditions, in addition to observation and curiosity and intelligence, and not just clever people observing nature. There were plenty of those before 1500 AD, despite what the chronological snobs might like to think. And so, what are these conditions?

2.12 I think answering this little hobbyhorse of mine, answering the Needham question, I think, will help us pump some intuitions and some hypotheses about, what are the conditions that we need for a scientific revolution to take off? And if we can understand what conditions need to be in place for the scientific revolution, we can start to begin to understand what it is that we're talking about, when we're talking about science. Any explanation of what science is, is going to have to make reference to the conditions that enable it to emerge and take off. So this is why we're going to focus quite a bit in this lecture on this question.

2.13 It will also show us that modern science rests on certain presuppositions and conditions that are not themselves scientific. Remember, part of the question that is negatively

motivating this lecture series is Stephen Hawking's question, or his claim, that philosophy is dead and that it's passed the torch over—"the torch of discovery," in his words—over to the sciences. I'm challenging that in this lecture series, and I'm pointing to all kinds of ways in which you can explain and understand science that don't involve appealing to science. The good thing about that is we won't be arguing in a circle like Hawking is.

2.14 First of all, I'm going to have a quick look, in this lecture, at what other civilizations have achieved—China, the Islamic world in particular. We'll see that they had sophisticated observation, technology, mathematics, even certain experimental methods. We'll then turn to look at what was unique about Christendom in the Middle Ages, the medieval world. It pains me whenever colleagues and students, friends even, very distinguished intellectuals, use the adjective "medieval" as somehow a synonym for barbarism, or barbaric, or primitive. Again, there's that chronological snobbery at work there.

2.15 Even the phrase Middle Ages, between what? Between what? What they actually mean is, well, we had this amazing Greek and Roman flourishing, efflorescence of civilization and culture and thought. And then all was darkness and gloom and confusion and delay and ignorance and barbarity, until there was a renaissance, a rebirth. What comes before a rebirth? Death. What comes before enlightenment? Darkness.

2.16 But what I want to suggest is, actually, it's in the Middle Ages that you start to see the seeds being sown. The Enlightenment didn't just come out of nowhere. The Renaissance just didn't come out of nowhere by accident. It didn't just fall from the sky. No, there were conditions in place that were set in the Middle Ages, despite enormous adversity—in material conditions in particular, enormous geopolitical unrest, conflict, and warfare, and so on. This is where, really, the scientific revolution is born. It's where it begins to be incubated and then comes to a culmination, a peak, if you like, in the 16th and 17th century.

2.17 So, very quickly, first of all, before we get on to the Middle Ages, let's just talk about China and the extraordinary inventions that would reach us in Europe so many centuries later. Gunpowder, of course, was invented, we think, round about the 9th century AD, extraordinarily early. It's being used in a military context, at least by the 10th century. It reaches Europe late 13th-14th century. It transforms European warfare, of course, but it's invented by the Chinese.

2.18 Printing. We know that woodblock printing is already happening in China in round about the 7th century AD. There's movable type, round about the 11th century. Then Gutenberg and Caxton are not till—what—1485, 1470s, 1480s? Neither Gutenberg nor William Caxton invented the printing press. That was 400 years after the Chinese got to it first. Paper. Paper is being invented, is around, we know, in the second century AD, under the Han dynasty. It doesn't really reach Europe until the 12-13th century AD by the Islamic world.

- 2.19** Similarly with the magnetic compass, the mechanical clock, the blast furnace, cast iron, porcelain, silk production, advanced agriculture, the seismograph, measuring earthquakes—that's the second century AD, sophisticated seismographic techniques, the wheelbarrow, suspension bridges, and so on and so on. Great leaps forward in astronomy, recording eclipses with precision, supernovae, comets, catalogues of stars, quite sophisticated astronomical taxonomies, in many ways superior to anything European astronomy comes up with until, let's say, the 16th century.
- 2.20** Mathematics, the decimal system. The Chinese had a decimal system very early on. Negative numbers, solving polynomial equations, very sophisticated forms of algebra. Pascal's triangle. Pascal's triangle is known of in the 13th century, so before Pascal is coming up with it in, what is it, 1630s, 1640s? Engineering as well, extraordinary leaps forward in engineering. The Grand Canal—1,100 or so miles long. It's the fifth century BC, that extraordinary feat. The Great Wall.
- 2.21** Very sophisticated ship-building, far more sophisticated than— the ships that they could produce were far, far bigger than any that Columbus used to reach America in the late 15th century. Similarly with medicine—acupuncture, herbal pharmacology, smallpox inoculation— that's happening round about, smallpox inoculation happening round about the 10th century, and it's not until the 18th, 19th century that we've got something like that.
- 2.22** So this is the puzzle. Why is it that the civilization that could produce all of this, that was so superior in so many different fields of human inquiry and practical, experimental, inventive success, why isn't it that China never produced a Newton, never produced a Darwin, never produced a Copernicus, a Kepler, never produced scientific geniuses that could trigger a revolution in human thought and in the natural sciences, in the understanding and the manipulation of the natural world?
- 2.23** China had sophisticated observation, record-keeping, a lot of knowledge, a lot of skill, a big bureaucratic state that could support learning and inquiry. But there are a few things that it lacked, and I think this is where we start to see the clues for the conditions that needed to be in place for the scientific revolution to get going.
- 2.24** It didn't have any very sophisticated theological framework. It certainly didn't have an idea that the natural world was the product of a supremely rational creative being with enormous powers, as is key to the Abrahamic monotheistic worldview, Judaism, Islam, and Christianity, according to which God decides, as an artist decides, "This is how the world is going to be." Just as you can look at a painting, or maybe not all modern paintings, but most paintings up until about 1920, you could look at a painting, and it's intelligible. You can understand what the artist is trying to convey.
- 2.25** Other things that it lacked—universities that were independent of the state, that were not under state control. Legal pluralism, different kinds of legal codes. Mechanical

philosophy, it didn't really have any philosophy for mechanism. It's the framework, bits of the framework, were not in place.

- 2.26** Similarly, if we look at the Islamic golden age from roughly 750 AD to 1000 AD, in fact, even leading up to the sack of Baghdad in 1258, let's say the Western Roman Empire falls about 476 AD, should we say? And really, the great civilizational glories and all the wonderful texts that come to us through Greece and Rome and through early Christianity, through Neoplatonism, get picked up and preserved by Islam, by Islamic scholars. There are some remarkable achievements.
- 2.27** Alhazen, for example, writes a book of optics, "Kitab al-Manazir." That's the 11th century—an extraordinary investigation into light, vision, mirrors, lenses, refraction. You can't read a page of that book and say, "This is not experimental science, or a form of experimental science." Al-Khwarizmi, 8th century, 9th century AD, effectively comes up with algebra. We get the word algebra, Arabic word, of course, from one of his famous works. "Algorithm" actually comes from his name, Al-Khwarizmi. It's through him that the decimal system is translated to Europe. Of course, we talk about Arabic numerals.
- 2.28** Avicenna, the great Avicenna, 930 to 1037 AD, his Canon of Medicine is a comprehensive medical encyclopedia, which is used in European universities all the way up into the 17th century. A very serious piece of work; it tries to synthesize Aristotle with Islamic theology. You've got observatories in Cairo, in Baghdad, in Maragheh, in Samarkand, astronomical observations and star catalogues, and improving planetary models, and so on and so forth. Medicine, chemistry, engineering, hospitals, surgical instruments, distillation, advanced architecture, extraordinary architecture, of course, in Islam at this period, complex geometrical designs, domes, arches, and so on and so forth.
- 2.29** Again, we can ask the question: what was missing, or what was getting in the way? There are a few ways to answer this. One is to point to a theological commitment that emerges within Islam, that emerges a little bit in Christianity in the 13th-14th century. It's not unheard of, but it never wins, this idea; it never wins in Christianity, but it does become the dominant view within Islam. This is a theory about the way in which God interacts with the world.
- 2.30** We might think, how does God interact with the world? He's outside time, he's outside space, he doesn't begin to exist, he doesn't cease to exist. He's a bit like numbers, you might say. How can a God that is so removed from space-time causally interact with space-time? This was a big problem. How do you understand how God interacts with the world? And also, if God does interact with the world, how is it that anything in the world acts without him? If God wants something, then God's energies and his causal powers just flood the world. How can anything actually act in the world on its own, if all the world depends upon, for all of its existence, on God?

- 2.31** So this is a huge debate that happens within Abrahamic monotheisms. It's a debate that happens among Christian philosophical theologians, happens among Islamic philosophical theologians, Jewish philosophical theologians. It's a problem that's part of that broader picture, that broader monotheistic picture. There's a huge debate between the Mu'tazilites and the Ash'arites. The Mu'tazilites wanted to push for a more—a picture of the natural world as being more autonomous. God creates it, but then goes off and has a cup of tea and lets it get on with its own, leaves it to its own devices, leaves it autonomous.
- 2.32** You can start to see, with that idea, science becomes a little bit easier, because what you're investigating is just the way the world is as it is, the patterns and behaviors that it exhibits. Whereas if you take the Ash'arite view, which is that God is acting at every single moment, for every single causal event in the world, it's God directly doing it, then it starts to become—the predictability and the patterns and the regularities in the natural world become a little bit more complicated, a little bit more speculative, because God can do what he wants, at any point.
- 2.33** You can't say that God has to make sure that the sun comes up tomorrow morning. Who are we to tell God what he's going to do tomorrow morning? This is the view that starts to win through. It actually makes a lot of sense. If you believe in an all-powerful God, who's maximally excellent, morally perfect, loves the world, why would anything else—why would you appeal to any other kind of explanation? If God is on the scene and God is omnipotent, who else is doing anything? God's power just floods the world.
- 2.34** Actually, the counterintuitive position is to say, no, God leaves room; he leaves space for the natural world to unfold according to its own patterns and its own essences and its own behaviors. In a funny way, I don't want to criticize the Ash'arite victory here. I think it's actually internally probably the more plausible view, but that's not the view that wins out in medieval Christian thought. The view that wins out there is a view much closer to the Mu'tazilite view that gets defeated by the Ash'arites in the Islamic golden age.
- 2.35** And that's the view, broadly speaking, between what is called, in medieval Latin, God's "potentia absoluta" and his "potentia ordinata." That is to say, God's absolute power, his total power, the power that he has no matter what, his absolute power, and on the other hand, his "ordinata" power, his ordained or ordered or constrained power. The idea that begins to emerge, after much dispute, is that God, in creating the world, as it were, constrains himself. He says, "I'm creating this world, and then I'm not just going to intervene randomly in it now."
- 2.36** Or there may be moments, there may be miraculous events, but those miraculous events will be freighted with a very, very particular theological significance. If prophets saying insightful things were regularly being brought back from the dead, then the world would be just a mysterious and obscure place. How can you start doing biology in a world like that? But if it happens just once, to that particular person, you can say,

there are very, very rare moments where God wishes to leave his authenticating signature on the processes of history in some miraculous intervening way. But it's rare.

2.37 Overwhelmingly, you can assume that nature will be left by God to take its own course, even if that means earthquakes and tsunamis and climbing accidents; that there is, as it were, a commitment by God to the regularity and the intelligibility and the predictability of the natural order that we can discover and rely upon. That's a crucial way in which what seems like a very abstract theoretical dispute about the nature of divine causation and creaturely causation has a huge downstream impact on the conditions for and the way in which we understand the world, the way in which we understand its legibility, its readability.

2.38 Under occasionalism—and occasionalism is the doctrine, the name, the label given to the Ash'arite doctrine—occasionalism because we might think that this baseball bat is hitting the baseball, but in fact, that's not what's happening; it's simply an occasion for God to move the baseball in a particular way, and to move the baseball bat in a particular way, in a way that is coordinated such that we think that there's a genuine interaction between the two. But it's simply an occasion for God's intervening power. So God doesn't create stable natural laws that then operate independently. God directly creates every event at every moment. It's a very, very radical view.

2.39 But again, once you've got an omnipotent divine being that's part of your picture, what else is going to do all the causing? What else is going to do all the creating? What is it that's driving all the different events? So you can see, you can see why that view took off. It was a way of, I suppose, expressing a piety to God, saying that nothing other than God has pure power, has causal power. If we want to say that nature, little old people like us, have got causal powers, then in some way that limits God's absolute power. This is the idea.

2.40 Now, there are other factors that explain why the Islamic golden age never quite led to the scientific revolution that we see occurring much later. The Mongol invasions, I've talked already about the heart-breaking sack of Baghdad in 1258. There's also other internal disputes within Islam, big debates between those who favored so-called "ijtihad," this idea of independent reasoning, in favor of "taqlid," this idea that you follow established interpretations of the Quran and the Hadith. Again, these are debates that happen within Christianity and within Judaism.

2.41 Should God's creatures, as it were, employ their God-given reasoning and their God-given perceptual powers to understand the world and to live their life by? Do you use the Book of Nature, or do you use the Book of Scripture? Do you just follow what's been revealed, what you believe to be revealed? There are good arguments for both sides—if you're in God-world, if you're in the theological world—because it's not like the Book of Nature is some rival to God. Within these systems, God creates nature just as much as he writes his book. They're just two books, so which are you going to pick? Sometimes you're going to have to, because if they conflict, you've got to choose.

- 2.42 So, these are not distinctive tensions within Islam, but they're resolved in a particular way that is distinct from the way in which Christianity resolves them. So across these different traditions—Chinese traditions, Islamic traditions, you could say even Babylonian traditions, Mayan traditions, Egyptian traditions— you've got a lot of ingredients in place that you might think would lead to a scientific revolution— sophisticated observation, accurate prediction, impressive technology, mathematical skill, institutional support, libraries, research centers, but still no modern science. Nothing close.
- 2.43 What seems to be the case is that modern science needs a very particular configuration, a constellation of conditions, to get going. Not any one single factor. Almost nothing in history can be explained by one single factor. Nothing in society, in fact, almost nothing can be explained by a single explanation. Always be very, very suspicious if somebody is giving you a uni-causal explanation for anything. It's almost always the case that there are multiple dimensions, multiple explanatory strands that go into why anything is the case, why anything happens, and then the question is working out what those are and what proportions they play, and bringing that thing about.
- 2.44 I think this constellation of conditions emerges, as I said, in the Middle Ages, in the ages of darkness and gloom and death. I want to just go through what I take those conditions to be, the conditions that led, as it were, to an Enlightenment and a Renaissance, before the Enlightenment, long before the Enlightenment, and long before the Renaissance. These are not accidental. These are what shape how Europeans understood, how they understood nature, how they thought about investigation.
- 2.45 The first idea, which is distinctive to all three Abrahamic traditions, but which gets interpreted in a particular way, as I've already alluded to, is the doctrine of "creatio ex nihilo" in Latin, creation out of nothing. Doctrine of creation out of nothing. Genesis 1:1, "In the beginning, God created the heavens and the Earth." It's a very, very strange idea. We're very familiar with it, but there's no other, there's no parallel to this.
- 2.46 Yes, there are ancient, Near Eastern, Mesopotamian creation myths, but they don't say there was nothing to begin with. There was something. There was kind of chaos, or there was some divine carcass leftover from fights between previous gods that then got molded into the world that we see, all these very strange ideas. Aristotle thinks there's no beginning at all, just the beginning of the world is—there isn't any beginning. The world stretches back infinitely into the past. This is the so-called steady-state theory, which was believed, as I said in the previous lecture, up until the 1960s. Very strange view. Very strange view, that.
- 2.47 I mean, if the universe began infinitely far back into the past, how long would it have taken for me to be speaking to you now? It's not a trick question. The answer? It would have taken infinitely long for us to get to the point where I'm speaking to you now. But

I am speaking to you now. I really am. I'm not a figment of your imagination. This is happening. Or even if I am a figment of your imagination, it's still happening in your imagination now. And therefore, it can't be the case that the interval between now and the beginning and, as it were, the origins of the universe, is infinite.

- 2.48** However long it might have been, and cosmologists might—maybe it's a trillion years. Maybe it really is 13.8 billion years or whatever. It's a finite interval because we've reached the end of it now. We're right at the edge of it, constantly. So space-time begins, reality begins from nothing at a particular point in time, and it's brought into being by a person, admittedly an unusual kind of person, but a person in the sense of somebody, a being who we can at least analogically speak of as having a center of subjectivity, a will, intentions, reason, possibly feeling, to the max or to the compatible maxes, the compossible maxes.
- 2.49** We're to think about nature in the world as if it might not have been what it is, as if it was just one particular configuration of conditions that was willed into being. Just as we might look at Monet's "Peupliers," those wonderful paintings he does of the poplar trees, and say, look, he might have painted willow trees; but no, he decided to do poplars. That's what he did. That's just what he decided. That's Monet for you. He just, nothing required him to do it. It's just part of his creative inspiration and his creative freedom, his artistic license.
- 2.50** Now, by way of an analogy, you could say that the Christian theology of nature that begins to get going in the Middle Ages—it's around before that, but it really starts to crystallize in the Middle Ages—is why dolphins? Why do they look like that? Why flamingos? Well, that's just how God wanted it to be. This is part of a creative blueprint. It didn't have to be the case that there were flamingos. It didn't have to be the case that there were dolphins. This is part of the creative freedom, the artistic freedom, of a divine artist. That's the way to understand it.
- 2.51** That doesn't mean we can't explore it. Quite the reverse—we can understand the patterns that Monet's painting exhibits. We can understand the impressionist style, what he's trying to do with perspective, and what he's trying to evoke in terms of phenomenology and perception. But this is nothing like Aristotle, who just says the world has existed forever, no matter what. There is a sort of a god up there, the unmoved mover, "nous, noesis, noeseos," thought, thinking itself.
- 2.52** He's a kind of cosmic narcissist. He just sits up there, thinking about himself, because he's the most perfect being ever. What else would a perfect being contemplate, if not the most perfect thing, himself? Not interested in what's going on down here, and nothing to do with it, won't have anything to do with it. Messy, imperfect. Just no connection. The rest of the world loves this being for Aristotle and is in the whole of—as it were, the energy, the animating force of the natural world is, in the end, directed towards this cosmic narcissist.

- 2.53** But there's no real relationship. There's no motive for saying, "Oh, we can understand nature in terms of the intentions of this cosmic narcissist," because he hasn't got any intentions for the natural world, or what Aristotle calls the sublunary world, or the world under the moon was a sublunary world, our world, and the world above the moon was a slightly different substance, and that's where the cosmic narcissist sat. Matter's eternal. It's just being continuously rearranged, and that's that.
- 2.54** At the edges of Plato's thought, there's this one dialogue, mysterious dialogue, called the *Timaeus*. Plato writes about 35 dialogues. To know more about Plato, I would refer you to the outstanding series, *Introduction to Plato*, for the Peterson Academy, delivered by none other than me, where I make the point that one of the frustrating things about Plato is that you never know what he's thinking, because he's writing in these dialogues and all these different characters, focused mainly around Socrates, at least in the early dialogues. But Plato himself is never a character.
- 2.55** In two of the 35 dialogues, he's present, but he doesn't say anything. One of the dialogues happens, "hypo ton Platonos", underneath the Plato tree. This is Plato's little joke, that, like God, I am everywhere because I am nowhere. I don't say a word in this dialogue, but the whole dialogue is happening between Socrates and Lysias, underneath the Plato tree. So it's very hard to pin down what exactly Plato thought.
- 2.56** There's this one dialogue that comes towards the end of his, we think, comes at the end of his career, called the *Timaeus*, which becomes a very influential dialogue. It's one of the few that really survives all the way through the first millennium, and pretty much, and at least commentaries on it, all the way through the first millennium and then into the Middle Ages. That was all that was really known of Plato.
- 2.57** But it's got this creation story, where you've got this, what's called a demiurge, like a carpenter god, like an architect god, who designs the world according to certain numerical patterns. But it's not really, "In the beginning, God created the heavens and the Earth." This is not Genesis 1:1. It's more like the Mesopotamian creation myths. You've got chaotic, messy, preexisting matter that this craftsman, this divine carpenter, molds according to already-existing eternal forms. This is not from nothing. This is not *creatio ex nihilo*.
- 2.58** So on neither of the dominant Greek views do you have this idea that God freely creates the natural world, the world of science, from nothing. So why does this matter for science? Why does this difference matter for science, this development within Christian theology? Well, the thought is that, if the universe is created from nothing by God's free will, it could have been contingent; or rather, it is contingent. It could have been something else, is what it is to be contingent. Not necessary.
- 2.59** Something's necessary, it's true no matter what. $2 + 2 = 4$ is necessary. It's not contingent. $E = mc^2$ could have been $E = mc^3$. Now, it's not. It is $E = mc^2$, as it relates to this possible world, this space-time. But we can say that there is no—just as we can say

that there's no possible world in which it is not the case that $2 + 2 = 4$, we can say that there are infinitely many worlds in which it is not the case that $E = mc^2$, in which it is not the case that the force of gravity is what it is in this world. It might have been slightly different. OK, a slightly stranger world, but you can imagine such a world. It's a possible world. You can't imagine a world in which squares are also circles. You can't imagine a world in which $2 + 3 = 4$.

- 2.60** God could have made different laws. He could have made different constants, different structures. Why is this important? Because it means that you've got to get up out of your armchair to find stuff out about the world. You can't just use logic. You can't just use reason. You can't just discover the truths, the answers to something, by just thinking in your armchair, as you can with maths, as you can with logic. God could have done different things. The world could have been arranged in all sorts of different ways. You can't just rely on just your understanding. You need sense experience to be married to your understanding, to find things out about the world.
- 2.61** This is the distinction that philosophers like to make between a priori knowledge and a posteriori knowledge. Philosophers like to use these fancy Latin phrases. A priori knowledge is the knowledge that you have prior to getting up out of the armchair, armchair knowledge, maths knowledge, logic knowledge. A posteriori knowledge is the knowledge that you get after you've got up off your posterior from the armchair and started inspecting the world, a posteriori knowledge. Two different, distinctive ways of acquiring knowledge about the world, attaining knowledge about the world, which comes to be contested, and we'll discuss that in a later lecture.
- 2.62** But this is the point for now. You can't just work stuff out about the world by sitting in your armchair and doing some sums. I mean, as we saw in the previous lecture, it turns out Peter Higgs did exactly that. So it is possible with other people's observations about the world to then work it out and do some maths on it, and then make some inferences. But broadly speaking, you need more than just pure reason to find out about the world.
- 2.63** If it is the case, if you believe that the world was brought into being freely in a way that might have been different, God might freely have chosen to create a world in which $E = mc^3$, the point is you can't find out from your armchair whether it's $E = mc^2$ or $E = mc^3$. You've got to get up out of the armchair, and you've got to go and find an Einstein.
- 2.64** This thesis, this theory about the fundamental difference between the Greek approach and the Christian approach, was made very famously by the French historian of science, philosopher of science, Pierre Duhem, the early 20th century. He wrote a famous book called "Le Système du Monde," the system of the world. And his view was, effectively, that a good scientific theory for the Greeks was a logical system, a symbolic system, that helps us to classify certain laws, helps us to make correct predictions, but it's agnostic about the underlying nature of the world. It's agnostic about whether there was a God that brought it into being, or whether a God didn't bring it into being.

- 2.65** This wasn't a question that mattered so much to Greek thought. For the Greeks, the world's structure, crudely speaking, could be deduced from necessity, rational necessity, geometric necessity, as we see in Plato's *Timaeus*. Circles are perfect, therefore the celestial spheres are perfect. We can think, we can imagine a circle. Pythagoras helps us to think about these pure circles in the mind, so that's what's happening out there in the celestial spheres.
- 2.66** I mean, they can't be ellipses. Why would you have a perfect circle and then a wonky circle out there? It doesn't make sense to them, that view. No need to experiment. Of course, it's a circle out there. It's a circle in here, it's a circle out there. That's crude, but that's, broadly speaking, the habit of thought within the Greek mind.
- 2.67** That doesn't work within Christianity. What we think of, what us finite, fragile, fallen selves, on the Christian view, happens to think about stuff, circles or whatever, may or may not map neatly onto how God freely decided to bring the world into being out of nothing. Who are little old we to constrain, through our thought, how God is going to design the world? That lacks awareness and a sense of humility and awareness of the enormous gap between creature and creator that simply doesn't exist, is inconceivable to the Greek mind.
- 2.68** So the point is that empiricism, that is to say, that word in philosophy that means a stress on the importance of sense experience and how we acquire truth and knowledge about the world, is non-negotiable. It's not optional. You have to get up out of the armchair. We've got to go and find out what God actually chose. We don't know. We've got to explore what he decided to do. The difference might be, for Aristotle, why do heavy objects fall? For Aristotle, it's in the nature of things to seek the center of things, the center of the Earth, Earth's natural place. That's logically necessary, given the nature of heavy objects.
- 2.69** The Christian view—heavy objects fall because God ordained the laws of gravity, and heavy objects obey the law. Natural objects obey the laws of nature that God has ordained. He could have ordained differently. The gravitational force might have been a bit weaker, might have been a bit stronger. We're not going to know until we get up out of the armchair. So this is a very important idea.
- 2.70** Of course, the idea also that, opening of John's gospel, "En archē ēn ho logos," in the beginning, in the originating principle, was the logos, was the principle of order, the ordering principle. God is supremely rational. He's not fickle like Zeus. He doesn't do random things like Poseidon. He doesn't have bad days. Supremely rational. Creation is an act of divine reason, not arbitrary whim.
- 2.71** You can trust that there's going to be, provided you haven't been drinking too much or whatever, if the irregularities are popping up, it's probably more to do with maybe you've been drinking or maybe you're not thinking clearly, than a problem with the world itself. We should expect to have confidence in the predictability and the

intelligibility and the orderliness of nature—very different from Greek and Roman polytheism.

- 2.72** Another important ingredient is that, with this doctrine of *creatio ex nihilo*, you can say that the natural world is secularized. This might seem a strange idea. How could a religion secularize the natural world? Well, once the doctrine of *creatio ex nihilo* is in place, then you're not going to appeal to Poseidon to explain the earthquakes. You don't need to explain strange happenings in the woods by talking about satyrs and nymphs and dryads. You don't talk about the change of the seasons because Persephone has gone off in a sulk. All of that just disappears.
- 2.73** There's one God with a rational plan. We don't know what it is. It's going to take a few millennia to work it out, but we can work it out by applying our intelligence, and experiment, and marrying our understanding to our experimenting and to our experiencing. And so the whole of the cosmos gets, you might say, disenchanted. It just gets—there's no more spooky stuff anymore. Everything is naturalized, everything is secularized.
- 2.74** There are these wonderful moments and very sad, poignant moments in classical literature—Greeks and Romans just not being able to cope with this. Their entire world was organized by all these different gods and all these sacred sites and all of these ways of explaining the world and understanding the place of, their place within it. All this gets swept away properly in the fourth century. Christianity emerges as the dominant belief system and worldview of at least the western half of the Roman Empire.
- 2.75** Only God is sacred on this view. There is no Mother Nature. There's a sister nature. Sometimes she's a beautiful sister, sometimes she's an ugly sister. She's definitely a misbehaving sister a lot of the time, needs to be controlled and tamed. And human beings, as it were, the older sibling with delegated authority from God the Father, to try and control and subdue and organize the world and name it, as we see in the early chapters of Genesis. Very, very different picture of the relationship between human beings and the natural world from what we have with the Greek mind and the Roman mind.
- 2.76** Now, one good thing about nature not being sacred is that you can cut it up. You can dissect it. You can put it on the rack, as Francis Bacon says later on. Test it. It's fine. You're not blaspheming. Poseidon is going to be OK. When you start trying to create some harbors to keep his storms at bay, it's fine. Because there's no Poseidon, and that's not the proper explanation of why there are storms and why you didn't make it across from the Peloponnese to Thurii or to Naples.
- 2.77** You can manipulate, you can experiment, you can intervene in the natural processes. Creation is a gift by God. It's a difficult text, almost like a difficult poem that God has given us that we need to work out. It's OK to get the highlighter out, and the pens and

the rulers out, and maybe to cut the pages up, and just to try and understand it better. This is not sacrilege. There's a demystification of the world.

- 2.78** There's something quite sad and poignant about that, but also it certainly provides the conditions for thinking about nature as something that we can test and experiment and manipulate and use and harness to our own ends, without thinking that we're interrupting some God's plan for what he wants to do with horses or trees or whatever it might be.
- 2.79** Many other important conditions emerge, of course. The medieval universities, which are important as they emerge in the 12th-13th century. Bologna, very much a university focused on law, in 1088. Paris, theology and arts university, the 1150s. Oxford, 1160s. Cambridge, 1209. And that spread all across Europe after that. Salamanca and Leuven and Padua, Prague, Vienna, Heidelberg, and so on. This is a new institutional form. It's not under the control of the state, and it's not under control, by and large, it's not under the control of the church, either.
- 2.80** They are intellectually autonomous and independent guilds, societies, institutions. They're self-governing. They confer their own degrees. They don't need the Pope's permission to confer a degree. They don't need the King's permission to confer a degree. They've got their own legal rights and so on and so forth. This is very important. It meant they could question authority, even if it's church authority, even if it's a royal authority. In principle, they could test it. They needn't worry about political or religious interference in free inquiry.
- 2.81** Very different from China, where you have Chinese academies basically controlled by the imperial state. The curriculum is there to really, the point of the university is to prepare future citizens, future civil servants, for these very difficult examinations, testing, knowledge of Confucian classics. The point is to train bureaucrats loyal to the Emperor. That's a very good end. It served them very well. But it's not the conditions that are propitious for free inquiry, testing received wisdom, experimentation, hypothesis, bold conjectures that are absolutely the lifeblood of scientific invention and discovery.
- 2.82** Same problem with the Islamic madrasas. Again, you've got a very clear structure of authority based on "fiqh," or Islamic law, Quranic studies, the Hadith, theological doctrines, the one that we referred to earlier, of course, occasionalism. Don't get me wrong, they could be excellent. The one in Al-Azhar in Cairo, of course, is arguably the oldest university in the world. It goes back, I think, to the ninth century. But no autonomy, certainly no legal independence from the caliphate. They could be shut down at any moment by a secular authority, the Sultan, or a religious authority, the Caliph.
- 2.83** All of these different factors flowed into this confluence of ingredients that, as it were, provided the seed bed for the scientific revolution. It's the emergence, the first sparks

of that scientific revolution, that we're going to touch on, begin to touch on, in the next lecture. Thank you for listening.

- 2.84** I've heard you speak in the past in the Exodus series that the spread of Christianity was almost unexplainable and an obvious pushing force behind development of scientific and the revolution, really. It makes me think: what about some of the first scientists, like Luke, who wrote the gospel of Luke? He was a physician and scientist. Obviously, he had the freedom, and he published it like a scientist. What did that have to do in regards to the spread of wildfire of Christianity, which then led to people like Galileo and Newton, and not the other countries like you mentioned, China, Islam, and such?
- 2.85** Yes. Thank you very much. So Luke, the author of the Book of Luke, and almost certainly the Book of Acts, we think was almost certainly a doctor, a medical doctor, a physician, but not necessarily, not a physicist. We know that from a lot of the vocabulary that he uses. The way he describes a lot of the miracles involving miraculous healing and disease suggests an unusually strong grasp of technical medical vocabulary.
- 2.86** We know he was a Greek speaker, too. In the first sentence of his gospel, one of the most beautiful sentences of Greek in all 27 books of the New Testament, you're right—he does set out his method when he's writing to Theophilus, and he says, "I've taken great care, Theophilus, to set out everything that—" "heurēka," I think he says, "everything that I have seen as I've witnessed it." And if I haven't witnessed it myself, then I've checked with people who have.
- 2.87** Now, you might say that that is science, but I would say it's more, there's a stress on the importance of eyewitnessing, of testimony. It's not really an interest he's got in the natural world, or patterns in the natural world, but really, patterns in human history, stuff that human beings have done, and particularly this one human being has done, and how to understand and interpret it. I think if you'd pressed him, he would have been a bit baffled.
- 2.88** He would have understood, I think, probably, actually, the distinction between history and science; that history is a form of human inquiry that studies the one-off, that studies what cannot be repeated. Mark Twain is probably right that history doesn't repeat itself, even if it rhymes. But it doesn't repeat itself in the way that laws of nature, rhythms and patterns in nature, repeat themselves. That's a very different kind of human inquiry, because once you've decided that $E = mc^2$, then you've discovered something that is reliably a regularity in the natural world.
- 2.89** If you've established that Caesar crossed the Rubicon in 49 BC, triggering all kinds of problems in the Senate that would lead to his death in March 44 BC, that's a totally different kind of thing. It's a different—but it also relies on eyewitness, relies on experience. It's not armchair knowledge. Historical knowledge and scientific knowledge are similar in the sense that neither are armchair knowledge. You can't reason your way to whether or not it's true that Caesar crossed the Rubicon in 49 BC, just as you

can't reason your way, just through thought alone and logic alone, to the conclusion that $E = mc^2$.

- 2.90** But certainly, this idea that goes all the way through Luke, that God acts for a reason, God isn't just arbitrary, you can see patterns in the actions of this man, or in the miracles that these Apostles performed, or in the way in which that Roman governor responded to Paul, that God's hand is in this somehow. And so you might say that there you've already got the mentality that is way, way different from belief in Zeus and Poseidon, and where he could be having a bad day any day, and he could be deciding to act for all kinds of capricious and incomprehensible, unpredictable reasons.
- 2.91** You could say, even in Luke, you start to see this idea that there's a rhythm to the world, or there's intelligibility to the processes of history, which you could then see moves quite easily into this idea that there's intelligibility in the processes of nature, of science, that you can also work out, because ultimately, the whole thing is part of the creative blueprint of a single rational being, supremely rational being.
- 2.92** My question is, you laid out some of the more practical elements in Muslim and Chinese cultures that would have prohibited the advancement of science. I was curious if you could maybe lean into or briefly explain the religious traditions of Indian and/or Asian countries at the time that might have also contributed to the advancement of science or the curiosity, as opposed to just maybe the more hierarchical prohibitions?
- 2.93** Yes, that's a very good question. Within Chinese thought, really, you might say that, well, it's inconceivably complex, both Chinese and Indian philosophies, from a very early stage. There are different schools, and very few of the schools agree with each other. But crudely, you could say that Chinese philosophy, the two dominant philosophical traditions within Chinese civilization is Daoism and Confucianism.
- 2.94** Daoism is certainly a philosophy that understands the natural world and our place in it in terms of coherence and continuity. Something like rationality, you might say, though that might be a little bit too strong, but harmony—harmony, coherence, flow, continuity. You might say that lends itself to rational people emerging, coming up with brilliant ideas about how we fit together with nature and how nature fits together with itself. But without producing the Darwins and the Newtons, without any single grand vision of this is actually one single blueprint that came into being out of nothing and could have been something completely different.
- 2.95** In Confucianism, you've got all sorts of things to say about that, but as stress on authority, tradition, reason, scholarship, devoting yourself to the life of mind, a life of scholarly work. That brought with it, you might say, all kinds of advantages to a civilization committed to that thought, deference to the wisdom of the past. That brings good things, as with all deference to tradition, that brings benefits but also disadvantages.

- 2.96** I tend to think that ideas that have been around a long time are probably ideas that are more right than yours, more right than the ones you've just come up with this morning. Not necessarily, but I would give the benefit of the doubt to the accumulated wisdom of centuries of my ancestors over and above what I thought about, this brilliant new idea I've had in the shower this morning. I tend to think that ideas age in reverse. As we get older, we get weaker. As ideas get older, they get stronger. Confucianism is a philosophy that embeds that and would have conferred a lot of advantages on Chinese thought.
- 2.97** Indian thought, where to begin? I suppose you could say the singular civilizational contribution and genius of Indian thought, at least in terms of thinking about the philosophy of science, would be mathematics. What is it in Indian philosophy that lends itself in particular to mathematical genius and mathematical insight? Well, this is speculation, and I'm not a specialist in Indian philosophy, but what does characterize large swaths of Indian philosophy—not all, and there are some that actively deny it— is a sense that the world of perception is somehow fickle and somehow not real.
- 2.98** What is ultimately real is Brahma, what is the underlying oneness of being, and that what we think we can see is actually not what is ultimately real. What we see is diverse; it is a world of multiplicity, of lots of different things, that, within much Indian thought, disguises the underlying unity of all reality in the one or in Brahma. Why is that relevant to mathematics? Well, because mathematics is that field of human inquiry that, more than any other, is removed from the fickleness and uncertainties of sense experience.
- 2.99** It's really with—this is speculative, and I haven't really thought about this before, but since you pressed me, I would say maybe that mathematical conjecture, that state of thought of thinking about mathematics, is probably one that is most likely to be getting to the root of being itself, the nature of being that lies beyond the messy world of sense experience that is constantly changing.
- 2.100** You remember Heraclitus and the river. "You can never step into the same river twice," and you can't even step into the same river once. There's something about the world of sense experience that's messy and uncertain and contingent. Maybe mathematics, mathematical inquiry, is that human inquiry which is most convergent with the theological and philosophical belief that's so characteristic of Indian thought, that all reality is one beyond the world of sense perception.

Lecture 3: Scientific Thinkers

At a glance — Key scientific thinkers who shaped the enterprise.

- 3.1** Welcome to Lecture 3 in this lecture series on Philosophy of Science. In the first lecture, just to recap, we've set the scene with this bold claim of Stephen Hawking, that philosophy had been made unemployed by science, that scientists had put the philosophers out of business. And so what's the point in having a philosophy of science? What's the point in thinking about the most fundamental questions that human beings can ask themselves in anything other than a scientific way?
- 3.2** We think we have minds. We think we might have souls. Philosophers have argued about it for millennia. Now we just know it's atoms all the way down. Because when we start doing things with atoms, all these amazing things happen, and we get these incredible inventions, and make extraordinary progress, and we start manipulating the natural world. So, the mind must just be another scientific problem that we haven't quite solved yet.
- 3.3** Morality must be just another scientific problem that we haven't quite worked out yet. We've looked for courage under the microscope. We haven't found it yet, but give us the money, the research funds, the labs, and we'll get there. We'll work out how to detect courage one day. This is the great imperialism, this colonizing spirit and momentum, not just of science, but of any human discipline that does well, that becomes successful. You might say the same of theology. I even say the same of philosophy. Its success tends to go to our head. As people like to say, to the man with a hammer, everything looks like a nail. To the person armed with the scientific method, everything's got to be a scientific problem that my particular hammer can solve, can nail.
- 3.4** We showed, really, how from the very beginning, thought that we can think of as scientific or as proto-scientific, it just is already, from the beginning, philosophical thought. It's a kind of natural philosophy. It's a kind of reflection, deep form of philosophical reflection on the world, about the methods of science, about what nature might actually be.
- 3.5** We saw the influence of Aristotle, and then we moved quite quickly into the first millennium AD, the late first millennium. We looked at the rise of Islam. We glanced across at Chinese civilization, very briefly at Indian ideas in the Q&A at the end of Lecture 2. We started to think about this great question of Joseph Needham—why, in that place at that time, a time and a place that you would not have predicted, why was it that the conditions were most propitious, that the configuration of conditions were most helpful and conducive, to the scientific revolution, that is to say, Europe from roughly 1150 up to 1400, 1500, 1600?

- 3.6** We're going to be looking at the, as it were, the sequel to that period that we were looking at. You might say the late Middle Ages, as we're actually moving now into the 1400s, 1500s, actually moving now into what is called, in that self-congratulatory way, the Renaissance, the beginning of life in European thought after all the stupidity and darkness of the Middle Ages. We're looking now, really, at the run-up to the scientific revolution. In fact, you could say, really, we'll be really wading into the scientific revolution in this lecture, from Copernicus through to Newton. We're looking at Copernicus and Galileo and Kepler and Francis Bacon and the great Isaac Newton.
- 3.7** I've suggested already that the way that we think about history is—the language itself is prejudicial. There's a bias in the language of the Dark Ages of religious dogma, Renaissance, Enlightenment, science breaking free from the church, and that's how modern science emerges. It's science versus faith, science versus religion, reason versus revelation, faith versus reason, et cetera, et cetera. I think, what I'm going to try to convince you in this lecture and the next is that those are very unhelpful, crude, and false dichotomies and polarities and pairings. Actually, the story is a lot more interesting and complex than that.
- 3.8** The evolution of human thought is, broadly speaking, a much more continuous one than the historians of ideas often lead us to believe. Sure, new discoveries and new theories are posited that are, by definition, by being new, discontinuous with what's gone before, but could never have arisen but for engagement in what had gone before. So these are the figures we're going to be looking at. These are the revolutionary breakthroughs. There were many others, but these are the ones: Copernicus, Galileo, Kepler, Bacon. We'll have a look at Descartes as well, in fact—more of a philosopher than a scientist, perhaps, but a very important figure—and finally, Newton.
- 3.9** But all of these figures never refer to themselves as scientists. As we'll see, that's not a word that emerges until the 19th century. It's coined by a guy called William Whewell, W-H-E-W-E-L-L, in, I think, the 1820s, 1830s. He was master of Trinity College, Cambridge. At this stage, all of these figures, all those six figures, saw themselves as natural philosophers. Remember Newton, the Principia, or the Principia Mathematica, 1687, but it's the, as I said in the first lecture, Philosophiae Naturalis Principia Mathematica, the mathematical principles of the philosophy of nature, a strand of philosophy.
- 3.10** And all of them, even Galileo, as we'll see, Copernicus, and Kepler, and Descartes, and Francis Bacon, and Newton, all of them saw what they were doing as basically an exploration of God's creation. They were theologically motivated, sometimes in slightly curious ways, sometimes in downright heretical ways, perhaps, in the case of Newton. But God is what they're thinking. In the famous words of Kepler, "I'm thinking God's thoughts after him." That was the mental framing.
- 3.11** Now, the textbooks don't like this way, to be reminded of this, because it doesn't fit with the darkness, Enlightenment narrative, the religion and then science dichotomy

and opposition. But it's true. They were all profoundly pious, religious, mainly Catholic figures, all of them except Newton, motivated by theological commitments, as well as philosophical commitments. And not only did they not separate the sciences from philosophy, they didn't separate philosophy from theology. I mean, wondering whether or not—the question whether God exists or not isn't exactly not a philosophical question. It's one of the great philosophical questions.

3.12 And the question of what God might be like, if there is a God, is also obviously a philosophical question. Plato, Aristotle, Aquinas—none of the great philosophers would have been able to understand the idea that this is, "Oh, this is just a religious theology question." No. I mean, look, it's a question, if true affects all of reality, it may be false. That God exists, it may be true that God exists, but the question itself is not just—that's not a religious question.

3.13 Just as trying to decide what the origins of the universe were before space and time began is not itself a scientific question, because science only starts when space and time get going. You only get laws of nature with space and time. If you're talking about what was the case before space and time began, which is what the Big Bang theory tells us, this is what Stephen Hawking and Roger Penrose tell us, science can't help, because we're asking about the conditions of the emergence of science. Science can't explain that. Or maybe we can't either. Maybe nothing can explain it. But it's not a question that falls within the domain of science either.

3.14 My thought here is that the scientific revolution is, again, probably a misnomer because it implies there's some rupture. And it's true that there were big changes in the way these philosophers thought about reality and the nature of reality and the nature of God's relationship to reality. But it's much more of a transformation, I think, than a rupture, and a revolution, or maybe a reformation, you might say, or an efflorescence, an evolution, you might say. It's a scientific evolution, might be a better way of thinking about it.

3.15 I'm conscious I'm fighting a losing battle with the historians and the textbooks here, but this is my counter-narrative, which is true, so you should believe it. Why would I be saying it if I didn't think it to be true? This common myth, that science succeeds only by freeing itself from religion and from philosophy, that as soon as scientists stopped listening to priests and philosophers, they started to make progress, the reality is that the great scientific achievements of the 16th and 17th century were initiated by these pretty devout men.

3.16 The structure of this lecture—I'm going to move quite quickly through these six figures. Copernicus. I'm going to talk about his famous heliocentric theory, the extraordinary idea that Ptolemy was wrong. Not everything revolves around the Sun—geocentrism—but actually everything revolves around the Earth—I'm sorry, geocentric model—but that everything revolves around the Sun, the heliocentric model. What motivated that?

- 3.17** Galileo's experiments, Kepler's laws of planetary motion, Descartes' mechanical philosophy, Francis Bacon's experimental method, particularly the inductive method. Finally, of course, the great synthesizing of Sir Isaac Newton, perhaps the greatest of them all, but still very much a philosopher and, even more so, a theologian. As we know, scholars have been looking through his copious writings kept in the Wren Library in Cambridge, at Trinity College. It's clear that his work on physics and mathematics was just a small fraction of his other work. He was obsessed with strange, arcane theories about theology and alchemy and so on and so forth.
- 3.18** Let's begin with the great Copernicus, Nicolaus Copernicus, 1473 to 1543. A Polish canon, he's a canon in the church. He's a church official. He's an astronomer, a physicist, a physician, a mathematician. The claim, of course, for which he's most famous, is that the Sun is at the center of the cosmos and that the Earth and planets orbit the Sun. The Earth rotates daily on an axis, and this is an extraordinary challenge to 1400 years of Ptolemaic astronomy, according to which it is the Earth which is at the center. It's the Sun and the planets which orbit the Earth. This is basically Aristotle's cosmology, which Ptolemy develops.
- 3.19** One of the arguments, or one of the theoretical advantages of Copernicus' theories, was their extraordinary simplicity. The Ptolemaic system had become unbelievably complex. This is something we're going to touch on when we come to look at how scientific pictures of the world can change, how scientific theories change. This will be later in the course. One person we're going to look at is Thomas Kuhn's, his famous book from 1962, "The Structure of Scientific Revolutions."
- 3.20** What is it that forces a civilization or intellectual culture to shift, to change its ideas very radically, about the nature of the world? One of the points that Kuhn makes is that, when a theory is getting more and more and more complicated, that's not a sign, as many people in universities seem to think, of the flourishing of a field of inquiry. We need to get more grants, more money, more labs, more seminars, more professorships, because there are all these different theories now.
- 3.21** Kuhn's great insight is, no, that's probably a sign that the theory is in trouble. Just as if you got a really bad cut on your arm, and putting lots and lots of plasters on, somebody might think, "Look, gosh, that wound has been treated so well. Look at how many plasters"—what's it called in American, plasters? Band-Aids—"Look at all these Band-Aids. There's so many of them. Look how well this person's been cared for." That idea of Kuhn's, that actually, this is a sign normally of a theory that's in trouble. There's so much shoring up that needs to be done. It's probably a sign that it's creaking and it's about to collapse, the opposite of theoretical flourishing.
- 3.22** One of the great virtues of Copernicus' hypothesis of heliocentrism is that it got rid of all of the epicycles of explanation that had attached like barnacles onto the hull of Ptolemaic astronomy. It's a principle in philosophy that is an important one. Simplicity in one's theory and one's explanation, all other things being equal, is a good sign. It's

not always going to be true, but very often the simpler explanation is the more likely explanation.

- 3.23** Occam's phrase is, if you ever said it, or said something like this, that "Entia non sunt multiplicanda praeter necessitatem," that entities should not be multiplied beyond necessity. Once you've got a theory that explains what needs explaining, then, or if you say you've got two theories that explain what needs to be explained, one of them asks you not really to have to sign up to very much at all, and the other asks you to, as the Queen says to Alice in "Alice in Wonderland," to believe at least six impossible things before breakfast, then you'll go for theory one.
- 3.24** It may be that there are a lot of—it may be that it's not slam-dunk, as it were, it's not dispositive, but chances are that this simpler theory is more likely, is more probable. It's a meta-theoretical advantage to a theory, to an explanation, if it asks you to sign up to relatively few things beforehand. So there's a great mathematical simplicity. Copernicus' model is very lean and elegant.
- 3.25** But it's not some secular break from philosophy or from theology. Part of his reasoning actually is quite theological. He says that the Sun is a much nobler body than the Earth. It's the source of all light and of heat; that is to say, it's a much more plausible symbol of God, the giver of life and being and of all illumination itself, than this crummy rock that we're on. And so, why not? What's so heretical about wanting the Sun, thinking the Sun should be at the center of it all? It's much more like God. Are we saying that the planet is like God, that this? So, it should be at the center of creation, not Earth. Earth is corruptible, base.
- 3.26** He says here, in his famous *De Revolutionibus*, on the revolutions of the celestial system, the celestial orbs, "In the middle of all sits the Sun enthroned. In this most beautiful temple, could we place this luminary in any better position from which he can illuminate the whole at once? He is rightly called the Lamp, the Mind, the Ruler of the Universe." This is a theological theory of beauty, of theological aesthetics. The Sun is in the middle; that reflects cosmic order, divine design. This is part of a harmonious picture of creation.
- 3.27** So Copernicus, my point here is that he's not just a secular scientist who's just become an atheist and suddenly started to come up with these brilliant scientific theories. He's using a theological idea about the analogy between the Sun and God and the Sun's nobility and the Sun's centrality. Sun is the giver of warmth and light and life. He's applying a medieval philosophical principle, Occam's theory of simplicity. He's seeking church approval. He's not trying to go against the church. He's not going to resign as a canon.
- 3.28** So yes, this is a big revolution, a huge revolution in astronomy, the first in 1400 years. But it's not some huge rupture. It's a revolution that occurs within a broadly Christian intellectual universe, still a form of natural philosophy, you might say, investigating the

cosmos primarily as God's creation. It's not yet some kind of separate discipline, in separate science. So that's Copernicus.

- 3.29** Galileo comes a little bit later, 1564 to 1642. He's born the same year as Shakespeare, 1564. He's an Italian, mathematician, astronomer, natural philosopher, professor at Pisa, then Padua, and is responsible for some extraordinary breakthroughs, really, if you take all of his findings together. Studies in motion, most famously of all, perhaps, overturns Aristotle's theory that I referred to in an earlier lecture, that the way to explain how heavy objects fall relative to light objects is just to say that it's the heaviness that explains how fast they fall, that the speed of their fall is proportionate to their weight. That's the basic idea.
- 3.30** Galileo questions this, correctly. But the question is—he didn't think this was right, but then he thought, "Well, how am I going to test this? How do I show this? How do I try it out?" Experiment, experimentum, experiment on it. They didn't have available instruments to measure speed accurately. No stopwatches, no precise clocks. So what does he do? He comes up with a very simple idea. He's going to get an inclined plane, like a ramp. And you roll the balls down the ramp at different angles. It's not a vertical drop, but you've got a variable now. Change the incline of the ramp, and you've got the deltas, you've got the changes, and you've got the numbers. That experiment is going to generate the numbers from which you can then extrapolate, it's 45 degrees and then 90 degrees. And so he starts testing his hunch this way.
- 3.31** What does he realize, of course? That when you roll balls down the ramp at different angles, you get this dilution of gravity, motion slows, and you can start to measure them. You can measure the angle of the incline, the weight of the ball, the surface texture, and so on. And so, all of those numbers help him to form a very simple formula that becomes, as it were, his law of motion, that distance is proportional not to weight but to the square of the time of the temporal interval of the fall.
- 3.32** So what does this mean? Let's try and explain it. Imagine dropping a ball. After one second, it has fallen a certain distance. Let's say 5 meters. Double the time. After 2 seconds, how far has it fallen? Not 10 meters. Aristotle is wrong. It's fallen 20 meters, four times as far. After 3 seconds, triple the time. Not 15 meters, as Aristotle's view about the proportion between weight and distance might mean, but 45 meters, nine times as far. So what's the pattern here? You double the time, four times the distance. You triple the time, nine times the distance.
- 3.33** And this is extraordinary. It seems as if the square function is wired into the fabric of the natural world. It's an extraordinary insight, extraordinary discovery here, that nature is mathematical, that it obeys laws that are expressible, not in just crude one-to-one, two-to-two, three-to-three, but the square relation. It's this—who could have predicted that? It's a precise mathematical relationship. A striking—why not cubed? Why not just a proportion of one-to-one? No, it's the square relationship. If it's seven,

it's 49. If it's 10, it's 100. 12, 144. Nature is mathematically legible. Stuff has mathematics, precise mathematical relationships, wired into it.

- 3.34** This is an extraordinary discovery. Not an invention, it's a discovery. We're not imposing maths onto the world. We're reading the maths, maths, off the world. Numbers are wired into the world. It's these immaterial, necessarily existing things, numbers, that little two is somehow wired into stuff. These two completely different things, mathematics and nature, are somehow completely intertwined.
- 3.35** So, why is this revolutionary? Well, how does he get there? It's not just through clever observation, but it's a controlled intervention, controlled manipulation of the world, changing the incline of the plane, changing the weight of the ball, changing the surface, the texture of the surface, and so on. He's testing the world not by observing it, but by manipulating it, changing the variables, testing it.
- 3.36** And so, he's combining these three great ingredients in science—empiricism, you are observing, he is observing, how things occur in the world; mathematics, this is the armchair, armchair thinking; and experimentation, he's doing stuff on the world. He's not just observing it, he's manipulating it, as I said.
- 3.37** Other important discoveries—that there are mountains on the moon. Nobody had considered this. But he comes up with this theory, that the moon has mountains. It has a rough cratered surface. You might think, well, of course, of course, it does. We can basically see it on a full moon. The moon's sort of close, particularly a desert landscape like Arizona. Of course, you can see that there are things up there, that it's not a continuous smooth surface.
- 3.38** But this was a profound challenge to Aristotelian astronomy and cosmology. The moon is a heavenly being. It's supposed to be perfect, completely smooth, unchanging, geometrically ideal and precise. And so, if it turns out that the moon has imperfections, just like little old us, like the world we're in, the planet we're on, then somehow it's the same sort of thing, and it's not perfect. And the whole of Aristotle's division between the sublunary world, everything below the moon, and the superlunary world, everything above the moon, disintegrates. The whole cosmology that had been in place for nearly 2,000 years can't be right.
- 3.39** That's not just a little detail. It's absolutely central to the two-tiered structure of Aristotle's cosmos. Awkward. Maybe the heavens and the Earth are made of the same stuff. This is the implications of his thesis. He discovers that Jupiter has moons, four moons. He calls them—and this is a very politic, diplomatic move—the Medician stars, named after four Medici patrons, just to make sure, if he was going up against the Pope, he needed some secular princes on his side.
- 3.40** Why was this important? Well, if Jupiter's got its own moons, then not everything revolves around the Earth. There's Jupiter-centrism, too. It actually has got four times as many moons as we do. It can't be that the Earth is—that there are orbiting dynamics

that orbit around something that is not the Earth. There are other centers of motion in the cosmos. Maybe we're not in the middle of it. Maybe we're not at the center of absolutely everything, after all.

- 3.41** Galileo is, again, in many ways, like Copernicus, begins to think that nature is written in a grand book. "La filosofia," rather, natural philosophy, "la filosofia è scritta in questo grandissimo libro, che continuamente ci sta aperto innanzi agli occhi (io dico l'universo)." Philosophy is written in this grand book, the universe, which stands continually open to our gaze. "Egli é scritto in lingua matematica." It is written in the language of mathematics.
- 3.42** "E i caratteri son triangoli, cerchi, ed altre figure geometriche, senza i quali mezzi è impossibile intendere umanamente parola; senza questi è un aggirarsi vanamente per un oscuro labirinto." It is written in the language of mathematics, and its characters are triangles, circles, and other geometric figures, without which it is humanly impossible to understand a single word of nature. Without these, without the language of mathematics, the scientist wanders about in a dark labyrinth.
- 3.43** This is mathematical Platonism, where the forms are wired into nature. And you can deduce physical truths from mathematical reasoning. God is like a rational geometer. Part of the creative plan and the creative energy of God is structured or is mirrored in mathematics. Now, Galileo famously gets into a bit of trouble, actually twice. The first time, in 1616, is more of a warning. A lot of these, Copernicus' books get put on the Index Librorum Prohibitorum, the index of prohibited books.
- 3.44** And one of the cardinals meets with Galileo and says, "Look, if you insist on this heliocentrism theory to do some useful calculations, we won't ask any questions, but don't say this is a proven physical fact, please. We need a better proof before you overturn the whole Ptolemaic system."
- 3.45** Now, is this a big clash between secular champions of science and reason, versus the obscurantist, tyrannical, authoritarian diktats of the Church? It's hard to see how. The debate is, remember, whether or not Ptolemy, nothing to do with religion or Christianity, just this Greek pagan philosopher, inspired by Aristotle, whether or not his astronomy was right. And if Galileo was right, then Ptolemy was wrong; heliocentrism was right, geocentrism was wrong. This isn't like the doctrine of the Trinity. It's not some central Christian belief. I mean, who knows how God structured the cosmos, if there is a God? Maybe Ptolemy was thinking God's thoughts after him; maybe Galileo was. There's nothing in the Bible that explains, that is that insistent. There are a few verses in Joshua, maybe, but it's not decisive. No decisive scriptural basis either way.
- 3.46** So, that's the first time he gets into trouble, 1616. Then again, 16 years later, where he publishes a famous, very famous work, beautiful work, called "The Dialogue Concerning the Two Chief World Systems." This is the book that really gets him into trouble, but it's a masterpiece of Renaissance literature in many ways. He thought the Pope had given

him permission. Pope Urban VIII, who was a friend of his, says, "Yeah, try it out. Why don't you just have this dialogue, like Plato did, get the different characters to approach the different models in different ways and see what happens?"

- 3.47** The trouble is, Galileo writes the dialogue, and it's very obvious that it's the character who's supporting his heliocentric theories that wins. Even worse, the character defending geocentrism uses the exact arguments that the Pope himself had made to Galileo in private. Awkward. Not diplomatic. So he's put on trial by the Inquisition, 1633, forced to recant, kneel, and state that the Copernican view, the heliocentric model, is false, contrary to holy scripture. His book is banned. He's put under house arrest—house arrest, he's not put into prison. He's living in a very comfortable villa, actually, near Florence, not a dungeon. He received guests.
- 3.48** He wrote another dialogue. It was quite risky, but he went for it, to two new sciences. Of course, there's the famous legend that, after he had been forced to recant, he muttered under his breath, "Eppure, si muove." And yet, it moves. And yet, it moves in a way that confirms my theory to be true. Probably false, sadly, but it captures that, his heterodox, defiant spirit.
- 3.49** So that's Galileo. And again, I don't want to belabor the point, but this isn't some surge of secular skepticism. It is still a theological disagreement about how to understand the creative order. Galileo's view is that we should really be reading the Book of Nature as seriously as we should be reading the Book of Revelation and the Book of Scripture. The two can be understood in a way that don't contradict each other. Both reveal truths. Both are authored by God. It's not as if God inspires the Bible, but has got nothing to do with the world in which the Bible arose. So that's Copernicus, Galileo.
- 3.50** Kepler, the third in this triumvirate, develops laws of planetary motion, and indeed does so in a series of brilliant moves, develops very precise quantitative mathematical laws, again inspired by deep theological commitments. Thinks of God as a geometer, as a musician. Creation exhibits mathematical beauty. God orders the world according to mathematical principles, mathematical harmonies, in particular. We don't need to go into too much detail about Kepler, just to say that he's a very important figure. Again, a pious figure, and somebody who takes observation and experimentation to new heights.
- 3.51** The second triumvirate, the second set of three brilliant men: Francis Bacon, René Descartes, and Isaac Newton. Let's start with Descartes. I lectured on Descartes in another Peterson Academy lecture series, in the Philosophy of Mind lecture series. He's a towering figure who appears in different philosophy courses. 1596 to 1650, he authors a number of very important texts, but "Meditations on First Philosophy," 1641, is probably his first most important and totemic text.
- 3.52** His significance in the philosophy of science is that he divides the world up into two fundamentally different kinds of thing, and understands that the first is what he calls

"res cogitans," the thinking self, what we as subjects do. Whenever we use the word "I" or we use the word "you," we're picking out the thinking thing. You might say, in another language, the soul. Then the rest of the world is made up of "res extensa," material stuff, extended matter that you can measure and quantify in space and time; you can apply geometry to, the objective world, you might say.

- 3.53** And that's it, that's it, nothing else. So there are no qualities, as Aristotle insisted, inherent in matter: no colors, no smells, no heat, cold, in matter itself. These just exist in the mind's experience when matter affects our sense organs in particular ways. But it's not something—heat and cold and colors and sounds are not wired into the world itself. It's just part of the way in which we interact with the world.
- 3.54** We might say there are no green apples. There are simply objects in the world, and there are certain objects in the world where, when light refracts off that object and hits the back of our retinas, we see the color green. A dog is not going to see the color green. It's not because, as it were, it can't see the greenness of the apple; it's simply that it's not in the appropriate relationship of perception to the apple. It's to do with its retinas, not just the apple, or not just the material that is conducive to light refracting off it in a certain way, that when it comes onto it, refracts onto our retinas, we see the color green.
- 3.55** All physical change, none of this Aristotelian stuff about potency becoming actual. It's just matter in motion. That's all it is: stuff plus chance plus laws. Mathematical laws. Just pure mechanism, pure mechanism. "Give me matter and motion," he's alleged to have said, "and I will construct the universe." Or everything physical is reducible to matter in motion, including animals, because animals don't have res cogitans. Animals are just complex biophysical machines, to be explained in terms of external stimuli and internal response and behavior and so on and so forth, as our bodies are to be understood. The difference is that we have within us a thinking self, a res cogitans.
- 3.56** There are no final causes in nature. It doesn't make sense to talk about our old friend the acorn being oriented towards the oak tree. Planets don't orbit in order to achieve anything. Apples don't fall to reach the center of the Earth. They're not seeking the center of the Earth. It's just stuff following mechanical, mathematically describable laws.
- 3.57** This is a metaphysical revolution in the sense that it overturns a lot more of Aristotle than Copernicus, Galileo, and Kepler had done. This strikes right at the taproot of the whole metaphysical architecture that had dominated Western thought, broadly speaking, since the fourth century BC. There are no longer four causes, as Aristotle said—a formal cause, material cause, final cause, and efficient cause.
- 3.58** Very quickly, for Aristotle, a full explanation of, say, David's statue of Michelangelo wouldn't just be the mechanical process of Michelangelo chipping away at the marble to make the statue. Aristotle says, "No, you've also got to explain the material cause."

The explanation of Michelangelo's statue of David has to make reference to the Carrara marble that it's made out of. Without the marble, there's no statue. So that's a legitimate explanation. You need that.

3.59 The formal explanation, that is to say, the idea in Michelangelo's mind, without which it would just be a lump of marble. It wouldn't have any form. It would have no shape. Maybe he sketched it out and hung up the sketch in his workshop. That's the form that he had in his mind as he was chipping away. Aristotle says, "You need to include that."

3.60 Then there's the efficient cause; that's actually the process of hammering it away and being in the factory and working at it. That's the efficient cause, actually the productive cause, causal explanation.

3.61 Then there's the final cause, the fourth cause, which is the final cause, which is, "Why did he make the statue in the first place?" Aristotle thinks this is the most basic cause. If you're not including why he's decided to make it, as you go up and get the marble from the Carrara quarry and haul it down and sketch it out and go through the whole difficult process, if he's not doing it for any reason, he must be doing it for some reason. Maybe he needed the money, and maybe he wanted to be nice to the Pope, whatever. There was some personal, some motivating explanation for why that statue ended up being what it is.

3.62 Descartes gets rid of three out of those four, straightaway, at a single stroke. The only legitimate scientific explanation is the efficient, productive explanation of causation. Actually, the hammering away and the chiseling away of the marble in the factory, in the workshop, that's it. It's not legitimate because it's just matter in motion. Matter in motion doesn't care about Michelangelo's relationship to the Pope or how much money he needed that year. Matter in motion doesn't care about the creative blueprint in his artistic imagination. Matter in motion doesn't even care about matter if it's not in motion. It's simply matter in motion. That's the only explanation that is legitimate.

3.63 This is a huge shift, huge shift. It enables much of what came after, particularly for Newton's mechanics. So this is a very rationalist way of thinking about nature; that is to say, it puts a primacy on reason over sense experience. Descartes says, "We can know things indubitably," for sure. He says even if there was a "malin génie," even if there was some evil demon, trying to deceive me about every single thought, there's one thing he couldn't deceive me about, and that is that I have thoughts.

3.64 Because if I had no thoughts, how could he deceive me? He needs to deceive my thoughts, right? He's got to lead me into some dodgy thinking. So I must have thoughts. Even if it's skepticism all the way down, even if I'm just a brain in a vat being manipulated by mad scientists, I'm still a brain in a vat. You can't take that away from me. It must be the case that I think. "Je pense, donc j'existe." "Cogito, ergo sum." "I think, therefore I am." If I am nothing else, I am a being that thinks. I'm a being that holds thoughts. And nothing can deceive me. Nothing can deceive me about that. Even

if I'm being deceived about the entire content of my thoughts, I need to have thoughts to be deceived.

- 3.65** Now, that sets the whole of Western philosophy off on a whole, very skeptical, down a very skeptical rabbit hole, unfortunately. But that's for another lecture series. The point for our purposes is that Descartes thinks he's found a foundation for certainty and thought within the self, not outside there in the world, but within the self, within the thinking self.
- 3.66** Then he tries all these different moves that are thought to be, by many philosophers, not terribly successful, where effectively he appeals to God and God's benevolence and God's love and his justice. He's saying, "Well, God being a God of benevolence and love and justice wouldn't want me to be deceived, anyway. So not only can I be certain that I think, I can probably be maybe a little bit less certain, but I can trust that God is, for the most part, going to underwrite or guarantee the reliability of my thinking about the world." This is where the argument gets a little bit weaker. God's not a deceiver. I can have a clear and distinct, "*une idée claire et distincte*," an idea that is clear and distinct, about God as a perfect being who will ensure that my investigations and my reasoning are, broadly speaking, reliable.
- 3.67** So, what can we say about Descartes? Well, he's a philosopher first. This is really the beginning of modern Western philosophy. It's the beginning of, if you like, it's the first time we really see a shift in the intellectual universe of the West, from being theocentric, centered on God, to anthropocentric, centered on human beings, centered on the subject, starting from what I think. "I think, therefore I am." "*Cogito, ergo sum*." Before that, it was really, "I'm thinking God's thoughts after him," as Kepler says. Our thinking is tracking God's thoughts. The certainty is not with me; the certainty is with God, and I will try to think his thoughts after him.
- 3.68** Penultimately, coming now to Francis Bacon, our first of two Englishmen, English philosopher, statesman. He's a Lord Chancellor, actually, under James I, a very important role in the English Constitution. He writes in 1620, famously, the "*Novum Organum*," the new instrument. This is nothing less than a proposal or a manifesto for a whole new way of doing natural philosophy, replacing all of Aristotle's *Organon*, the six logical works of Aristotle.
- 3.69** Part of what Aristotle insisted on was that, when we're doing science, when we're investigating the world, we start from supposedly self-evident principles. "*Ta endoxa*" is what Aristotle calls them: essences, definitions, axioms, rules. Then you deduce the conclusions from them. It's, you might say, the armchair method. In many ways, it could work very well.
- 3.70** Bacon says no. First principles are often wrong, he insists. Hasty generalizations, limited experience, wishful thinking. If you start with false premises, first principles that are false, you're never going to get to conclusions that are true. "Garbage in, garbage out,"

you might say. Bacon says no, you've got to put nature on the rack. You've got to put nature to the test. That's where you get your principles: from testing nature, not from starting out with your ideas in the armchair, and then going out, and then trying to impose your armchair ideas onto the world beyond the armchair. You've got to come up with a systematic collection of observations. You record things carefully, comprehensively, and then you don't jump to hasty conclusions or hasty generalizations. You collect data first, then you cautiously generalize.

- 3.71** For Aristotle, you could say that the goal of philosophy is a kind of contemplation of the world, "a theoria" in the Greek, contemplation of divinity, understanding for its own sake, wisdom, contemplation of truth. For Bacon, there's a whole shift in posture. For Bacon, natural philosophy is about the mastery of nature, controlling nature, technological control, practical benefit, improving the human condition. This is a very different way of thinking about man's relationship to the natural world.
- 3.72** "Scientia potentia est." Science, knowledge, "scientia" just means knowledge in Latin. Knowledge is power. That's what natural philosophy should give us. It should give us power over nature, not just understanding. It's got to be useful. It's got to have practical benefits. This is a revolution of sorts. This is a very dramatic break, moving away from what is the ultimate nature of reality, the classic philosophical approach to how do we gain reliable knowledge about the world, and how can we use it to make life better for ourselves? How can we control it? This, I think, is an important shift in the journey from philosophy to science, from natural philosophy to the natural sciences.
- 3.73** Finally, to conclude with the great Sir Isaac Newton, 1643 to 1727, I think he's born the same year that Galileo dies. He's a natural philosopher, a mathematician, an alchemist, a theologian, and seems to have taken his alchemy and his theology much more seriously than his mathematics and physics. He really did seem to care a lot more about them. He develops, of course, in the Principia Mathematica, in 1687, three laws of motion. The first law, the law of inertia. Second law, you might have done any physics at school, force equals mass times acceleration, $f = ma$. It's quantitative, mathematical, precise.
- 3.74** Again, those mathematical relationships that Galileo saw, when he saw the relationship between distance and the square of the temporal and time squared. If you've got the mass and the acceleration, you can work out the force. If you've got the acceleration and the force, you can work out the mass, and so on. There are mathematically precise relationships that are wired—that are laws that affect stuff, mass.
- 3.75** Not, as Aristotle says, it's the laws of natures, not the laws of nature. Everything has a nature, and it has its own little set of laws. The acorn has its own little set of laws that are particular to it, included in which is something involving an oak tree. This is a completely new shift. It's not the laws of natures, it's the laws of nature, these laws that govern all of nature. These laws don't care about the different kinds of stuff and essences, whether it's acorns or walnuts, whatever it might be, it doesn't matter. This is

just matter in motion, subject to universal laws that are blind to the kinds of things that they are.

3.76 Then, of course, a third law, that every action has an equal and opposite reaction—very often true to life, true in life as well. The universal law of gravitation, which states the law of gravity, where the force between two objects will be inversely proportional to the square of the distance between them. So again, a very striking, mathematically precise relationship, not too far off Galileo's. Why squared and not cubed? And that same law affects the apple falling from the apple tree, as the moon orbiting the Earth. It doesn't matter what object the moon is. It doesn't matter what object the apple is. The laws govern them, no matter what. This is the big shift.

3.77 What these six figures give us is an unprecedented degree of predictive success when we're applying these hypotheses and laws and theories and mathematical equations to absolutely anything. As I said, $f = ma$. You can just go around, and you can just keep applying, and keep finding that same configuration of mathematical relationships. As with Galileo, the universal law of gravitation, everywhere you look, it didn't matter what—two objects, didn't matter what those objects are. But every mass attracts every other mass, as I said, whether it's an apple or a moon, with a force that's proportional to the inverse of the square of the distance between them.

3.78 This permitted all kinds of important breakthroughs, thinking about planetary motions, the tides, comet trajectories, variations in the moon's orbit, and so on and so forth. Everything could be described by these mathematically precise laws. Even Newton's obsession with alchemy needs to be understood in this context. The modern view is that, what was he doing wasting so many years of his life, writing books about how you could make gold out of nothing, out of rock, trying to transmute base metals into gold? This is the philosopher's stone.

3.79 But I think the way to look at it is, he's investigating God's powers in matter. How does God structure copper in a way that allows copper to be transformed into other things, can be transferred into, transmuted into other things? And of course, this is before there's a complete understanding of the nature of elements versus compounds and molecular compounds and so on and so forth. So it's not a—alchemy is not irrational, given the presumptions and given the context of the 17th century. It's just a different framework. It shows that, for Newton, still, philosophy, theology, it's all part of a single quest to try to understand God's creation, God's action, the way God works in matter.

3.80 So, there we have it. And we've seen in these six figures huge leaps forward in experimentation, in the inductive method, the post-armchair method, as opposed to the armchair method, of Aristotle, the role of mathematics, and the way the language of mathematics, scritta, "egli é scritto in la lingua matematica," everything in nature is written in the symbolic language of mathematics.

- 3.81** Huge changes in substantive contributions to science. Heliocentrism: the Sun is at the center of the cosmos, not the Earth. Laws of motion, with Galileo and Newton. Universal law of gravity, with Newton. The mechanical philosophy in Descartes and so on. The unification of the heavens and the Earth against Aristotle. But actually, no, the moon is the same kind of thing as the Earth, at a single stroke, overturning Aristotle's two-tier cosmology. The elimination of final causes, the importance not just of experience, but experiment, manipulating and trying things out on the world.
- 3.82** This really is the scientific revolution. In the next lecture, in Lecture 4, as we move to the end of the first half of the series, we'll start to look at how this scientific revolution culminates in the 18th and 19th centuries, and actually, into the 20th century, too, to the point where it begins to make sense for Stephen Hawking to declare that philosophy is indeed dead, and that science claims the entire domain of factual knowledge. Thanks for listening.

Lecture 4: Origins of the "Scientist"

At a glance — The origins of the very idea of the 'scientist'.

- 4.1 Welcome to Lecture 4 in the lecture series, Philosophy of Science. We're coming in Lecture 4 to the end of the first half of our journey, of our inquiry into the nature of science—what science is, what it's for, how it helps us or doesn't help us understand the world. We're coming to the end of, really, the historical part of the story, and in the next lecture, we'll start to look at specific conceptual problems and debates and flashpoints in the philosophy of science, as science beds itself down as its own freestanding discipline, its own freestanding way of thinking about the world.
- 4.2 We'll just chart the emergence of that science as we understand it now, the science as we understand it today, the professionalization of this part of philosophy, natural philosophy, from gentleman amateurs to seriously brilliant researchers discovering new things about the world and forging extraordinary progress in medicine and in engineering and in understanding the very nature of the world. These are enormous leaps forward in the history of ideas.
- 4.3 We'll look a little bit at that. How did that happen? I mean, remember, we were talking before about that decisive century, the 17th century, the so-called, the self-congratulatory Enlightenment; and really, a lot of the big ideas about the scientific revolution was unleashed, but a lot of the ideas sustaining it were really beginning to bed down and become much less heterodox and controversial.
- 4.4 Oddly, though, another thing that the Enlightenment brought with it was the emergence of a philosophical skepticism about the very building blocks of science itself. David Hume, 1711 to 1776, famously developed a skepticism towards causation itself, the glue between objects that explains why things knock into each other and have a causal influence over each other—cause and effect, the most basic idea in the whole of nature, you would have thought. Hume starts to introduce, as we will see, some skepticism about that.
- 4.5 This is an irony about the Enlightenment and the scientific revolution: as the scientific revolution was taking off, a lot of the seeds for philosophical skepticism were being sown at the very same time, skepticism that we can really get to know the world at all, that maybe it's just a construction of our own minds, because there's been a big shift from a God-centered way of thinking about philosophy, to a subject, individual-centered way of thinking about reality and autonomy. We lay down laws for ourselves and we make our own decisions and so on and so forth.
- 4.6 So, how does this shift happen? There are certain crucial moments. I've mentioned already Isaac Newton's famous "Principia Mathematica" in 1687. But probably the next shock, as it were, to the old way of thinking about science and doing science, was 1859, with the publication of Charles Darwin's "The Origin of Species." There are other similar

figures like this, in thinking of Lyell and his extraordinary discoveries in geology in the 1820s, overturning many, many assumptions about the history of the Earth and the nature of geological change and so on and so forth.

- 4.7 But it's really Darwin's *Origin of Species* in 1859 that naturalizes so much of what had gone before. It feeds into—he really seems to have decisively put away with theologically motivated appeals to design in nature, or Aristotelian ideas about there being purposes in nature, that the acorn, as we've discussed, has as part of its destiny to be an oak tree. All of that vanishes, and the vanishing accelerates with the publication of Darwin and the broad acceptance of Darwin's theory of evolution by natural selection.
- 4.8 Another factor to bear in mind is the Industrial Revolution. So, as it were, the real-world impact and astonishing success of this kind of human inquiry was beginning to become overwhelmingly clear in terms of all of the inventions and everything that emerged in the 19th century, in the Industrial Revolution. So, enormous and highly productive factories. The railway, of course, the steam engine, the emergence of the telegraph—it was extraordinary. Imagine going from a point when getting a message from London to Delhi took three weeks to five minutes. We talk about the internet being revolutionary, but that single change, suddenly, you could communicate, albeit in a limited way, by cable, with a telegram, to Delhi in the space of a few minutes and back.
- 4.9 So the world is shrinking in lots of ways. It's getting a lot more interconnected in the 19th century, mainly through the so-called Pax Britannica. What this did was, it gave science, it invested science with an authority that was hard to deny. It's the scientists that are helping us get these steam engines together. It's the scientists who are discovering this about the nature of ultimate reality in the laboratories and so on and so forth. So, they've got a significant authority by this period, by the second half of the 19th century. It's much more a profession, with university courses in it, university-funded research, peer-reviewed journals, conferences, all of the characteristics of a guild, of a guild, of a collection of human beings dedicated to serious inquiry into an important aspect of the world. So technological success has a big factor to play.
- 4.10 Then you have the emergence of some skeptical philosophical positions which say, really, science is the be-all and end-all. The philosopher's got nothing to say. You should just stay in the backseat of the car, occasionally comment, maybe, but not on the directions. That's the idea that begins to emerge now, that philosophy is something that has had its day, it's had its millennia, it's had its 3,000 years, but it's time to retire it, and it's we, the scientists, who are going to step in and take over. We're driving now.
- 4.11 Think for a moment about the word scientist. We might think of it as a word that has been in use for hundreds and hundreds of years. In fact, that's not true. The word is first used by a guy called William Whewell, who I've mentioned once in the lectures before now. He was Master of Trinity College, Cambridge, one of the largest colleges in

Cambridge, certainly the wealthiest college in Cambridge, founded by Henry VIII. It's won more Nobel Prizes than France. It's been a very impressive incubator of scientific genius over the centuries, including, most famously, Sir Isaac Newton.

- 4.12 You're seeing the emergence of an infrastructure for these different places for science to flourish. William Whewell coins the word "scientist" in 1833. Some people resisted. They thought it sounded too much like dentist, but eventually it was accepted. Really, by 1900, that's the word that everybody's using pretty much to describe the people we would describe with the word scientist.
- 4.13 Why does this matter? Well, language is always a good indicator, or a lagging indicator, as the economists like to say—there's always a bit of a time lag—but language is a lagging indicator of changes at the level of ideas. And so the fact that this new word is catching on—scientist—meant that it had something to catch on to. There was a phenomenon that it was describing, namely an increasingly professional class of experts who are not gentleman amateurs, who are not doing natural philosophy in the bath on a Saturday morning or writing it or whatever. But this is a profession.
- 4.14 This is a class of society that has important things to communicate for the flourishing of our society, for the future of our society, for the prosperity of our society. There needs to be infrastructure around that to recognize it and embed it within society. You institutionalize it. You have journals, you have chairs at different universities, you have maybe senior advisory councils of scientists helping with the government, particularly over something like the COVID-19 pandemic, and so on and so forth. This is the transition that we see very dramatically in the 19th century.
- 4.15 You can say that natural philosophy starts to fragment and break down into the different disciplines that we now recognize: physics and biology and chemistry, of course, huge leaps forward there. Geology, too, I've mentioned already. That's starting very early, actually, back in the 1830s. Then there's more and more sub-specialization under each, all of those different—you think of physics, there's now all kinds of different corners of physics, different research projects like string theory. It's a very long and expensive research project that doesn't seem to be getting anywhere, but the infrastructure has got to be able to make room for that, risky experiments and theorizing.
- 4.16 The whole architecture of this space of human thought is becoming much more precise and fixed in the 19th century. You start to get, interestingly, lots of, in London in particular and in Edinburgh, gentlemen's societies, clubs, so famously the Royal Society, which is 1660, very early. Its Latin motto is, famously, "Nullius in verba," which basically means, in Latin, don't take anybody's word for it. Don't take anybody's word for it. That's the motto. That's the spirit of the Royal Society of scientists and its very, very distinguished membership still, and it's still going strong. There were lots of equivalent clubs and societies all around Europe and America.

- 4.17 You start to see these philosophers emerging who are beginning to say, "It's time to make ourselves redundant. It's time to make ourselves unemployed. We're going to retire philosophy now, and we've got to, as we go, work out a philosophy that explains why philosophy needs to dissolve itself in the face of the overwhelming power of science. We need a theory about this." It's a good little example that you can't stop doing philosophy. Even if you're trying to bring the whole thing, all the philosophy down, you need philosophical arguments to bring philosophy down.
- 4.18 But figures like Auguste Comte, the French, not a lunatic exactly, but certainly a very violent and sometimes slightly disturbed Frenchman, developed a very interesting age of thought thesis about human beings, going back to human origins. To begin with, there's the theological phase, and then there's the metaphysical phase, and finally, there's the positive sciency phase. This is how we're evolving as a species.
- 4.19 So, the theological phase is: Zeus sent the rains, or Poseidon started the earthquake or the sea storm. We talked about these appeals; we talked about this reasoning in the very first lecture, where we were talking about the emergence of a skepticism towards that point of view. Or maybe it's just water. Maybe the world is all just made out of water. Maybe we don't have to appeal to Zeus, and I'm wondering why he's having a bad day today. We've got no way—what Zeus is going to do is always going to be unpredictable, as we know from Homer. So that's already a big shift.
- 4.20 And so, Comte is not locating it there; he thinks it happens more recently. But then we move to the metaphysical stage, the ideas that Plato and Aristotle and others began to develop, which is the idea of essences, the idea of purpose and function and form, and those ideas. Then, finally, he says there's going to be this enlightened, positive, sunlit uplands of this trajectory of human thought. That's really the scientific stage, we can just describe all regularities, all particles, make predictions. We don't need to appeal to anything other than that. We don't need to appeal to theology—phase one—and we don't need to appeal to philosophy, metaphysics—stage two.
- 4.21 You've got to stop all this looking for absolute notions like the origins of the universe and what causation between things actually is, how to understand it and so on. What is life about? What is the universe about? Was it designed? What is life's purpose? None of that matters. It's just the business of science to say, you describe the phenomena, you observe patterns, observe regularities, look at the mathematical laws, that's it. That's all it is.
- 4.22 No more talk about meaning and mind and morality and all those complicated things. We've done away with childish things, as St. Paul says in another context. We've grown up now, we've left home, and we can explain everything. And anything that we can't explain, we'll just explain away. We'll say that at some point we'll discover a scientific analysis of courage, or the number two, or God. It's all in the science.

- 4.23 What that's led to is the emergence, as I said, of a philosophy that is almost an anti-philosophy. It's a philosophy that says that we shouldn't be doing philosophy. Philosophy should be retiring and allowing the scientists to take over. And so, a big philosophical battle blows up about this. The name for this camp, the camp that believes that all truths are scientific truths, is naturalism, or you're a naturalist. Very important to make the distinction there between the way in which the word is used in philosophy and the way in which it's used elsewhere.
- 4.24 These are not philosophers who like to throw off their clothes and run around, like Diogenes the Cynic liked to do, to shock, "pour épater les bourgeois," to shock the bourgeois. Naturalism is broadly this view; there's lots of different kinds of naturalism, but basically, the idea is that all reality is explicable in scientific terms. The only truths that you can know are scientific truths, maybe some logical ones that are just true because they're true. But that's got to be the foundation. That's got to be the foundation. Everything is a scientific truth, or you can collapse it down into a scientific truth, or explain it away in terms of some other underlying, more basic scientific explanation.
- 4.25 Sometimes you hear the word "scientism" being used. Scientism is the idea that all we can have—that science is the only source of knowledge. It's the only real source of knowledge about the world. It's a very popular view, but in many ways a very radical view, a very radical view. Because in saying that all truths are scientific truths, as I've mentioned in a previous lecture, you've got to think, well, you've got to deny that mathematical truths are true. Or you have to explain why they are ultimately to be understood in terms of science, in scientifically explicable terms, but what would it even mean to explain maths in scientific language? It's just a completely different domain of human thought.
- 4.26 Similarly, it's a very high price to pay, the naturalist has to pay, when it comes to morality. It seems as if all truths, whatever else moral truths are, they're not scientific truths. There's nothing in the science that tells us how we ought to behave. It just describes reality. It just deals with "is," not "ought." That's just the way the world is. There are these behaviors, and there are those behaviors, and there are these developments, and this is the most comprehensive and deepest way we can understand what's happening descriptively, scientifically, making no comment on what ought to be done and all that kind of thing.
- 4.27 That's quite radical, that there are no moral truths; they're just maybe feelings, or they are conventions, or they're rules that a given society decides to come up with, or a parliament decides to come up with, or a president decides to pass. That's all they are. They're just to be understood in scientifically acceptable ways. You can't evaluate, really, these judgments, moral judgments. So that's a big shift with the emergence of naturalism.

- 4.28 Of course, there's questions about mind and consciousness. Does neuroscience do away with the idea that we are conscious, free subjects that experience the world in a way that no mere object could? There's something distinctive about it that is unique, private, that only you have private, privileged access to, namely, what it is like to be you. That's not something that any scientist could describe, because science just deals in "he, we, they, one," what anyone can read and understand and publicly test.
- 4.29 But what it's like to be you could never, by definition, be publicly shareable, because if somebody else read it in the same way that you understood it, then they would be you. But they're not you. There is something that it is like to be you that can't be translated into third-person public language; there's something irreducible there. Or at least that's a very plausible intuition. It's not an intuition that the naturalist, however, can make any sense of, because, precisely, because scientific truths are not interested in the subject. There is no subject in science. It's just objects. It just studies objects. That's what it studies.
- 4.30 But such is its success, philosophers have a counter view. Well, yeah, that's all it studies, and that's all there is. It's all that there is, that science is studying. You actually get these philosophical views, which are very contentious. They may be right, they may be wrong, but they are certainly contestable, and they've certainly not been accepted for many, many centuries in the history of Western philosophy leading up to that. It's a very unusual view, actually, a really reductive materialism.
- 4.31 Epicurus, in the third century BC, you might say, was this way inclined. The Stoics as well, who emerged around about the same time as well, but certainly not a dominant view. The dominant view was, if anything, matter was the least real stuff. It was because it was here today, gone tomorrow. It's always changing, it's rusting, blowing up, or whatever it might be.
- 4.32 Whereas the real things are the angels, numbers, spirits, gods. I mean, they don't rust. So they're more real. That's their view. Whereas for us, anything that you can't see and touch and smell may exist, but not really sure and can't really see it. What's the evidence? So it's a complete, as it were, 180. It's a flip-around in terms of the different emphases that an intellectual culture and a popular culture places on the relationship between reason and sense experience. There's always that balance that has to be struck.
- 4.33 So, questions of meaning seem to go out the window. Questions about freedom. Am I free? That becomes very difficult within a philosophical world where it's just scientific truths. Why? Because what it means to be free, what it is to say that I am free, in philosophical terms, is to say that there is an agent that is causing things in the world, in a way that is not determined or shaped by the complete set of causes and effects running up to this moment where the agent initiates an action in the world.

- 4.34 You can explain the agent in neuroscientific terms, in terms of heredity and genes and all the different causal webs that he's been in over the course of his life, and still, you would not know what he's going to do. You could have complete knowledge of all the science leading up to this action, this one person takes. The person who's committed to the idea of free will and freedom is going to say, the person who knows all of that is still not going to know what the person is going to choose; that there are truths that the omniscient neuroscientist does not know.
- 4.35 That's one fairly standard way of thinking about what it is to be committed to the idea that we're free. So the claim that we're free just is the claim that science is explanatorily inadequate when it comes to predicting and describing agent behavior, that is to say, human being behavior in the world. It's not saying, "Oh, there's this mysterious thing within human beings that we haven't detected scientifically yet, called freedom." And so, when it comes along, it's fine. It's not going to overturn your entire worldview.
- 4.36 No, it's just the claim that science will never be able to fully know and predict patterns of causation that are not causation in the scientific sense. They're causation by agents intervening in the world, persons, subjects. And as I said before, science is blind to subjects. There are no persons. There is no "my point of view on the world" that can be translated into scientific terms. It's always just "this is the world."
- 4.37 So this is hugely significant, of course, also for morality, policies about penal policy, incarceration, criminal justice, all these questions. If you take a naturalistic perspective, then you're pretty much committed to some determinism. Therefore, it's quite hard to make sense of moral culpability, the idea that weak human beings can abuse—agents—that can abuse their free will. All of that goes out the window. So whatever side of the fence you're on, these are big, dramatic changes in the world of ideas that have very concrete effects and impacts on public discourse and debate.
- 4.38 This is the dominant camp that begins to emerge in the 20th century. The first phase of it occurs in Vienna in the 1920s, inspired by the publication of Ludwig Wittgenstein's *Tractatus Logico-Philosophicus*, which was an extraordinary attempt to argue that the world is a totality of facts, of states of affairs that we put together through logic, in terms of understanding the world. It's not made up of things, says Wittgenstein.
- 4.39 This text says, in effect, that the entire world is constructed by languages that describe the logical structure of our language. The world is part of this logical structure. What makes the world is the logical structure that we can give it. The idea is that you can't jump outside your linguistic skin; you're always bound to your logic and to your language. That is what renders the world meaningful. "There's no deep mystery to it," says Wittgenstein. "I've solved all philosophical problems by showing you that it's just this you need to look at, language and logic, and that's what creates the world, and everything else is confusion."

- 4.40 He's so convinced that this is the most successful philosophy, this is the final word on philosophy, that he retires. He's barely 30, and he retires because he says, "I don't need to be a philosopher anymore. Nobody needs to be a philosopher anymore. I've solved it all. I've shown the fly the way out of the fly bottle," he says. He goes off and has a series of sometimes disastrous careers elsewhere back in Austria. But he comes back to Cambridge in 1929.
- 4.41 John Maynard Keynes, the economist, famously says, "I met God at the station today." He was on the 5:15 from King's Cross, London, and Wittgenstein was very full of himself as a personality, larger than life. He realized he might have been wrong, and so he comes back to Cambridge and develops a very different kind of philosophy that attacks and undermines his earlier one.
- 4.42 But his thought inspires a group of mainly Viennese scientists and thinkers and philosophers, academics in Vienna, the so-called Vienna circle—Rudolf Carnap, Otto Neurath, Moritz Schlick—very much influenced by Wittgenstein, by Russell. The idea is to really get to the point where we think of philosophy as, strictly speaking, meaningless. All you do, whether you want to know if a sentence has meaning, you can simply ask two questions. If it doesn't qualify for either of the categories that those questions relate to, it's a meaningless statement. It has no meaning; it doesn't describe the world in any way. It's just meaningless.
- 4.43 The Vienna Circle says, this is the verification principle. This is to verify every statement. And that's all you need, just verification of the words that you use. And they say, so either the statement is a statement that is true because of the meaning of the words that are being used, like "all bachelors are unmarried men." All bachelors are unmarried men. You can say that's true, just in virtue of—once you've understood the dictionary meanings of each of those words, you can see that it's true.
- 4.44 It's trivially true. It's not something you could go out into the world and say, "Oh, actually, I might meet one married bachelor. You never know." No, the person who thinks that hasn't understood the meaning of the word bachelor. It's just true; it's almost circular in its truth claim. It's just unpacking something. It's not adding anything new in terms of observation from the world. So, those statements have meaning.
- 4.45 Then the only other category, the second category, is, if it's empirically verifiable, if science can verify it; experience and testing and hypothesis can establish it as true. Again, if it's not one of those two, it's, strictly speaking, meaningless. It has no meaning. So, "God exists." "There are substances." "I love you." That can't mean anything. It's not analytically true in terms of the words used. What would it be to come up with a scientific description of the sentence "I love you"? You measure the firing of neurons in the brain when the statement is made. Is that the same meaning as the statement? No, of course not.

- 4.46 So that's the view of the Vienna Circle. This way of thinking, this science-first philosophy, or science against philosophy, you might say, a young English philosopher called Freddie Ayer, A.J. Ayer, A-Y-E-R, goes over to Vienna. He has heard about these guys, manages to get along to a few of the meetings—
- 4.47 More than Wittgenstein managed; he just managed one. The great master arrived at the seminar. Remember, the Vienna Circle is partly inspired by Wittgenstein's work. It's a disastrous meeting. Wittgenstein storms out, says, "You haven't understood any of my philosophy. You're useless, I'm not interested in coming back. You've completely misunderstood me." A little bit upsetting, I would have thought, but they persisted, they carried on.
- 4.48 Then another man came in from England, Freddie Ayer, and spent a few weeks and months with them, learnt German, and really started to try to understand what this science-first way of philosophy, naturalism, really was. He comes back to London. He sits, like Karl Marx did 100 years before, in a bar in Soho and starts writing and writing and writing; and I think in about three weeks wrote a book that was one of the most influential and best-selling philosophy books of the 20th century in English: "Language, Truth, and Logic," 1936.
- 4.49 Although there weren't radically new insights from Ayer in this book, it was the first time that this new philosophy, this way of thinking about reality, thinking about philosophy, first emerged really in a major way in English-language philosophy, and to some extent, the educated English-speaking public. This view dominated for a long time. Non-scientific claims are literally meaningless.
- 4.50 Over the years, it faltered and failed. In fact, there are interviews with Ayer in the 1970s where he's asked—there's a video, I think, on YouTube about this that shows this scene—he's asked something like, "Well, logical positivism didn't really succeed, did it, Ayer?" He said, "No, it didn't. It was a failure." "Well, why do you think it was a failure?" "Well, its central premise turned out to be false." There we go. He was very relaxed about it. The major philosophy that he's most known for, in his 60s and 70s, he says, "Yeah, it was a little mistake at the beginning, and so I got it all wrong."
- 4.51 What was the central premise? Well, the central premise is this premise of verification. These are the two ways to verify any statement. If a statement can't be verified in one of those two ways, and it can't be both, then it has no meaning, and it has no purchase at all.
- 4.52 Well, the question then was, what was asked by the critics was: which of the two categories did this verification principle go into? It turns out that there's no way of expressing the verification principle in the way that it's a sentence like, "All bachelors are unmarried men." It's not that kind of sentence, so it's not the first category.
- 4.53 But then, what would scientific evidence look like for the claim that statements are only meaningful if they are, one, analytically true, true in virtue of the way that the terms

are used, and secondly, true in virtue of which they've been empirically or scientifically verified? It seems that the verification principle itself, the central premise of this logical positivism school, can't meet its own test. And as it's said, the only test that matters.

- 4.54 So, we get to the awkward conclusion that the verification principle, if it's meaningful, is meaningless. If it's meaningful, if it's true that a statement can only have meaning if it's either analytically true or empirically true, and the verification principle doesn't meet that, either of those, it's meaningless. It's meaningless if it's meaningful. Something's gone wrong. Something has clearly gone wrong. At least Ayer had the courage and modesty to admit it and not fight to the bitter end about it.
- 4.55 Various other problems as well: the verification principle seems to be very constrictive and restrictive about what we can call, what we want to call scientific entities. A good example of this would be quarks and bosons and neutrinos, or black holes. Before we had actual evidence of black holes, it was just a speculative hypothesis, where the maths and the effects on the understanding of the physics was such that it seemed like there was this gap that ought to be filled by something with this description of a black hole.
- 4.56 Stephen Hawking famously pushes in his famous book of the late 1980s, "A Brief History of Time," one of the most best-selling books—one of the most best-selling and least-read books in history, I should say. I think a lot of people like to have "A Brief History of Time" by Stephen Hawking on their coffee table to make people think that they are super clever and can follow any of it. The problem is that the verification principle doesn't seem to be able to bring in these theoretical posits that we're assuming exist before we discover that there really were black holes. It's like, well, there's got to be something like a black hole.
- 4.57 On a strict application of the verification principle, no, that doesn't mean it's a meaningful statement under the terms of making it into category two, because we don't actually have any direct evidential, empirical science evidence for it. This is just theory. You're positing that this thing might exist, in order to make sense of everything else that you can see, but that's not allowed.
- 4.58 So, it's both very restrictive, but also very narrow. As we've seen, it evacuates meaning from statements about morality, statements about meaning, statements about free will, consciousness, the subject-centered view of the world—those are not meaningful statements. That's a very high price to pay for a theory, for a philosophy.
- 4.59 Remember that philosophy is trying to make sense of the world, not just on its own terms, but also in ways that make it intelligible for how it's our world, how we exist in it. I mean, are we really going to say our intuitions about right and wrong and good and evil are just firings of our physiology, or our synapses in our brains? That seems—maybe that's true, but it would involve a recognition of such a fundamentally mistaken way of thinking about the world as meaningful, namely that it's a moral world, that

there's right and wrong and there's good and evil. Any philosophy that says, "That's a complete figment of your imagination, this is just illusion, it's not real," I mean, that's still—maybe they're right, but it is a big, enormously consequential philosophy to hold when it comes to thinking about those questions.

- 4.60 OK, so that's the verification principle and why it began to unravel. There were a few developments over the years of the 20th century in the post-war period, and particularly in the '50s and the '60s and the '70s. There emerged the word "physicalism." The idea here was, well, OK, we're not really sure what nature is. We're not really sure what matter is, so we won't call it naturalism. We won't call it materialism. We'll just say, what we believe is that everything that's real can be described by physics.
- 4.61 The reason for this, of course, is that at the level of the very, very small—quantum physics—and at the level of the very, very fast—theoretical physics, theory of relativity physics, as it were—we're getting all these competing explanations and theories and schools, and it's not even clear that we know what matter is.
- 4.62 Matter is actually—stuff is much, much weirder than you might think. Enormous debates as to how to understand the subatomic world. It's not the sticks-and-balls model that we're used to from school, and actually still seems to be taught at school. Physicalism is—the language keeps changing to track the degree to which the world is becoming more mysterious, the more science we do on it. It's becoming more mathematical. It's becoming stranger. Quantum physics, quantum mechanics, you've got four or five very different conflicting schools as to how to interpret the evidence from the quantum world.
- 4.63 All of these different theories, as we'll see, explain the same phenomena. It's not clear as to how you choose between these different schools. That seems to be a philosophical judgment. There's nothing—the science is silent on which of the models best explains the scientific evidence. It's just not what science does. But philosophy keeps creeping in through the back door. These judgments just are going to be philosophical judgments, metaphysical, meta-scientific judgments.
- 4.64 That's something on the implications of naturalism as a comprehensive position. There's no more moral philosophy. We can just understand that through evolutionary psychology and Darwinism in terms of survival-seeking and wanting to improve your adaptive value. That's the only thing, the only ultimate actual explanation of why human beings behave in the world. In terms of consciousness and the sense that there's an I and there's a you, no, neuroscience explains all of that, gives the proper explanations, the only true explanations and so on. Anything else is just fiction. It's just imaginary, and so on and so forth.
- 4.65 All of these different disciplines, areas of philosophy, get reduced. And so this is very much a default position among professional Anglo-American philosophers today across

the English-speaking world, particularly the great faculties, philosophy faculties, probably Oxford, NYU, Berkeley, Stanford. Harvard, historically, has had some very strong philosophers, perhaps not so much these days. Cornell: very strong. Yale: very strong. Cambridge: pretty strong as well.

- 4.66 In all of those faculties, the most, the top 10, 20 philosophy faculties in the English-speaking world, this would be, I think, a clear majority position, naturalism. It would be a dominant, not universal, but it would almost always be the leading way of thinking about reality, that really philosophy is a backseat driver to whatever the scientist is deciding to go. It can't give directions, and it's got no say on the destination, but might comment on the scenery and the landscape along the way. That's the broad position at the moment. Scientism, naturalism, physicalism.
- 4.67 And so, this is the dilemma for philosophy. Should it dissolve itself, or has it got knowledge, substantive independent knowledge, things that it can say about the world that science can't, that are true, that are true ways of describing the world, even though they are not scientific ways of understanding the world? It does mean something to ask the question, "Why is there something rather than nothing?" That seems a coherent question, maybe a very deep and difficult question, but it's a question we can understand.
- 4.68 Now, how do you answer that in a way that is scientific? Or how do you make sense of it in a way that is scientific? It's not a scientific question because it's asking about, "Why science at all?", when there might not have been, or it might have been a completely different kind of science or a completely different world to describe. That can't be a scientific question. You can't justify that science just through a scientific statement. Otherwise, you'd be arguing in a circle. You could ask questions that say, well, "What came before science? Where did science come from?"
- 4.69 The fact is, science has got nothing to say about what happens at the moment of the Big Bang, or let alone before the Big Bang. Because before the Big Bang, it's not even clear how we can use the word "before," because the Big Bang theory says that that's when time begins. So how can you have something that comes before time begins? It seems a bit—how do you make sense of that? Or even at the same time as time begins. It's beginning, too.
- 4.70 What does a scientist have to say about what's simultaneous with the Big Bang and what comes before the Big Bang? Nothing, because the Big Bang just is the theory about how space-time came into being, and science is the study of the contents of space-time. The contents of space-time are the objects of science, because space and time is how we understand the world through space and time. There are laws of nature that apply only to the container, the space-time container, and that's what science studies.

- 4.71 Maybe there's another container, but there's absolutely nothing in science that tells us that it would be the same laws, it would be the same science. Who knows? You could guess, but it wouldn't be a science. There's nothing in science that can tell you what's happening in the science of one container that can tell you what's happening in another container or what's happening outside the container. It's not a limit of science. It's just a limit of the universe.
- 4.72 Sorry. You've got the whole universe to study. Don't start complaining that you can't explore and study beyond it. You can, but not as a scientist, not as a physicist. You are a metaphysicist. You're asking questions about the conditions of the possibility of physics, the origins of physics, why physics, why this physics when it might so easily have been some other physics? There's nothing in physics that said it would always have to be that, it would always have to be the particular physics that it is, that the physics of our world happens to be.
- 4.73 So you have this peaking, this moment of triumph for the 19th century through the Industrial Revolution, through extraordinary technological progress. When the theory hits the ground, extraordinary things happen, world-transforming things happen. But as it were, it's the old story of the Greek myths of hubris and nemesis. It also carries with it, as I said, the seeds of skepticism about itself. Because if thinking about science is just the random moving around of synapses and blood and synaptic electricity in my brain, well, who cares what that brain decided was the case? It's just physics happening. It doesn't mean anything. It's not true about anything; it's just the way the world is.
- 4.74 So, this is the irony—that science achieves a kind of extraordinary transformative, world-transforming success, but with the presumptions, skeptical assumptions, skeptical posture, a testing posture towards reality that bleeds into philosophy itself, including the philosophy that would justify the nature of science and help explain all those other things that science can't explain, like the mathematical and the moral and the meaningful and so on and so forth.
- 4.75 So, yeah, this is the irony of the last, let's say, 150, 200 years, as we've seen this big turn towards something like naturalism in particularly Anglo-American philosophy, but also very much on the continent as well. So that's, as I said, this is the last of the historical half or the more historical half of these lectures. And we've come to the end of describing the evolution of natural philosophy, thinking about the world, and then its rapid transformation in the 17th and 18th century through the scientific revolution, and then the emergence of a professional class called scientists in the 19th century.
- 4.76 And then the transition into the 20th century that shows how the impact of all this science and technology-focused intellectual culture began to bleed into philosophy and began to sow seeds of skepticism, that if all philosophy and all human beings and decision-making and freedom and morality is just to be explained by science, it can only be meaningful, only be meaningful if it's a scientific claim, then all of that seems to disappear.

- 4.77 So, philosophy comes back in, as it were, well, philosophy sort of dissolves itself, and that's the interesting final coda to this. That brings us up to date, brings us up, roughly speaking, to the 2020s in philosophy, from having started in Lecture 1 in 600 BC. So, not bad. In four hours, we've covered a significant chunk of time, just about 2,600 years, 2,600 and 50-odd years.
- 4.78 So, for the second half of these lectures, we'll be turning from the historical to the conceptual. We'll be looking at some of the live contemporary issues in philosophy of science that really would lay the foundation for a deeper quest into philosophy of science. In the next lecture, we'll be looking at the so-called problem of demarcation, what counts as scientific, what counts as not scientific. We've already seen that problem arise in terms of the verification principle being both too restrictive, too narrow.
- 4.79 The problem of induction, whether we can say that there are actually patterns in the world that we are describing, so that we can—from the past behavior of an object, we can predict into the future of it reliably. Hume famously comes along in 1740s and says, "No, can't verify that. There's no actual empirical evidence that there's this thing called causation and laws of nature in the world. Can't see them." So, what happens then? This is inductive skepticism. How do we make sense of scientific statements, the logic of scientific statements?
- 4.80 Then we'll look at how scientific theories change and what's happening when they change. Is it the case that our understanding of the world has got more precise, in such a dramatic way that we need a whole new way to describe it, as some people think, or is it the case that the world is just the picture that we make of it? Sometimes that picture will get dropped, and we'll take up another picture. But it's just a picture. It's a picture that is useful to us. Understanding the world in this particular way is useful to us, but it's not true about the world. It's just true about our way of making sense of the world.
- 4.81 It's much more—you imagine you're wearing sunglasses, pink sunglasses; it's more that we see the pink because that's on our side of the equation. The world is not pink. And similarly with one scientific theory, you say the world is not that scientific theory. This is just the sunglasses we can see them through most clearly now. And next year, we might have another set of sunglasses, but we don't even know what—it's just a different set of sunglasses. It's not better, it's not worse. It's just a different theory.
- 4.82 That's the difference between Einstein's theory of relativity and Newtonian mechanics. Say, well, they can't both be true. Oh, no, they're just different sets of sunglasses. That's quite a radical view. We'll be exploring that in the next lecture, and then moving through in Lectures 6, 7, and 8 into other debates and ways of exploring, undermining, consolidating the scientific method and the project of science. Explanation, unity. We'll look at some applications as well in the last lecture. Thank you for listening.

- 4.83 I have a question regarding, you mentioned Darwin. He was said to be the quintessential naturalist with his theories of evolution, yet he was a deeply Christian man. It's said that he was so distraught over his treatise and book, that it was going to undermine the church, that he had pathologic anxiety to the end of his life. Is it your opinion that he really discovered microevolutions, say, versus macroevolutions, which is the argument you would hear today? I'd like to know your thoughts on that.
- 4.84 Yes, absolutely. Thank you for the question. First of all, the way that the word naturalist is used in the mid-19th century is still, the dominant use is, people who explore nature and catalog it and start the Natural History Museum in London. Darwin wouldn't have thought of himself as a naturalist in that sense. I don't think he would ever have thought of himself philosophically as what we now come to call a naturalist. Probably if you pressed him, maybe, but he didn't have, really, firm philosophical convictions.
- 4.85 Some very fascinating letters where he asks his correspondent, "How is it that we could say that adaptive fitness and evolutionary theory, natural selection, is guiding our thoughts?" Because aren't our thoughts in the business of tracking truth, rather than survival-seeking? I mean, if we're just designed to seek survival, then sometimes we'll come up with false beliefs that will confer adaptive value. Sometimes we'll come up with true beliefs, but that's not—so if that's the whole story, says Darwin, in this fascinating letter, how could I trust my own thoughts, even my thoughts about evolution?
- 4.86 He studied theology at Cambridge, came 10th in his year at Christ's College, Cambridge, nearly became an Anglican priest. So he did have a strong faith in the early phase of his life. But at the age of 10, the tender age of 10, his daughter Annie died. This, I think, is around about 1851. And this had a terrible impact on him, psychologically. He stopped accompanying his wife, Emma, to church. At that point, she was an evangelical, Emma Wedgwood, granddaughter of the great Sir Josiah Wedgwood. He would always put on a comforting face to her on the question. He didn't want to upset her too much.
- 4.87 But he wasn't a philosophical atheist at all. I think he probably was, maybe towards the end of his life, you might have called him a Christian agnostic, somebody who's still very much in that world. But yes, he certainly wasn't—it wasn't a manifesto for the end of religion in any way, and it was not intended by him to be that. But the effect of it was to really completely reconfigure, change how we think of ourselves and our relationship to the world. We just emerge from it, emerge from the animal kingdom, as opposed to all of the narratives up until then that said, "We have some authority over this world and freedom that escapes just pure determinism."

Lecture 5: Scientific Reasoning

At a glance — Scientific reasoning — induction, deduction, and inference.

- 5.1 Welcome to Lecture 5 in this eight-part lecture series, Philosophy of Science. Today, we begin the second half of the course, and we're going to be looking now at debates, at issues, topics that we've been looking at the history of, you might say, in the first half.
- 5.2 The two main questions we're going to be looking at in this lecture. First, what is science? How do we distinguish what's scientific from what's not scientific? The so-called demarcation problem. Then secondly, we're going to be looking at the problem of induction, how we make sense of scientific logic itself, scientific theories themselves. There are all sorts of strange problems that come up when trying to justify the method, the inductive method, so-called, that science uses.
- 5.3 First of all: the demarcation problem. How do we lump scientific things, objects of science and scientific inquiry, into one category that differentiates that content from non-scientific content, like mathematics or history or literary theory or poetry? These are just non-scientific avenues of human inquiry, but perfectly legitimate, but not scientific. Also, how do we differentiate that silo of scientific content from pseudoscience, from claims that purport to be scientific and to belong and to be in good standing in the science silo, but which, for whatever reason, we haven't yet accepted or we long ago abandoned or dismissed? And so the question is: what are the tools that we can use? What's the sorting mechanism that we can develop that puts different statements and claims and bits of evidence into those three different silos correctly?
- 5.4 We saw in the previous lecture, just right in the 20th century and onwards, this interest in the verification principle. Well, the way to get the right content into the science silo is verification. It's the verification principle. And anything that's not scientific and also true, we've got a method with the verification principle to get that in the right silo as well. And then everything else is in the third silo. It seems simple.
- 5.5 But then it turned out that the verification principle unravels. It turns out, awkwardly, that the verification principle itself can't be included in the first category of things that are just true by definition of the words used, or in the second category, namely, a truth that you discovered through doing science on the world. If the principle that you're presupposing to do science on the world is itself a scientific one, then you've got a problem; you're arguing in a circle. If it's not scientific, but not true in virtue of the terms used, it's meaningless. The basic principle is meaningless. I'm sorry to recapitulate that material, but it's an important weakness, Achilles' heel, to remember with the verification principle.
- 5.6 So the answer to this came in the form of a Jewish-born Austrian philosopher who grows up in Vienna, is a very important intellectual figure in the Viennese scene in the

1920s, and then escapes the rise of the Nazis in the 1930s, heads to London, and develops some very interesting ideas, in particular the logic of scientific discovery, where he tries to set out how you go about checking what should be in the science silo, what shouldn't be, the demarcation problem.

- 5.7** He said that actually verification has failed. It's beginning to be abandoned by the '50s and '60s, but there's something, a different principle that we could apply, in many ways the opposite principle, that will work. This, he says, is falsification. Falsification. The idea here is that a scientific statement is true not because it can be verified or it can be subjected to the verification principle. We can say that a scientific statement is true for as long as nobody's managed to falsify it. It's not false. If it's not false, it's true.
- 5.8** That's how truth works. There aren't degrees of truthness, truthiness. There aren't degrees of pregnancy. There aren't degrees of truth. It's either true or it's not. If we can say, "Oh, it's not false," that's progress. We're not making any grand claims about how we can verify it, how the verification principle works. Just say, "Look, maybe have a go. Have a go at making it false. Convince us it's false. Anyone in the world can have a go." Popper's view is, look, for as long as nobody's had a go successfully, it's true. It's not false. It's true. It's therefore true, for the time being, at least. That's the idea.
- 5.9** Here the thought is that what science is doing is, it's making predictions. It's making claims about the world, it's sticking its neck out. When Peter Higgs comes up with all the maths that says there's going to be this God particle, there must be a God particle, the Higgs boson particle, he's making a prediction. If it turned out that there wasn't one or it was disproved, then he'd be wrong. He made a prediction, and sure enough, 20, 30 years later, the Higgs boson particle was found. They discovered it. He was right.
- 5.10** If you think of pseudoscientific theories, non-scientific theories, they don't have that advantage. Think of psychoanalysis, Freud, Marx. Think of that in terms of whether it's sticking its neck out. A therapist, a Freudian therapist's analysis of why everything's going wrong or why everything's going right is not making predictions. It's just not testable. You can't falsify it. How would you falsify a therapist's analysis of how you're feeling, what you're feeling? What would it be to contradict that?
- 5.11** Or you get another therapist's report. He's got a completely different account of why you're feeling the way you're feeling. OK, well, we'll just work out which is right. We'll just settle, which is more accurate? Which is more accurate? Well, they both explain how I feel, but they can't both be right because they conflict, contradict each other. Here, here, here, here. It was my mom, it was my dad.
- 5.12** The point is, science shouldn't be like that. It should be converging on some single underlying reality. There shouldn't be these silos of completely different dialogue, different discourse, different language, different high priests, whether it's Freud or Jung or Lacan or whatever. Psychoanalysis has had 150 years or so to show itself, to prove itself as a science like physics or chemistry or biology, but there's still like five different

schools, all of them believing completely different things about mind, the unconscious, and so on.

- 5.13** Can we say it's a science at the level of physics? Can you have a Nobel Prize in psychoanalysis? It doesn't make sense, because there's a feeling that it hasn't bedded down into a discipline, an avenue of research and thinking, that bears the same, what we'd expect an actual science to have. You know, broad consensus. Yes, disagreement, but not tectonic disagreement. You can still talk to each other. You're making predictions, you're constantly testing your claims. There's an agreement as to what the success criteria are and the failure criteria are.
- 5.14** We just don't get that with something like psychoanalysis because every therapist's report is unfalsifiable, and therefore we shouldn't treat it as true. Well, we shouldn't treat it as not false, we should treat it as not falsifiable. We'd never know if it was false or not. If we can't ever know whether it's false or not, if it could be true every time, then it's never true at all. What it's telling us is useless. It's telling us what it would say of any given set of facts.
- 5.15** You might say the same is true of Marxism, where the idea is, the revolution is always around the corner. It's always going to be utopia. If the revolution hasn't happened, it's because the sense of alienation and class consciousness is overwhelming. When there are revolutions, it's because it's not overwhelming anymore and it's broken through. When it's not a utopia, it hasn't been tried in quite the right way, and it should have been tried in this way, and then we would have got it. There's nothing in the world that can contradict the theory.
- 5.16** This is why people who believe in false conspiracy theories, some conspiracy theories are true, there just needs to be a conspiracy, and either there is or there isn't. If there is a conspiracy, like a planned, organized, orchestrated attempt to achieve some goal X, then a conspiracy theory is true. The theory about that, those conspirators, is true, or it's not. One of the challenges of any really good conspiracy theory is that you can't falsify it.
- 5.17** You try and convince somebody out of a conspiracy theory, suddenly you're part of a conspiracy theory. "Well, of course, you would be saying that. Of course, you would be telling me that what I believe is a conspiracy theory, a false conspiracy theory, because I know you, they've told me about you." And so you've got a theory for the attempt to falsify, which will always protect you from falsification. Because, actually, the person contradicting you, saying that you're wrong, that your statements are false, is actually proving that they're true. Because you've got a whole theory that these people would be coming along, claiming that what you're saying is false.
- 5.18** This is where the conspiracy theories are in a spiral, this epistemic spiral, because everything that confirms a conspiracy theory confirms it, and everything that disconfirms it, confirms it as well. The idea is that science shouldn't be like that. It must

be taking risks. The theories, the hypotheses, they've got to be making some claim about the way the world is, which could be shown to be false. This is actually quite a very interesting development by the great Karl Popper, the logic of scientific discovery. Yes, sorry, "Forschung" is the German word, the logic of scientific discovery.

- 5.19** So, this distinction helps us to distinguish, tell the difference between, say, relativity theory, psychoanalysis, astrology. Astrology would be a good example of, was it a science once upon a time? Yes, it was. There were all sorts of beliefs about the stars that might have been true, that was seen as scientific, but had been falsified.
- 5.20** Or rather, the claims that astrology makes will always be vague, reinterpretable. Never specify what would count as falsifying it. You can never read an astrology column where you could actually, they're saying, "Oh, it's telling me that that's going to happen to me next week. OK, that's a prediction. If it's false, then I know that astrology is not true." But it will never make those predictions. It will always say, "This will be an interesting week. This may be an interesting week." Well, you get to Sunday. Well, it wasn't, actually. It was a pretty boring week. But it did say "may." It didn't say "would be." It's just "maybe."
- 5.21** And what's "interesting" anyway, I mean, maybe I'm just a dull person, so I think that interesting things are dull. It's just not falsifiable. Any way the week goes, I will be able to say it was interesting, or if it wasn't, I've got a different view of what interesting is, from the person who said it was. Who am I to say what's interesting? Did say "may," didn't really commit himself. He was very polite. Didn't want me to have a bad week or get nervous about it.
- 5.22** But there's no predictive power here. There's no explanatory power. There's a principal distinction between the claims and the predictions that the scientist is making, that the astronomer is making, that distinguish it from the claims that you find in an astrology column, even though ostensibly they're about the same kind of thing. They're two competing theories about what the stars are and what their power is over our lives, et cetera, or not, as the case may be.
- 5.23** So, falsification comes to be a relatively dominant and well-received theory. It's a good way of making sense of our intuition that the claim that "all swans are white" is, at least was, a true statement. It said, "All swans are white." Apply the verification principle to "all swans are white." Have you gone through every corner of space-time, finding every single swan that you could? Have you seen every single swan ever, and have you confirmed that it's white? Can anybody make that claim?
- 5.24** No. It may be the case that there are non-white swans. Just because you say all the swans that you've seen are white does not mean that all the swans that there are, are white. So how can you verify it? Now, Popper comes along and says, "We verify it because we can't falsify it." We say that it's true. All swans are white because that belief has had a long time to be tested and shown to be false. Have at it. Go and find us

a pink swan, and then we'll say it's no longer true. You've made it false, and that's it. It's just open season on this claim.

5.25 Then, black swans were discovered in Australia. So it turned out, it was, it seemed to be true in a more plausible way by the falsificationists than it was seen to be, couldn't really be shown to be true by the verificationists. So it was false. It turned out to be a false statement. But the reason it was true before it was falsified is because it hadn't been falsified. Whereas the verifiers just couldn't ever say that all the swans we've ever seen are all the swans that there are. In fact, it turned out to be false. All those swans that we'd ever seen were white, but there were some non-white swans.

5.26 This, as we'll come on to see, is the great problem of induction. Induction. Induction is to be understood really as the opposite of deduction. Deduction is where it's kind of armchair logical arguments. You might say a classic example from the textbooks of a deductive logical argument, a form of a logical argument, would be to say: "All men are mortal"—premise one. Premise two: "Socrates is a man." Conclusion: "Socrates is mortal." You don't have to go and meet Socrates, talk to his doctor, work out anything about him. You can say, look, if it's true, if that is true, if all men are mortal, and if Socrates is a man, that's the information. Those are the inputs. This must be true. This must be the conclusion. That's deductive. It has to be that.

5.27 An inductive argument would be one that puts in inputs that are themselves contestable. Claims that would have to be established through scientific inquiry. Truths that there would be no way, as you're sitting in the armchair, to see it as true, no matter what. So, for example, the scientifically laden claim that "the sun will rise tomorrow." The sun will rise tomorrow. You say, well, why? The premise would be, well, I've seen it. I've seen the sun come up every day, every day, and everybody has, for years, decades, millennia. That's what the sun's always done. Why would you think that it wouldn't come up tomorrow?

5.28 Because, I mean, surely it must. Then you say, well, the reason you should think that it's going to come up tomorrow is because all these past instances have ended in it coming up in the morning. So this future one, the next link in the chain, will also happen. The great David Hume, in 1748, in his great work, "An Enquiry Concerning Human Understanding," says, "No, you're talking about two completely different things: the past and the future. Who knows what's going to happen in the future?"

5.29 And if you say, "Oh, but look at all these threads connecting the past. That's why we should believe that this is what's going to happen in the future," you're going to say, "Well, hold on. Why?" You're saying that we should use what's happened in the past to predict what's going to be in the future. That's the logic for why we should expect the future to look like the past, because that's how the past has always looked. And it's, how exactly? What gives us confidence in that? It's problematic, philosophically.

- 5.30 Why believe that the future resembles the past? Or at least how can you justify that by talking about how the past is grounded in the past? These are two completely different ways of—two completely different ways of describing the world. One, you've got experience, you've seen what's happened in the past, and the other, zero experience, by definition, because it's in the future, it hasn't happened yet.
- 5.31 This is this idea that all inferences from experience suppose as their foundation that the future will resemble the past, and that similar powers will be conjoined with similar qualities. If there be any suspicion that the course of nature may change, and that the past may be no rule for the future, which is Hume's view, all experience becomes useless. It ceases to be a guide. We can't infer anything from it. We can't conclude anything from it.
- 5.32 If you say, "Oh, we've got arguments from experience as to why the future resembles the past," you say, "No, you haven't got any arguments from experience to say that the past resembles the future, the future resembles the past, because the future is in the future. Nobody knows yet." Or the future you might have been talking about is now in the past. Why is it in the past? Because we've seen it. We can maybe see patterns retrospectively, but we've never successfully taken from the past to predict the future, because the future is the future. Nobody could ever have done that; otherwise, they wouldn't be in the future.
- 5.33 This is the Humean, this is the skepticism about the very logic that science seems to need, to go from patterns in the past to the way things will, to predict how things will happen in the future. And the argument is, that can't be a scientific thing, prediction, because you're grounding it in the future states, which don't exist. Therefore, science can't describe what doesn't exist. There's no science of unicorns. There's no science of the future. There's no logical contradiction, says Hume, in supposing the future not to resemble the past. It's not a contradiction like suddenly discovering that there are square triangles.
- 5.34 Hume says, for all we know—I mean, he doesn't say this, but the position is—for all we know, it's as likely that the sun will turn into a pink elephant as it will come up as it's always come up tomorrow. It's just, we don't know. There's no scientific proof for what hasn't scientifically happened. "Custom," says Hume, "is the great guide to life." We have a custom. These are not laws. It's not cast-iron predictions. The mind, he says, has a tendency to spread itself out upon the world and to just assume that the world is pixelated just for our iPhones. That's just the pattern.
- 5.35 We spread it out, and we put different filters on, and we forget that the filters are filters. Just thinking, "Oh, that's a great mountain shot, or this is a great valley shot, but look how vivid I made it with my filter on my iPhone. That's how it is. It's how vivid it was." That's the view for Hume. We think that the world is structured and ordered in the mechanical way that Newton has shown us, but really that's just our mind spreading itself out upon the world to try and make sense of it.

- 5.36 Another person in a more contemporary argument against induction is developed by a guy called Nelson Goodman in a famous book, "Fact, Fiction, and Forecast," 1955. He develops this idea of a new way of trying to worry about inductive reasoning. He says, imagine that you've got green emeralds and "grue" emeralds. Green emeralds and grue emeralds. They both look beautiful green when they're put on rings, beautiful. They look like green rings. Beautiful green emerald, a beautiful green gem.
- 5.37 Well, what's the difference, you might ask, between the green gems and the grue gems? Well, the difference is that green emeralds are emeralds that are green before New Year's Eve, 2025, and green after New Year's Eve, 2025. Oh, well, what are grue gems? What are grue rings? Well, grue emerald rings are rings that are green before New Year's Eve, 2025, and blue after New Year's Eve, 2025. They will become blue at a certain point in the future. So what's going on here?
- 5.38 The person who believes in, is committed to induction, is going to say, "All emeralds we've ever seen are green gems. Therefore, all emeralds we will ever see at any point in the future are this color, green." Now, what will the skeptic, the inductive skeptic, say in response? He'll say, "There's absolutely no basis that you can appeal to that can convince me that these are green emeralds, not grue emeralds." We're explaining the same data. I'm not saying it's not a green ring, not green now. I'm just saying that it might be, for all we know, a ring that turns a deep sapphire blue on New Year's Eve, just after New Year's Eve, 2025. The data is identical.
- 5.39 The inductive skeptic is saying, "You're making unjustifiable, unevidenced claims about what this ring is going to look like for all time." We haven't seen "for all time" yet. We've got no basis for saying, you've got no basis for saying this is a green ring, not a grue ring. This is another way of thinking about the problem of induction.
- 5.40 Some standard response is, "Oh, no, green is natural, naturally occurring, and grue is just this clever idea that you've come up with in the philosophy seminar." Green is natural. OK, why is it natural? Well, because it's the thing that nature comes up with. What's your basis for thinking that nature will come up with this again? Because look at all the times in the past when it's done it. That just restates the problem that the skeptic is pointing to, namely that your definition of what's natural, your argument for why green is green, not grue, is based on the very faulty reasoning we're trying to point out—namely, that there's a natural way that everything is in the past, and therefore it's that same thing in the future. And that's just making the same appeal to circular logic.
- 5.41 Maybe the world, maybe nature just doesn't like us very much. Maybe there are all kinds of weird tricks it's going to play on us in the future. Why assume nature is going to be kind to us? You spend a fortune on a beautiful engagement ring, and then suddenly it turns this horrible color that you can't stand. It's like, "Well, but why is nature so mean to me? I spent so much money on that emerald ring, and it turned out, this beautiful green ring turned out to be gray, dull gray." Well, that's just the way

nature is. How is nature supposed to behave towards you? How would you have directed nature to be towards you and so on and so forth?

- 5.42** So the point is, science can assume some claims about what's happened in the past. You could say that some predicates, maybe some ways of describing things, are natural ways to, and others are not. That doesn't always work. But the point is, if you do think that the past can underwrite or can support claims about the future, then what you're saying is that there are puppet threads all around nature that are just there no matter what, and that there are laws, there are patterns and processes, and there's order such that the future will resemble the past.
- 5.43** Fine. It seems as if science really needs that move to be possible. You'll say, OK, fine, you can have that, but don't call it scientific; don't call that a scientific claim. That's just a hunch. It's a philosophical claim about the nature of the world, including the future. It's a view that doesn't mind about whether or not the claims it makes are supported by scientific evidence. There could be lots of other claims that you could make that—historical claims, or claims about a poem, a judgment about a poem. They can be true even if they're not scientific, and so on and so forth.
- 5.44** So that's the puzzle. It seems that you can never do philosophy-free scientific argument. You're always going to be appealing to some philosophically contentious position. Either there are no patterns out there, there's nothing that justifies us thinking that the future will resemble the past, or we say, yes, there is something. There are the laws of nature, the threads that run through the whole of space and time. And that's what we're talking about, when we're talking about a law of nature; we're talking about something that applies to the past and the future, no matter what.
- 5.45** You can disagree over that, and philosophers do. But science can't settle it. The scientists can't come in and say, "Oh, hold on, I've got this evidence about how the future resembles the past." Then we'd say, "No, you haven't." There's nothing science can help us with here, because everything that science has found, it has found. It's in the past. This is a philosophical argument, which if either side is right, the facts stay the same. The evidence is the same, whoever's right. This is the challenge. This is, it seems we're not doing science.
- 5.46** But we're doing something. It may not be an argument that thrills your heart, but it's an argument. It's a serious dispute. Are there laws in nature or not? I mean, that seems like a pretty serious and worthwhile philosophical debate. Not a trivial one, certainly. This is the great problem with induction. Induction has worked before. This way of scientific reasoning from the past to the future has worked before. So this is going to work. It uses induction to justify induction.
- 5.47** "Nature is uniform." How do we know? As I said, maybe it turns out nature's playing tricks on us, lulling us into a false sense of security after many thousands of years, and suddenly all kinds of strange Alice-in-Wonderland things are going to start happening,

and it's just, we thought it liked us, but actually it really didn't like us. It was lulling us into this false sense of security, so that the shock of discovering the world is actually a very strange place is even more intense.

- 5.48** Science has been successful. Will it be successful? AI has been so successful, therefore I'm going to put all my life savings into Nvidia. Just look how well it's done. If I bought then, just imagine if I bought then and what I'd have now. So I've got to put in everything now. The person who's skeptical about induction is saying, "The problem with overconfidence in the sciences is a little bit like the problem of overconfidence in Wall Street," that this stock is going to go up forever, because it's just been going up forever till now. Of course, it's going to keep going. It might be bumpy here and there, but of course, it's going on forever, in infinity.
- 5.49** Humean philosophers want to say, "No, there's just no reason for saying that. You have no evidence that there isn't going to be a massive crash in the stock, that there's this huge AI bubble that's overpricing AI stocks by roughly 20%, and that's going to go overnight when it goes, and maybe it's tomorrow." "Oh, no, but look at yesterday. It just kept going up." "Yes, yeah, but we're not talking about yesterday. We're talking about tomorrow," and there's something very different about tomorrow and yesterday.
- 5.50** Something quite fundamental when it comes to establishing knowledge claims about the world—namely, that yesterday has happened, and we can experience it, we can get reports about it, we can reach consensus, maybe a bit of disagreement, we know what happened yesterday. We have no idea what's happening tomorrow. No evidence from tomorrow of how it's going to turn out. The trader, addicted to this one stock going up forever, is not being a serious trader. This is the criticism.
- 5.51** There haven't been many really convincing responses to this. It's still, this skeptical atmosphere, attitude, is still, I'd say, pretty dominant among philosophers of science all around the world at the moment. Are there, are there rival theories to it? I mean, also different versions of it. If you're committed to evolutionary theory as the comprehensive account of the origins and development of human beings and human cognition, having a theory of rationality, a theory of how it is that we can generally find out truths about the world, seems to be undermined.
- 5.52** Because natural selection doesn't care about whether we form true beliefs about the world. Sure, sometimes you're more likely to survive if you can work out that there are two tigers in the next-door valley and two tigers in the other valley. You'd better get out of there. Maybe you could take on one tiger; you're not taking on four. You form a true view, you act in the right way, you survive. The hominids who tended to do that tended to outlive the ones who thought, "We'll take them all on. Four is nothing."
- 5.53** But also, you could imagine systematically false beliefs about the world also conferring survival fitness. After all, often atheists will say that the emergence of religion is the emergence of a systematically false set of beliefs, but that must have some

evolutionary explanation. Religion comes from somewhere. I mean, it exists. Maybe God doesn't exist, but religion exists. Well, what is it? Well, the evolutionary biologists, Darwinians, will say, "It was to help stitch the group together, or these beliefs conferred a consoling power on parents mourning their children. It was a way of keeping the society together."

- 5.54 It'd be all kinds of just-so stories, explanations consistent with Darwinism as to how it is that we behaved in these different ways, why it is that we believe these false things. The point there is that survival value, Darwinism, is not interested in conferring survival on the people who can do their sums. It's not interested in that. It's interested only in whether particular beliefs that human beings form, whether those human beings survive more, survive better.
- 5.55 There's no theory of why, on this comprehensive accounting of evolutionary theory, why we should believe any of our thoughts to be true. They could just be there because they're helping us survive in the world, not that they're true about the world. This is a very interesting skeptical argument, as it were, from Darwinism.
- 5.56 There are various responses to it. One is probability theory, or sometimes called Bayesianism, named after Thomas Bayes, who was a church minister. He said, really, you can solve a lot of these skepticism worries by saying we have certain prior assumptions about the world. They may be wrong, they may be right, but it's probably better to think of them in terms of degrees of credence, degrees of credibility. We just have these prior assumptions, and nobody is free of any of these prior assumptions. Everybody's got some.
- 5.57 As we go through the world, as we reason about the world, and as we experience different things in the world, we keep tweaking our priors. We keep changing them and saying, "Oh, I haven't seen this done before. I didn't think that could be done before. Oh, I'm going to change my prior assumption of what could be done in this area." Bayes argues that we just go through life like that, just adjusting and, gradually over time, calibrating how to understand the world, the different lenses through which we should understand the world.
- 5.58 Again, at any point, you could say, "Oh, my prior about this may well be wrong, or maybe I didn't put enough belief into this happening in the future because of how often I've started to—how frequently I've started to see it happen." So probability theory is not saying that this is going to or will happen. It's more that, based on my priors and based on the different weights I put on different beliefs that are relevant to forming this judgment, I'd say there's a probability of more than half of it being true. I'm not, won't say more than that. I'm saying it's rational for me to assign a fairly high mark to that happening again or happening at that point in the future.
- 5.59 That's been one attempt to resolve skepticism. We're not being too dogmatic here, so you don't need to be too skeptical. It's just, given what's happened in the past, I think

that this is more reasonable for me to think that the sun is going to come up than it turns into a pink elephant. I'm not saying I know that. Just saying, I've been shaped to think it's not going to be pink elephant day. Or just pink elephant. There wouldn't be a day anymore. It'd just be pink elephant world.

- 5.60** Now, the trouble with that response is, the inductive skeptic can say, "Well, what are your priors, except the accumulated wisdom of everything that's happened to you in the past?" You're saying that the accumulated wisdom of everything that's happened to you in the past, that has made up your priors, this complex architecture of epistemic priors, prior probability assignments, is going to resemble the future to different degrees.
- 5.61** But that's exactly what we're saying is not, you don't have justification for. You can't just turn it into clever probability theory and then think that you've solved the problem. What is probability theory but saying there are degrees of likelihood that the past will resemble the future in these different ways? That's exactly what the inductive skeptic says you haven't got any right to claim. It's as likely that the sun turns into a pink elephant as that the sun rises.
- 5.62** You can't assign a probability because you haven't got—the only track record you've got to form the probabilities that you're going to assign to the future have already happened. They're in the past. You can't project them forward. Every time there's been an attempt to respond to the skeptics, the Humean camp has normally had an answer.
- 5.63** Now, the question is whether these conversations are turning into something like the conversation you will have with somebody who's fanatically committed to a belief in a false conspiracy theory. Every attempt you come up with to say that it is reasonable to think that the future may resemble the past and the past may anticipate the future, they'll be able to explain no. They'll always be able to say no.
- 5.64** Maybe, just as with a conspiracy theorist, a false conspiracy theorist, the best solution is therapy rather than philosophy, or is to just shake the dust off one's feet, walk away, and just say, "Look, there's just a dialogue of the deaf here, because any evidence that I bring to try to confirm or to disconfirm your view, you take as evidence in favor of it." Or, saying that you can simply restate the problem.
- 5.65** So there isn't really a particular resolution there. There are, as we'll see in future lectures, very powerful arguments defending scientific realism. That is to say, the view that objects that we describe in the world are, roughly speaking, there in the world and the way we've described them. There really are laws out there. There are patterns, there are rules. We'll have a look at some of those next time.
- 5.66** But this skeptical atmosphere and attitude has become a very dominant one, even after the failure of logical positivism and the Vienna circle and verification principle, and even after people came to have difficulties with Popper's falsification principle. And new versions emerged of anti-realism that said, really, we're just trying to create a map

of reality. You can look at a map and say, that's not— look at a map of Arizona, you can say, well, that's not Arizona. We say, well, no, it is Arizona. Look, there on the map, all the places are organized with the right distances, proportionately. I could see where Camelback Mountain is, and that's Arizona.

- 5.67** There's loads of information that is true about Arizona in this map than there is in any rock in Sedona Valley. It's a map. It is, of course, it's Arizona. It's not France. Maybe that's what science is. It's this really detailed map of reality. You say, "It's not reality. It's just a theory." I mean, yeah, it is just a theory in the same way that the theory of reality is that a map of Arizona is about Arizona. Our theory about reality is reality. It's about reality. It's just arranged in a way that helps us understand it and be guided through it and so on and so forth.
- 5.68** It's not a crazy idea, this view that science is really just in the business of theory, and it doesn't really matter what it would even mean to say that there's a world beyond the world that we can have any knowledge of. What does it even mean to say, "Oh, reality is actually like that, if we only knew." If we weren't hampered by our science and our senses, that's how reality actually is. It doesn't make any sense either.
- 5.69** As Wittgenstein says in another context, "A nothing will do as well as a something about which nothing can be said." A nothing will do. If you're saying that there's something about which nothing can actually be said or known, a reality beyond the reality that we can perceive and think about, it might as well be nothing. It's nothing. And nothing will do as well as a something about which nothing could be said.
- 5.70** It's not a ludicrous view, the map of reality view of science. It's plausible. Just as we change maps, change ways of doing our different maps over the time as things get more refined, the map of Arizona now is not the map of Arizona when Arizona was probably not quite. Yeah, it's all very different in the way the different towns and cities have grown and so on and so forth. So it's sensitive to change. It's not just making things up as it goes along. It does reflect things that have happened in the world, and it condenses them and conveys the information. So that's the anti-realist view. That's the more skeptical view.
- 5.71** So to recap, what have we discussed? What have we thought about? What have we talked about in this first lecture of the second half of the series? We've looked at two of the first really big philosophical problems and puzzles that occupy people when they're thinking about what the nature of science is. First of all, the demarcation problem. If you're going to have a philosophy of X, it would be helpful if you could define X or differentiate it from Y. Maybe that's not a bad place to start. We want a philosophy of X. Shall we work out what X is?
- 5.72** As we've seen, it's not so easy to work out what science is, in a way that makes very precise cleavages and distinctions between what's science and not, what's scientific and non-scientific, or what's scientific and pseudoscientific. What's scientific and what's

not scientific, but not trying to be scientific, and what's not scientific, but trying to be, trying to invest itself with the authority of science, as in, for example, psychoanalysis? So this is the challenge.

- 5.73** Now, another response has been to say, well, what we should look at is not induction, but something called abduction. This is not what philosophers do when they go to CrossFit or the gym, looking in vain for their six-packs. Abduction is a new principle, logical principle of scientific truth, of discovery or theory, justifying theories about the world. And what it says is, well, look, maybe a better method, rather than testing every single claim and making claims that because this has happened to it in the past, or this is the way it's behaved in the past, it's definitely going to behave like this in the future.
- 5.74** Maybe a better way of going about things is to say, let's put all of our different theories on the table and say, which is the best one? Which is the best theory? The criteria of optimality, the criteria of what counts as a good theory, there'll be some agreement about. Then we can apply—for each theory, we can say, does it explain what needs to be explained? Then can we tick off on our shopping list of meta-theoretical advantages, does it do this? Does it do that?
- 5.75** Then we can say, look, this theory does it, not perfectly, it's not a slam-dunk, but it's on the whole a more elegant theory than any of the others. Or maybe it's a more simple theory than any of the others. It asks us to sign up to fewer impossible things before breakfast. It makes sense of other areas. It's a theory about, we thought just about this one corner of science, but it actually turns to be consistent with or support something that's completely disconnected that we weren't even looking for. That happens quite a lot in science.
- 5.76** Then you can think, well, that's a good theory. I mean, if it's even helping us explain things we didn't even ask it to explain, or we didn't design it to explain this, but it also makes sense of that, so, OK, it feels like we've discovered something rather than that we're just inventing something. This is a form of confirmation.
- 5.77** So, there are all of these different theories, or criteria for what counts as a good theory, that you can develop, different ways of shaving, you might say, with Occam's razor. Occam's razor is the idea that you shouldn't multiply things that you're appealing to, to claim that something's true, if a leaner and meaner theory could do the same explanation just as well. You should shave away with Occam's razor all of the extra bits that you don't really need, that this other theory can do without.
- 5.78** Now, one way of shaving with Occam's razor is to say you can— a theory that invokes fairies and goblins is— you should probably go with the theory that shaves away the fairies and goblins, that gets rid of them. It's contentious. You're setting a very high threshold for getting people to believe in your theory. It's not unreasonable.
- 5.79** It may be that you can say, you don't need to shave away just ordinary scientific objects that you're appealing to. No fairies, no goblins, just rocks and electricity, nothing extra

you've got to believe in, but there's lots of it. It's a lot of electricity, lots of different rocks, feature in this explanation. Some people say it doesn't matter. We've already got rocks, already got electricity. I'm not asking the person to believe in my theory, to believe some new category of being exists, like the fairies and the goblins. So, I don't really care how many, as long as I'm not having to multiply the categories of reality. You can have as many rocks as you like. You can have as much electricity as you like. No need to shave it away. It's still a simple theory. It's the same stuff; it's lots of it, but it's just the same stuff.

5.80 You're telling me that there are fairies and goblins. You're asking me to fundamentally change my shopping list of what I think is real and what counts as reality. I'd much rather not do that, if at all possible. If there's some other theory that makes do without them, I'm going to go with that theory. I don't even need to know what the theory says. I'll probably go with the non-fairy, non-goblin option. Thank you very much. That's Occam's razor in qualitative terms and quantitative terms, as it's sometimes described. There's also Occam's laser. William of Occam is a 14th-century English philosopher, so he wasn't familiar with the notion of lasers. But adapting it from the razor, you could say, actually, no, what's most important about a theory is what's happening at the foundations of the theory. If you laser down, can you laser away all additional, all multiplication, or wherever there's too many, too many things going on at the fundamental level of explanation?

5.81 So if you think it's atoms all the way down, you've got almost infinitely many tiny little units that are at the fundamental layer of all reality. That's what's basic. You might say, Occam's laser should zap those, if there's some other theory that puts at the bottom the foundational ultimate explanation of the whole theory, less than that.

5.82 The monotheist might say, "Well, hey, actually, I've got an account of what's fundamental to all of reality that's less than all the atoms in reality. I've just got God. Just God, one, just one." I don't need more than one. It's more than zero, less than two, one. How can you get fewer than that, without it ceasing to be any kind of explanation, because nothing is not an explanation? You've got all these millions of atoms that interact in ways that are mysterious, but it's just basic. I just got one, admittedly, quite an impressive entity with enormous rational powers and causal powers and feelings and knowledge about all of reality distinct from himself. But still just one. That's all I need. You can say, well, Occam's laser is going to prefer that theory than the one that says you've got to believe that there are all of these things that are fundamental.

5.83 So, how does the theory look? Is it elegant? Is it beautiful? What is it to say that a theory is elegant and beautiful? Well, mathematicians who explore non-applied mathematics, that is to say, theories that will never have and are not claimed to have any application to the real world, what's going on there? Well, very often you will see mathematicians talk about exploring these different worlds using aesthetic language. "It was so beautiful when I finally saw, when I finally cracked Fermat's last theorem."

Andrew Wiles almost burst into tears. Partly, it's such an enormous moment for him, but it's just so simple, so beautiful.

- 5.84** "I went for the more beautiful option." You hear scientists being asked, "How did you even come up with this hypothesis?" "Well, I just thought it was elegant, it was simple, it was so beautiful." You really do get—I'm not saying that this is true of every scientist, but I'd say with many, disproportionately many, they often use this aesthetic language, a sense of awe at discovering nature's hidden script, hidden language.
- 5.85** That does, again, suggest that the world is there to be understood. The world is intelligible. It didn't have to be beautiful, didn't have to be simple, didn't have to be intelligible. Chances are, it wouldn't have been. So how do we understand that? How do we make sense of it in a way that isn't just pretending that it's all part of our projections?
- 5.86** So, as we look ahead to the next final three lectures, we'll move on to some of the other big issues involving scientific explanation, particularly the unification of scientific explanation, what it is that we're describing when we're describing explanation. We'll look at some of the arguments for realism, for the view that the world is actually as we describe it, as science describes it. Then in the final lecture, we'll look at various applications of science, different kinds of science, within the context of the philosophy of science, as we have come to understand it by the end of Lecture 7. Thank you for listening.
- 5.87** Can you elaborate on nature's hidden language?
- 5.88** Yes. In the medieval period, many philosophers used to talk about there being a Book of Revelation and a Book of Nature, and that they were both authored by God because the world was legible. You could read it, you could understand it, you could make sense of it in the way that you can make sense of blank splotches on a white page, like a language. The words, it's just ink on a page. No, but it's not ink on a page. It's configured in a particular way that it has meaning. This ink and paper has human meaning in it. This is puzzling.
- 5.89** And so, the idea is that, just as we wouldn't find a random book on a desert island and open it and say, "Oh, that's interesting, that this paper and ink has magically come together in a way that I can understand, I can follow. But it's a shame I'm still on a desert island that nobody's ever been here before." No, you'd say, "My gosh, there's no plausible natural explanation as to how this book has just popped up. It's got meaning. I can read it. I can understand it."
- 5.90** The thought is that the science, when we're studying reality, that we can understand it, we can make progress. It seems we can make predictions. It seems we can come up with theories that accidentally explain some other area of reality that we weren't even looking to try to explain, but it also explained that. That means science feels more like a process of discovery, rather than a process of tweaking and revising our existing

theories to make sure they're more useful to us or something, which is the map view, saying, "No, it's really out there. We're correctly describing the landscape." Yes, that's the idea, I think.

Lecture 6: Progress in Science

At a glance — Progress in science — how and whether science advances.

- 6.1** Welcome to Lecture 6 in this eight-part series on the Philosophy of Science. What we're going to look at in this second, more conceptual or systematic lecture, having done in the first half mainly historical material, the development of ideas about science, what we're going to look at in this lecture is how science changes. Does it make progress? What does it mean to say that science makes progress? How is it that scientific statements that we think to be the yardstick of what counts as true and real, how is it that those statements are constantly being revised and constantly being changed?
- 6.2** So this is the great paradox at the heart of the scientific method, that on the one hand, science has authority. It's got to be certain and sure, solid. On the other hand, for science to move forward, we've got to be skeptical of what seems solid and certain. Well, how about a slightly different theory that might make it even more solid? And so, this is the task, this is the challenge, of doing science and of thinking about science. When should we respect the authority of science? When should we be skeptical and say, well, it's the nature of science to continually test different hypotheses, whether we're trying to falsify it or whether we're trying to verify it.
- 6.3** That's going to involve different considerations. Some people will say that it doesn't make sense to speak of progress. That was just one map. Now you've got a different map. You might prefer on Google Maps to see the real street view. You might prefer just the map that shows you where all the hills are, the map that shows you where all the cafés are. It's just, do you want a coffee? No. OK, I don't need that map. Do you want to be able to see whether the neighbor's building another shed in the garden? OK, I need that particular map, the photo map.
- 6.4** It's the same with our scientific theories. We just have different maps of reality, depending on how we're trying to navigate it, what we need from the world.
- 6.5** But it's our objectives. It's our ends. It's not nature's ends. It's not God's ends. It's just theories that are useful for us, maps that help us understand the terrain in the way that we need to. It's not that one is better than the other, objectively. Science doesn't make progress like that. It's just one map, then another map. What are you going to say? Why is this map better than that map? Well, they're just maps of reality. You can't have a map of all the maps explaining why some maps are accurate, and others are not, sort of maps all the way down. No one map has a greater claim on being, as it were, anchored in reality, because we can't see reality. We've just got the maps. It's a hall of mirrors, a hall of maps.
- 6.6** The person most associated with this view—we're ditching one map, we're taking up another map, and just society has changed, our intellectual culture has changed; it's not that one is truer than the other—the person most associated with this view

probably is Thomas Kuhn, who wrote in 1962 a very famous book, one of the most, possibly the most famous book in philosophy of science in the 20th century, "The Structure of Scientific Revolutions." The Structure of Scientific Revolutions.

- 6.7** We talked about Karl Popper's famous work, leading to his theory of falsification, falsifiability as a criterion of truth or justification. That's "The Logic of Scientific Discovery." This is "The Structure of Scientific Revolutions," 30 years later. Kuhn, in this book, challenges the idea of scientific progress. He challenges the idea that what's happening is that we're getting a better map each time, that science grows organically. It accumulates more and more truths about the world, then eventually, its theories need to get revised, and you need new theories, but theories get better. They get more accurate.
- 6.8** Kuhn is resisting this view. He's just saying that it's a different map, but there's no way to judge between the maps, because then you would be jumping outside reality, or maps of reality, and be trying to compare two maps, and you've got no sense of the actual terrain, you haven't been on the actual terrain. What Kuhn says is that science is more like a paradigm, a paradigm, like a theoretical domain or world, a paradigm for understanding the world. Fundamental assumptions, methods, ways of approaching solutions, ways of seeing the world.
- 6.9** We talked about the astronomy of Ptolemy, the old Ptolemy from the early centuries AD, and how he gets overturned when Copernicus comes along and Kepler comes along. It just gets—that whole picture is ditched. The Ptolemaic map of reality is ditched. You say that's a paradigm. And Copernican heliocentric cosmology began to emerge as the better map of what's actually happening.
- 6.10** But Kuhn says this is just a paradigm shift. Paradigm shift. You might have heard that phrase in other contexts before. This is where it comes from. This is the book that it comes from, The Paradigm Shift. He says that all that's happening in a paradigm shift is that one map is replacing another. As I said, it's not better than, it's not worse than, there's no objective standards by which to measure which is better, which is not. It's just a different paradigm. There is a paradigm shift.
- 6.11** There's normally a moment of crisis within a theory, says Kuhn. Lots of shaking on the inside as the theory is struggling to cope with new information coming in, as predictions are getting weaker. There are all these anomalies that it can't quite explain correctly, as was the case with Newtonian mechanics. Eventually, says Kuhn, it's a project that runs out of steam. It was all going so well, but it starts breaking down and dissolving from within, and then another paradigm takes its place; a new way of thinking about the world takes its place.
- 6.12** In this case, we went from the Ptolemaic to the Copernican to the Newtonian to the Einsteinian to maybe one of the revolutions in quantum mechanics, several of which say Einstein was wrong. In his view of what the truth was about quantum mechanics,

that would be a shift. Kuhn's point is that, again, it's what's called the incommensurability of the paradigms. You can't measure them against each other. You can't evaluate them against each other. They're totally different kinds of things, totally different things.

- 6.13** One of the challenges is that often the theory, the competing paradigms, will be explaining the same thing. So Ptolemy had an explanation for why the sun rose, why the sun rises. And it turns out that Copernicus also had a theory about why it looks like the sun is rising. It still looks like the sun is rising. The data is the same. Whether Ptolemy is right or Copernicus is right, whether Ptolemy is right that the whole of the cosmos revolves around the Earth, or whether Copernicus is right to say the whole of the cosmos revolves around the sun, what would dawn look like on each of those two theories?
- 6.14** The answer is the same. It looks like the sun's coming up. Well, actually, it's not because Copernicus is saying, "No, no, actually, it's not. It looks like it, but it's tricking you. It's actually you, it's us that's moving. It's us that's rising." You say, "Well, that's a very complicated theory. I don't think it'll ever catch on. Look, it's going up. It's down, and then it's up. It's rising." This is a much simpler theory. My sunrise theory is much simpler than your Earth-rise theory or your Earth-lowering theory.
- 6.15** And so, this is the problem, that scientific theories can cover the same data. There's no scientific way, or at least immediately, to distinguish between the paradigms. We still talk about sunrise. We don't talk about Earth lowering. "I'll see you when the Earth is lowering tomorrow." No, "I'll see you at sunrise. We'll have a sunset cocktail." We're not going to say, "No, we're going to have an Earth-heightening, an Earth-lifting cocktail." No, it's the sun's going down. It's a sun-downer. Who cares what the scientists say? This is what it looks like, and that's all that matters.
- 6.16** The world is the same place, whatever theory we apply to it. We're seeing it in the same way. Kuhn's interest is: how do these paradigms shift and change? Is it a passing convention, the way society is developing? How do you choose if you can't measure it, you can't compare them, you can't evaluate them comparatively?
- 6.17** Kuhn's view is that, look, you can still apply all of the theories that we discussed in the previous lecture. You can look at the simplicity of the theory. You can look at the power and the scope of the theory, see whether it explains other things, things that we didn't expect it to explain. See if it makes predictions, which we can confirm or deny. See if it sticks its neck out as a theory, and saying that this is going to happen, this is going to happen then, or this is the kind of thing we would expect to happen, as I said Mendeleev did with the periodic table, or our old friend Thales from Lecture 1 did, when he predicted that eclipse in 585 BC. He got it exactly right. So he stuck his neck out, and it wasn't an astrological projection.

- 6.18** This is a problem. When you've got a theory that is underdetermined by the evidence, the evidence doesn't settle which interpretation, scientific interpretation, you're going to take either way, then there's a headache. What do you do?
- 6.19** A bit like, as we saw in the last lecture with green emeralds and grue emeralds, now we've got two theories. One is that emeralds are green, the other that emeralds are grue, that is to say, they're green now, but they will turn blue on New Year's Day, 2026. How do you tell the difference between green and grue emeralds? Well, you can't. Same observations, different predictions. And the current observations don't tell you which prediction is correct. For sure, you may think, you may lean towards green, but maybe the world is a mysterious place. It's a point from the previous lecture, of course.
- 6.20** You have this issue probably in the most interesting corner of the natural sciences right now, namely, quantum mechanics. Really, the philosophy of the very, very, very, very small, the subatomic world, where all kinds of—it seems that nature behaves in all kinds of unpredictable and puzzling ways that Newton can't make any sense of, not even Einstein could make much sense of. There have been a number of different theories that have emerged over the years, particularly over the last 100 years, to explain some of the paradoxes that we discover in quantum mechanics.
- 6.21** The basic problem is that it involves all sorts of paradoxes, like, for example, the paradox between two conflicting accounts of what light is. On the one hand, when we're doing some mathematics and doing some equations, you can only think of light as a wave function. In other areas within quantum mechanics, you can only assume that light is particular. Light has particles, not a wave, particles. If you don't assume that, then everything goes wrong. All the theorizing and the mathematics go completely wrong.
- 6.22** So what is it? Is light a wave, or is it a particle? What is it actually? Well, what it actually is, is both a wave, in order for all this to be true, and also a particle, because otherwise all this would be false. So it's not a contradiction because each half makes things true, but they can't be both the same, or maybe they are both the same. Maybe a paradox is not a contradiction; it's kind of a tension. We don't understand it, but it seems to be both. It's one big puzzle of quantum mechanics.
- 6.23** Another big problem is that it seems as if our observing the quantum world can influence the behavior of the quantum world. This is probably the most deeply puzzling feature of contemporary science. Before we measure the mass or velocity of a particle, it seems as if it doesn't have either. It doesn't have a particular position. It's as if measuring the particle collapses the wave function into a particular position.
- 6.24** One possibility becomes actual at the moment that we measure it. This overturns all of our priors and assumptions about the way reality works. Reality was not supposed to interact with us in that cognitive way, so that just by measuring something, we can—thinking about something in a particular way, we can change, we can actually make

something happen in the world. It's like telepathy. It's like telepathy. Like the paradigm example of pseudoscience: that minds can make stuff happen at a distance. That seems to be what our best physics is telling us is happening. We weren't expecting that.

- 6.25** People who think that the past is always going to resemble the future when we're doing science, you weren't expecting that to come along, were you? So we were right to be skeptical. So the role of the subject, the thinking subject, the measuring person, on reality, at this most fundamental, tiny, tiny levels, it's one thing having a wand and being able to do amazing things to buildings, but to have an impact on a world as infinitesimally small and undetectable to us as the quantum world is very strange. Just by taking snapshots of the way the world is, you can actually impact that world in some way.
- 6.26** It seems as if the claim that science shows that everything is determined by particular laws and by mass and by energy and so on and so forth, you can predict, if you've got all the data, you can predict how it's going to be. It's just, no, you can't. It's completely indeterminate. It depends on whether a measurer comes along or not. Until it's measured, the particle is in what they call a superposition. It could be in position—it could be in one position, and it could be in another position, one of the two.
- 6.27** This is where we have the example, for example, of Schrödinger's cat. Is Schrödinger's cat, is it alive or is it dead? You imagine a cat in a box with a quantum device that might kill it at any point. This theory will say, well, before we've opened the box, the cat is both dead and alive. Before we've measured or seen, observed, whether it's alive or dead, it is alive or dead. It's in this hovering between being alive and being dead. It's only when we get involved, we open the box, and we see it, that it's dead or it dies, or it stays alive, or it emerges alive. The observer plays some fundamental role. There's no reality independent of our observing reality. This is the thought.
- 6.28** Einstein hated this, hated this view. "God does not play dice," he famously says, that the world is not, at its most basic layer, indeterminate, just dependent on whether we decide to measure things that morning or not. There's got to be rules and laws. Do you really believe the moon stops existing when nobody's looking at it? Does it depend upon people observing it and seeing it for its existence? This seems to be the case of this theory in the quantum world, that it's somehow observer-dependent is very strange.
- 6.29** There's the famous limerick describing Bishop Berkeley's view that all reality depends upon our seeing it, that to be is to be perceived, "esse est percipi" in the Latin. The famous limerick is, "God must find it exceedingly odd to find that this tree continues to be when there's no one about in the quad." God must find it very strange that the tree just carries on existing when there's no one around in the quadrangle to perceive it and see it and hold it in existence, on this observer-dependence model.

- 6.30** The response is, "Dear sir, your astonishment's odd. I am always about in the quad, and that's why this tree will continue to be, since observed by yours faithfully, God." So Bishop Berkeley's response is, it doesn't matter if there's no one around in the quad, because God is always around in the quad, and God is everywhere. And he's always perceiving everything that is in reality, and there it is; that's why the tree is just always there. God is the creative intelligence and power behind all of reality; he's sustaining it by holding it there as an object of his intent and action.
- 6.31** Today, that will seem a very desperate move to save this idea that all reality is observer-dependent, but it features a lot in how to understand different theories of what's going on in the quantum world. The orthodox view is the so-called Copenhagen interpretation of what we're finding in the quantum world. Wave functions are not describing reality, they're just describing probabilities, probabilities of whether Schrödinger's cat is going to be alive or dead. We don't know at this point. It's both alive and dead, or neither alive nor dead at this point. It's in its superposition. It has 0.3% chance that it's going to survive, it won't die, or it will turn out to be dead or whatever it might be.
- 6.32** So probability is not certainties. And crucially, the cat is neither alive nor dead. It does exist in some form, but it's not alive and it's not dead. It's not as if it's one or the other, and we discover that it's one or the other. It's our discovering makes it either alive or dead, one of the two. You can see how science isn't getting easier as we find out more about the world; it's getting very mysterious.
- 6.33** It may well be the case, as we were discussing in the context of the inductive skeptic, that reality doesn't like us very much or doesn't really care about us having neat theories about itself. It's always going to be throwing surprises, and we can try to rationalize them away or come up with ever more elaborate theories that accommodates all of these different exceptions, but like barnacles on the hull of a ship, you're going to think, "I don't think this ship's very good. It's got all of these barnacles we can't really get off."
- 6.34** That would be a claim that the theory needs to change. We need a new map because this one is less and less accurate. We don't think it's accurate compared to what others could be. That would be a theory against Kuhn, that progress is possible. We want this map because it's a better map than the previous one. This is the problem, again, of thesis underdetermination, the underdetermination of theory by evidence. Evidence just looks the same. There's nothing within the evidence that can tell us which of the theories to pick.
- 6.35** Another historic example of this that's worth mentioning is theories about what space and time are. This is quite an old debate that goes all the way back to the 17th century. The question is, what is space and time? Do space and time exist independently of objects? If you emptied space and time of all objects and all particles, would there still be this container—space—and this container—time? This is Newton's view, that space

is a substance. It's like this big container. And time is like a big container as well, four-dimensional. That's the way to understand it; it's the way to understand space-time.

- 6.36** Leibniz says, "No, that makes no sense at all. There would be no space if there were no spatial objects." It wouldn't make any sense to say, talk about distance or extent or shape or surface area. It's not a surface area of nothing. It's not the distance between two non-existing things. It's a distance. That's how we start thinking about space or building the coordinates of space in our mind. But it's not like this container that we're thrown into. It's all this dense web of connections between countlessly many objects in the world.
- 6.37** That builds up this architecture that is just—you might think it looks like a container, but it's actually not. What you're seeing is the web, the big, vast web. Leibniz has this strange argument. He says, look, if space is a container, God could have created a universe three feet to the left. You've got space, all in this container. That's what space is. It's this container. Leibniz said, well, God could have, without breaking any laws about politeness or manners or the laws of nature or maths, "Now, I'm just going to put it three feet to the left."
- 6.38** Leibniz says this makes no sense. How can you have space that is non-space, something outside space, that is three feet to the left of space, when the whole idea of three feet depends upon space? What do you mean, three feet to the left of something that's not in space? Or three feet to the right of something that's not in space? It's not in space. It's not three feet to the left of anything or in any direction.
- 6.39** Leibniz says, God must always be acting for a reason. There must always be a sufficient reason for why anything exists in the world or why anything behaves in the way it does in the world. There would be no reason for God to do this. There'd be no reason for him not to do it. This must be the wrong way of thinking about what space is. You can see how almost there is a theological motivation for his point of view.
- 6.40** So, for Leibniz, space is not real, or rather, it's not a container. It is real, but all it is, is relations between objects. They're the real things. It's the connections, the web, the structure that's real, not the stuff that the structure encompasses, not just the container and everything inside the container. The idea here is that space doesn't exist independently, and it's the same—you can run the same arguments with time, but we don't need to go into that just now.
- 6.41** The point, though, again, is that, what does space look like if Newton is right versus what does space look like if Leibniz is right? The answer is: exactly the same, exactly the same. The world would not look any different if one of them was right, whichever of them were right; it would still look the same. How does science decide which theory is better, when the objects of science are identical for both theories?
- 6.42** Again, it feels as if we're having to appeal to something that's not, some non-scientific basis for choosing between two theories. As we talked about in the last lecture, maybe

aesthetic features, to do with simplicity and explanatory power and scope and so on and so forth. Both look identical. Both, as it were, are addressing the same appearances, the same facts. Similarly, when we look at the history of mechanics, physics, Newtonian mechanics, in particular, since the 17th century, we see different ways of understanding what motion exactly is and how we understand mechanics.

- 6.43** Newton himself, it's primarily in terms of forces, $f = ma$, for example, the gravitational force. That's Newton's way. That's what he insists is the case. The moon is pulled by the sun's gravitational force, and so on and so forth. But other theories don't use the idea of force at all. Field theory, for example, in the 19th century, doesn't invoke forces so much as fields filling space, so-called field theory. The field is what mediates action at a distance. It's a completely different way of thinking from Newton, even though it's describing the same findings.
- 6.44** Or you have Hamiltonian physics, Hamiltonian mechanics, which doesn't use forces or doesn't appeal to fields; it appeals to energy. Energy is a distinct principle that's fundamental. Kinetic energy plus potential energy—that's the fundamental way to explain the behavior of the physical world. Different mathematics, same predictions, same evidence.
- 6.45** Again, it's not clear that evidence can really—there are scenarios, there are situations where the evidence can't decide. Science can't decide. Science, as science, can't decide. We've got to go upstream of science, and we need to find different ways of settling between the theories.
- 6.46** As I said, once we've got to that point, we've got our theories on the table. As I said in the last lecture, we can then start to apply the different criteria of theoretical success—simplicity, power, consistency, scope; even elegance and beauty, I referred to last time as being something that is often reported by scientists and mathematicians when they've hit on some new theory that turns out to be true. The advantage of that phenomenology in mathematics, that way of perceiving things, understanding what the truth, discovering the truth-seeking process, is that theorems are theorems in mathematics. If they're true at all, they're true no matter what.
- 6.47** That's never the case in the sciences. It'll always be the case that some theory we have, some scientific theory we have, might be, very likely will be shown to be false or not quite accurate in certain respects in 100-150 years' time. It's highly likely that most scientific textbooks will be shown to be wrong in that way, or just not close to being fully right. And that seems to be how science develops.
- 6.48** Whereas with mathematics, we can hit on something, we discover it, and then we've cracked Fermat's last theorem. We've cracked it. It's done. It is what it is. It's true. It all works. There's never going to be a day when the solution to Fermat's last theorem is not, in fact, a solution to Fermat's last theorem. It's all done. It's a theorem, not a theory. A theory can always be tested. A theorem is true, no matter what. It's been

established to be true, no matter what. That's why, in mathematics, you have theorems, and in science, you don't. You can't. Again, to stress the role of philosophy in all of this.

- 6.49** Then just to turn to the final phase of the lecture, we're coming back to an issue that has begun to bubble away at the edges of the previous lecture, this question of realism versus anti-realism. I've used already repeatedly this metaphor of this, do you take the map-making view of science? Or would you say, no, the map is a guide to the terrain that we're going to go and explore and discover, and how could we make maps of terrain we hadn't seen, we didn't know anything about, we just had other maps of? How would we ever know it's a map of what it's saying it's a map of, if our only confirmation for it is another map?
- 6.50** We have an infinite regress of maps going all the way back to, when? Who came up with the first map? And how do we know that the person who came up with the first map had access to, had even been to Arizona? Or knew anyone who had been to Arizona? How could it—we could say that it's a map of Arizona because it was a map of another map of Arizona and so on and so on and so on. As I said, a hall of maps, hall of mirrors. You don't know where to look; you don't know what's real.
- 6.51** At one point, it was very plausible to say, can we believe that electrons are real? We can't. For a long period, we assumed they existed. We posited that they existed. It helped the theories to assume they existed. But we didn't have any evidence. We couldn't detect electrons for a long time. We just assumed that there must be negatively charged particles to make sense of other—it was a simple and elegant explanation that made sense of everything that we could see, even if we had to posit this one thing that we couldn't see, this negatively charged particle that we called an electron. It was called, baptized an electron in the 19th century.
- 6.52** Or as I mentioned before with black holes, for a long time we had no evidence that there were black holes. But by assuming that there were such things, it made so much sense of the rest of the theory. It made sense of everything else that explained things that we could see. And then, sure enough, the black holes turned up. I think 2019, we had the first definitive imaging that there really are black holes.
- 6.53** Now, on the realist view, on the view that the maps are about a particular terrain that we are directly seeing and forming judgments about, that does make sense. It does make sense that we can be making these predictions that the world is—our theories are making predictions. Our theories are sticking their necks out to say that the world is probably like this. We'll probably discover, even though we haven't yet, that it's like this.
- 6.54** Electrons become to be detected. A black hole is discovered. The Higgs boson particle is discovered in the particle accelerator in Geneva, 40, 50 years after Peter Higgs predicted it. Well, it does seem like this is more than just map-making. This is making

claims about the world that turn out to be confirmed, turn out to be right, true. And this doesn't make sense if electrons and black holes were just useful fictions that we were fitting into our map to make sense of them, to make it a better fiction. It does seem as if we've discovered something true about the world.

- 6.55** The argument that, for realism, the argument against the map-making view of science is, in Hilary Putnam's famous words, the great Harvard philosopher from the 1970s and 1980s, just died a few years ago, great philosopher of science, he develops this argument called "the no miracles argument" against the map-making view of science, the no miracles argument for realism, that reality is real, the view that reality is real.
- 6.56** He says, realism is the only philosophy that doesn't make science's success, the success of science, a miracle, a constant miracle. The idea here is that, the view that there are laws of nature that explain why the sun is not going to become a pink elephant tomorrow morning, but it's going to rise, or rather, fall relative to the Earth or whatever, the reason that that's true is because, if we didn't think that it was true, if we denied that there are such laws really out there in the world that explain these things, the sun coming up every day would be a miracle.
- 6.57** Every day, not one day. It could have turned into a pink elephant at any point, on any day, in the last 13.8 billion years. And it hasn't. But the skeptics tell us it could have become a pink elephant, just as likely to become a pink elephant. Every day, the miraculous happens. Every day, it just doesn't turn into a pink elephant, but it does the same as it did before. And Putnam says, are we just going to say it's just a miracle every day? This one-in-countless-trillion chance of it doing something other than coming up in the morning—the pink elephant, green elephants, etc.—and it could be anything; and yet, every time it's that.
- 6.58** And if there are no laws out there that explain why it's always doing that, then it would be a miracle every morning. This is what happens. Every time, it was just the probabilities of it happening every time, every 24-hour cycle for 13.8 billion years, and no pink elephants ever, or no any other, nor any other elephant, it was always that, that event happening, it would be a miracle, if there were no universal principles making that repetition the case.
- 6.59** So Putnam says, look, it's more reasonable to believe against perpetual, not to believe in perpetual miracles, than it is to believe that this is a miracle that's happening every single day. Putnam would argue that we predicted the existence and location of Neptune for all kinds of reasons. It made sense that this was not a moon, that actually a planet. Before Neptune, is observed, 1846, 1840s at least, its existence and its location are posited. Again, it would just be a miracle if there was nothing to our calculations, nothing to what the wiring of our guess is.
- 6.60** It was that we made this prediction, but in fact, it wasn't a prediction about the way the world is. It just so happened that the world showed its face to us in a particular way

that was exactly how we predicted it, namely, that there would be this planet, Neptune. And Putnam says, we can't live with the idea that this prediction is not a prediction, even though it says exactly what happened. It just got lucky. It's just lucky. No, that would be a miracle. It's just a miracle.

- 6.61** I knew nothing about—none of this math works, none of this science works. None of this stuff, the study of the planets we've been doing for hundreds of years, works. But I'm just going to guess this. Oh, the guess is correct, but it has nothing to do with my reasoning. I just got it lucky. I just got lucky. I got lucky predicting that there would be a Higgs boson particle. I got lucky predicting there'd be this black hole. I mean, it was a black hole, but nothing in my reasoning. I was just shooting in the dark, shooting into black.
- 6.62** Putnam says that this doesn't ring true. It can't be that this is just scientific predictions, and insights are just flukes. Theories describe reality. And so the best explanation for why these predictions turn out to be correct is that the theories are approximately true. They correctly describe, maybe not perfectly, but well enough, they describe the world in such a way that we can assume that that thing exists, or that it will behave in this particular way.
- 6.63** It's much more probable that theories are actually true, rather than—that our theories are true, rather than they're just correct maps for us—the no miracles argument, the explanation of novel predictions, the way in which we can posit certain reality being in a particular way, and that then not just being confirmed, but actually helping to unify what we can actually see.
- 6.64** Also the convergence of different sciences, the way in which philosophy—the way in which physics doesn't tend to conflict or collide with chemistry. Neither do they conflict much with biology or astronomy or cosmology. There is this convergence of different sciences, which, again, would suggest not just that the physics have the physics maps, and the biology people have the biology maps—they all converge, and that would suggest the best explanation for why they converge, in a way that psychoanalysis doesn't converge or astrology doesn't converge in the same way, is that there is just one underlying reality that these sciences are describing different aspects of, but they're compatible. They don't contradict each other. That suggests, again, that there's something out there; there's a single reality out there to be described.
- 6.65** So this is the attempt to push back against the map-making view of science, the map-making view of understanding what's happening in scientific theorizing. In some ways, the fictionalist way of thinking about science, it can make sense. You might say, scientific statements are neither true nor false. It doesn't matter they're neither true nor false, because we can still, within a theory, within a world, apply standards of correctness and incorrectness.

- 6.66 So, is it true or is it false that Sherlock Holmes smoked a pipe? Is it true or is it false? It seems like a very easy question. Of course, he smoked a pipe. He didn't smoke cigarettes. Actually, in some of the stories, he smoked cigarettes, but he's a pipe-smoker. The pipe is eternally associated with Sherlock Holmes. That's what he smoked. That was his preferred means of smoking. But of course, it's also not true that Sherlock Holmes smoked a pipe, because there was never ever any such human being as Sherlock Holmes. He doesn't exist. Never existed.
- 6.67 Well, is it true? Is it false? No, it's all false. Just a fiction. It's just a story. It's just a story. Yeah, but it's still true that he smoked a pipe. Well, he didn't smoke cigarettes. So, in what way is it true, if it's a completely false, fictional world? Well, we can say it's correct to say that he smoked a pipe. It's not incorrect to say that. If you answer that at a quiz show, you should be very angry if they turn around and say, "We're not giving you the prize because actually it's neither true nor false that Sherlock Holmes smoked a pipe." If you were playing Who Wants to Be a Millionaire?, you'd try to sue. You'd be very angry if they asked you a question that was a true or false question, that actually turned out to be neither true nor false.
- 6.68 And so, the map-making view of science might appeal to this escape route. We're not saying, when we talk about Newtonian motion in terms of forces, we're not saying it's true about the way the world is, but it's correct. We've got this big, impressive theory of motion as a mechanism. And within that theory, there are certain things that are consistent and correct, correctly stated within the theory, and there are certain things that are not, that just don't fit the theory.
- 6.69 The theories that cannot cope with the claim that we should understand motion in terms of fields rather than forces, that's out. It's incorrect. It's correct to say that it's forces. But it's not true any more than it's true that there was a person, Sherlock Holmes, who smoked a pipe. It's just, this is what's most compatible with the map that we've made of the world, the map that we've made of reality. It's either correct in terms of the rest of the map or it's not correct.
- 6.70 And so, we don't have to worry about these big claims about true and false, and it's not just everything's up for grabs. We can still get a sense of what's an appropriate scientific statement, what's grounded as a scientific statement, and what is not grounded, without saying that we've got a map-free perspective on reality. That's the thought.
- 6.71 And again, both fictionalism, both the correct-incorrect view of scientific statements and the true, the realist true-false view of scientific statements, both are consistent with the way reality is. Neither theory changes or can be vindicated by the actual evidence that the world would appear to be the same way, whichever theory was true. So again, we're trying to do science. We're trying to strip, see what happens when you strip philosophy away from science, and it turns out, we always end up back in this philosophical pit again. We can't seem to escape it.

- 6.72** In the next, concluding two lectures, we'll be looking at more ways of understanding successful scientific theories in terms of explanation, what explanation is, what unification is in scientific theorizing. Then, in the final lecture, we'll be looking at some applications of all the principles that we've learned in the seven lectures running up to the eighth and final one. Thank you so much for listening.
- 6.73** You've spoken a lot about science stripping away philosophy from science. What about things like authoritarian regimes stripping the truth from science? Like Galileo and the Catholic Church, or one of the most sinister ones, would be Stalin and Lysenko, leading to their false genetics and the death of millions of people. I know they're two different subjects, but I would love to know your thoughts.
- 6.74** Yes, sure. Well, this raises an interesting question as to what the ethical status of science is, what the moral status of science is, and I think you might say that all those science, doing good science relies on certain principles of virtue, like collaboration, paying a researcher when they've done the research that they've done, philanthropists being generous and putting money into a completely untested, wild scientific claim that might have a huge payoff. These are good. This is all virtuous behavior that is needed for science to progress, you might say.
- 6.75** But that's not the same thing as saying that the conclusions of some scientific inquiry have a normative or moral angle either way. It's just, science is what it is. And very often states, not just Stalin, but every state is investing in R&D. Every sensible state is going to put a significant chunk of, relatively significant chunk, of its budget into its best scientists and researchers, exploring new technological frontiers that could help a nation rush forward in terms of prosperity and flourishing.
- 6.76** So yes, there was Lysenkoism. Also, there was the Manhattan Project. Well, now you might say the Manhattan Project was a disaster and it ushered in nuclear war and all this kind of thing. But actually, that was a role, that was an example of government and scientists working, generally speaking, very successfully. The point is, it's neither good nor bad, I suppose. It's just, science is what it is. I think, if you're an Aristotle, you say it's a good thing to learn about the world.
- 6.77** I talked about, in a previous lecture, the case of Galileo, his initial arrest in 1613 and then his second house arrest in 1632, and described it as not really a clash between an authoritarian church and a secular hero who was managing to free himself of the irrationality of religious belief. No, this was an internal debate within Christendom about how to understand the Ptolemaic cosmology.
- 6.78** Galileo said, "It doesn't work." The Church said, "Well, actually, this is the model we've been using for a thousand years." Now, was there anything in the Bible that says, is it in the Ten Commandments, "Thou shalt worship no other astronomers than Ptolemy"? No. It's not some, it wasn't some hill that the Catholic Church was going to just die on

because no Ptolemy, no Jesus. It was actually—it wasn't this great clash that textbooks and popular culture sometimes make it out to be.

6.79 I guess there is always that tightrope between science is authoritative, you've got to believe it, you've got to trust it, follow it at all times. On the other hand, there's the recognition that the only reason that we're trusting science is that it's constantly updating itself and is receptive to falsification, attempts to falsify it and to incorporate new information. And so, that's the challenge when the state gets involved.

Lecture 7: Philosophy in Science

At a glance — Philosophy within science — where the two meet.

- 7.1 Welcome to Lecture 7, the penultimate lecture in this eight-part series, Philosophy of Science. We're getting to the point now where we're concluding our examination of the different concepts and principles and flashpoints in contemporary philosophy of science.
- 7.2 Last time, we looked at the question with all the different spectacles that the different theories give us, with all the different maps that the different scientific theories give us of the same terrain. Can we ever be sure that we can actually get through to the terrain, or that the map is reflecting the terrain? Questions about how theories change. Does science get better? Does it progress? Or is it just ditching one map and it's just got a different map that shows different things, but neither is better than the other?
- 7.3 The three areas we're going to look at in this lecture are causation, the question of causation, the question of laws of nature, and thirdly, what a scientific explanation really is. We're going to approach it, those two topics first. It'll be laws of nature first, then causation. Once we've got an understanding of just how much disagreement there is among philosophers of science on those two most fundamental questions, causation and laws of nature, and this is the cement of the universe, as one philosopher has called it, causation. The laws of nature are the architecture of physical reality, the principles, the axioms.
- 7.4 What are the different competing conceptions of these two features of reality, laws of nature and causation? It turns out that both laws and causes feature a lot in what counts as a scientific explanation. One of the key distinguishing features of why a scientific explanation purports to be a scientific explanation, as opposed to a non-scientific explanation, a historical explanation or a personal explanation, and also that distinguishes it from pseudoscientific statements.
- 7.5 Pseudoscientific statements don't actually explain, or they explain everything, and therefore, they explain nothing. Something that explains everything explains nothing. That's what we're going to be doing. That's going to be the battle plan for this lecture. Then that will set us up perfectly, as the previous two lectures will have done, for our final lecture, which will be looking at four case studies, four ways in which philosophy and science intersect in our world today in very practical ways.
- 7.6 So first of all, the question of laws of nature. We're using laws the whole time. You will have heard it in almost every lecture. I'll have been talking about certain laws. Of course, we talked in the final lecture of the historical first half of the series, we talked a lot about Newton, of course, and his laws of causation, the law of universal gravitational attraction, and so on and so forth. We've also touched on problems to do

with causation, how to be absolutely sure that a billiard ball, when it knocks into another billiard ball, is causing that second billiard ball to move.

- 7.7** Remember, this skepticism about how we can't just assume that the past will resemble—that the future will resemble the past. We can maybe form a habit about it. But the one thing we can't do is make causal claims and causal explanations about the future. So, the question of laws of nature really begins, really emerges as an idea in the 1600s with Descartes and with Newton.
- 7.8** Both of them have different ideas about laws. For Descartes, he's thinking a lot about mathematical laws. Newton is beginning to think more in terms of, these are the forces, and the language of forces is the language of Newtonian mechanics. They are to be understood as laws because there's, as it were, a divine lawmaker at the top. It makes sense that, just by analogy, we've got, as it were, a divine legislator for nature, setting down the laws, the laws of nature. But that's quite a new phrase. It's not around before 1600. Before that, people talked more in terms of the powers that particular objects have, the power of salt to be dissolved in water, and so on and so forth. But that's an Aristotelian framework that is, in some ways, on its way back.
- 7.9** So, what are we talking about when we're talking about a law of nature? Well, intuitively, we appeal to a law of nature when we want to explain some regularity in nature, something that just keeps on happening, or that we've established keeps on happening in a universal way. It's all over space and all over time. There's a principle. It seems that it's not just accidental. You can invoke a law, and it explains things. It helps us also form the basis of making predictions about how the world might be in the future. It gives us confidence if we believe in these laws. Of course, Hume doesn't, and that's why the skepticism starts to come in. He says, "You can't see these patterns."
- 7.10** So, laws help us to explain the world. They help us to predict what might happen in the world. They also explain these tricky sentences that philosophers spend a lot of time worrying about, counterfactuals. Counterfactuals. "If I had done that, this would have happened." "If I dropped this glass, it would smash." What makes that sentence true? Because it hasn't happened. The glass is still in my hand, but we have this strong sense that there's something that grounds that counterfactual as true. So what could it be? Why? What's the explanation for why this would happen if I were to do that?
- 7.11** Intuitively, a law of nature helps us understand that. If I were to do that, because of the law of gravity, this is what would happen. It's true that this would happen and false that it wouldn't. And so, that's another fairly technical area in logic, you might say, in semantics. But if a theory helps us to make sense of ordinary ways we're talking, then that's to its advantage. And so, the idea that there are laws makes sense of tricky sentences like that, makes them true.
- 7.12** So this is what we want laws to do. So we really want there to be laws to do all the explaining. So it's uncomfortable, you might think, that many have followed David

Hume's skepticism that there are any laws at all. Many in the world of philosophy of science take the view that there aren't any laws in the sense that we understand, might intuitively understand.

- 7.13** Some will say it's a hangover from a much more theological age, when it was natural to talk in terms of a divine lawmaker at the top, manipulating his creation, pulling all these puppet strings. But we've got rid of that. Not only do we not think there's a divine legislator at the top, lawmaker at the top, but we don't think there are even any puppet strings. We've got rid of the divine legislator at the top. And says Hume, there are not even the puppet strings between different objects—causation. They don't exist. You can't see them.
- 7.14** And so, you might ask the Humean, of which there are so many today, well, what are they? What are laws? When we say that, we talk about the law of gravity, we're picking something out, we're picking out a particular principle that explains regularities and behaviors. Hume's view, the Humean view, is that laws are just regularities. They're just regularities that we describe, a constant conjunction. There's no miraculous thread going through all of these events. It's just a regularity that we apprehend and record and see, and that is useful to us in our theories. And so we call it a law.
- 7.15** But there's nothing law-like out there in the world. There's no magic oomph that just appears between two objects. There's no invisible puppet strings left by the God that we got rid of in the 17th century. There's nothing there at all. As I said in a previous lecture, Hume says, "The mind has a tendency to spread itself out upon the world and to turn the world into something that is intelligible and human and perceptible." Just constant conjunctions, just regularities, that's it.
- 7.16** What's the problem with this view? Well, even though it's a dominant view, it obviously, to my mind, doesn't do the job that we want a law to do, which is to not just state that there are regularities, but to explain why there is that regularity. In particular, that explanation is going to help us distinguish between a law-like regularity, a lawful regularity, and an accidental regularity. What if some of the regularities we're observing are just accidental, they're not governed by any law at all?
- 7.17** We want to be able to say, even though that's a regularity, it's not a law. It's got to have something more than that. The famous example here is that it is a regularity in the world that there are no gold spheres greater than a mile in diameter. It is a stated regularity. It's a regular feature of the world. It's never been done.
- 7.18** You can also say that it is an observable uniform regularity that a sphere of uranium wider than a diameter—wider than a mile in diameter has never been a feature of the world. It's a regularity. This has never happened. And so, on the Humean view, both are laws. Both are universally regular features of the world.
- 7.19** But the objection is that there's plainly a difference between these two scenarios. In the first case, the regularity is an accidental one. You could just about, there's not that

much gold in the world, but you could just about—it's not against, it wouldn't breach any, break any laws of nature, to get a huge ball of gold, wider than a mile in diameter, together. There's nothing in the world that prevents that from happening. It would be impossible to do that. It would break a law of nature to create a sphere of uranium that's the same size. The laws of nature prohibit it.

- 7.20** We want to be able to tell the difference between these two regularities. One is a lawful regularity, the other is just an accidental one. The worry is that the Humean view can't really cope with that, can't really make a valid distinction between those two scenarios. So that's one view of laws, really, that there aren't any laws. Laws are ways of speaking or labeling certain regularities and generalities that we've observed. You might say that's the no-laws view, but they still like to think of themselves as offering a theory of the laws of nature.
- 7.21** There's another group that emerged in the late 1970s in many American universities, people like Fred Dretske, David Armstrong, Michael Tooley, who accepted the criticisms of the Humean view and tried to develop an account of what it might mean to say that there really are law-like patterns; there really are puppet strings out there to be understood somehow—obviously not by reference to God. This is a very secular intellectual culture.
- 7.22** But maybe there are other ways that we could appeal to that are, broadly speaking, consistent with naturalism, consistent with the natural sciences, but that help us do the explaining when it comes to the laws of nature, and help to ensure that the laws of nature offer proper explanations for seemingly non-accidental regularities in the natural world. They developed a number of different positions. We don't need to go into the details, but very technical metaphysics. This is a technical philosophy, looking at this deep question of what a law of nature is.
- 7.23** In effect, as I said, we don't need to go into all the details, but in effect, what they posited was something like a universal causal relationship between particular things in the world. And that relationship is a relationship between what's called universals. So a universal, this is an idea, obviously, that goes all the way back to Plato. So the idea is, the problem is, how do we say that the white plate and the white cup both share in being white? They both have the same property. These are not two different properties, not two different ways of being in the world. There must be something in the world in virtue of which it is true to say that both those objects are white.
- 7.24** Now, what could that be? Well, it's a very, very long debate in Western philosophy, but broadly speaking, it's a debate between those who say there really is something that is the case, that is being white, which both of those objects have or display in some ways, whether it's part of their nature, whether it's essentially, or whether it's just accidentally. They participate in or they instantiate the property of being white.

- 7.25 The question is, do we go with Plato, and albeit in a different way, Aristotle, to say, yes, there really are these universals: being white, being square, being good? All of these are universals that particulars exemplify to different degrees. That's what makes it true to say that they resemble each other and so on and so forth, and to ascribe the same word, the same predicate, to both the plate, to both the bowl and the cup.
- 7.26 On the other side, in the other corner of the ring, as it were, you've got the so-called nominalists. They say that there's no such thing as being white. There's no such thing as being good. Those are just names, "nomina" in Latin, that we attach to these different objects to, as it were, help us navigate the world a bit. Our language, our nomina, our naming and our labeling, is what imposes the grid of intelligibility onto the world.
- 7.27 That's how we—that's what we're doing. We're projecting these patterns. We're projecting these similarities. These are similarities or resemblances that might be relevant to us. That's how we're thinking about it. There's no Platonic heaven. There's no Aristotelian forms and essences, none of that. We've put away childish things. That was from centuries ago. How could it possibly be true?
- 7.28 So these figures brought in the idea that there was some causal relationship, or necessitation relationship, between the universals, wherever they might be, whether it's in Plato's heaven or whether it's wired into all of the different Aristotelian essences. The idea is that you've got causation down here, and then you've got them between two particulars, and the particulars share certain universals. Then up with the universals, you've got a relationship of necessitation between the universals.
- 7.29 That's where you get the universality of laws from, and also where you get the causal, you get the causal energy. It explains why things behave in particular regular ways. So this was their idea: put the puppet strings back in. Don't worry about the fact you haven't got a puppet master, a puppeteer. That's in effect what is going on here, in many ways. They're trying to recover the explanatory power of the theological model, without the reputational costs within a secular philosophy faculty, maybe, of bringing the whole legislator back in.
- 7.30 Those, for many years, were the two basic camps: the no-laws view, and then the causal principle between universals, necessitation view. Then I would say, in the last 20 years, a third view has emerged, which is really a recovery, a retrieval and a reworking of Aristotle's view on these things. Broadly speaking, Aristotle is not, in the historical section of the course, I noted that a lot of the emergence of the scientific revolution was a repudiation of Aristotle, and in particular, a repudiation of Aristotle's doctrine of essences, essences and powers.
- 7.31 Really, the new dispensation said, we're not interested in these mysterious essences; the question, what is X? What we're interested in is, what does X do? What is its behavior? What is its observable behavior? We don't need some mystical account of the inner essence of an acorn. It's just, what do these things normally do? How do we

classify, catalog them, etc.? Describe them as dispassionately as possible and in as philosophically neutral a way as possible.

- 7.32** It's something of a surprise that this Aristotelian view has come back. But I think the position will say something like—this is the so-called powers view—will say something like, well, look, in your desperation to make sense of why laws have explanatory power and why they feature so strongly in scientific explanation, your attempt to shore that up is already getting pretty desperate. You've brought universals back. You've invented this strange necessitation relation between universals.
- 7.33** Do you believe in Plato's heaven? Or if there is no Plato's heaven, what exactly are universals? If they're down here, as it were, with us in space-time, are universals multiply localized, multiply instantiated? You get all of these questions start to emerge. The Aristotelian is effectively saying, well, look, we've got a more elegant theory here as to how these regularities occur, that things have essences, and part of their profile, part of what is essential to them, are properties that can be exercised in different circumstances, powers that can be exercised in different ways.
- 7.34** It's not so much a law of nature that you're seeing when you put some salt into a glass of water, and you see the salt dissolve. It's not a law of nature that is driving that process. No, it's more the laws of natures. It's a law, as it were, onto the sodium chloride molecule in salt, that it has certain powers, dispositions, to act on the world in particular ways, and certain passive dispositions, passive properties, that will ensure that it behaves in particular ways, if it's acted upon by another power. As it were, powers and liabilities are connected, obviously, in that way.
- 7.35** But it's to do with the molecular structure of water and the molecular structure of salt, and the powers that those have as essences, as natural essences, the essence of water and there's an essence of salt, and it's to do with their intrinsic powers to relate with each other in particular ways that explains the regularities. Wherever you see salt, you've got something that has the essence of salt. You've got all the explanations that you need.
- 7.36** That view has come back to quite an impressive degree, given that, for 500 years, really, Aristotle has been banned since the dawn of the scientific revolution. To see him coming back in the beginning of the 21st century was not something one might have predicted 30, 40 years ago.
- 7.37** So again, we've come to see that, even where you've got a fairly settled consensus as to what something crucial to science involves, namely the laws of nature, there's not much disagreement as to what counts as a law of nature and what doesn't, what counts as a scientific statement of law and what doesn't.
- 7.38** Even among the philosophers, there isn't really any disagreement about that. All the philosophers accept that these are laws. But there is a huge variation of opinion and a lot of conflict among philosophers of science when it comes to explaining what on

Earth laws are. You have at least one camp that doesn't seem to believe in them at all. You have another account that is having to introduce certain new hypotheses that make a lot of people very uncomfortable. It does seem to be like bringing back the puppet strings, but without the puppet master. There's signs of desperation in trying to shore up that, to shore up that realist account of how laws have an actual impact on the world that aren't just accidents.

- 7.39** Then there's a third conception that also doesn't believe that there are laws. It says that it's powers, and it's the powers that particular things in the world have and the categories that those particular things belong to—that's what explains how they behave in the world, act on other things in the world, and are acted on by other things in the world.
- 7.40** There's a huge contention underneath the bonnet, underneath the hood, as you say in American. There's a lot of dispute and conflict and disagreement happening beneath the surface. Science doesn't really help resolve— again, science doesn't really help resolve this issue because the facts are what they are, no matter what. Each theorist, each perspective here, is just assuming that the scientists—taking the scientists at their word as to what counts as a law of nature and what doesn't and so on.
- 7.41** The second area in the course, in this lecture, that we're going to look at, connected, of course, to laws of nature, is causation. It's even more fundamental, you might say. What is it that laws govern? Well, patterns of particular kinds of events, particular kinds of events, and how should we analyze an event, something happening, anything happening?
- 7.42** Say that A causes B. What exactly do we mean by that? We could say smoking causes cancer. Well, yes, it does. It has a tendency to. Does it cause it every time? No. How do we understand that statement? It's intuitively correct, but also you realize it's false, as it's stated. How do we make sense of degrees of causation and different contributory strands to a causal event? We think a phrase like "Smoking causes cancer," for example, to be very different from "Gravity explains motion." Gravity governs motion. Gravity that you might say causes motion.
- 7.43** So causation is central, like the laws of nature, to how we explain things, how we predict things, how we make sense of things, how we can assign control and responsibility in some scenario. Hume's great criticism in 1748 in "The Enquiry Concerning Human Understanding" is really one that comes from his radical empiricism, this radical stress on the view that, really, the benchmark of what's real and true is sense experience. It's not the reasoning, it's not the rational filters through which we seem to make sense experience intelligible; it's just sense experience comes first, and then our ideas and our thinking emerges out of contact with what we see in the world, what we see and sense.

- 7.44 And so, he imagined a billiard ball knocks into another billiard ball. And Hume says, "Well, what do you see?" We might say, "Well, yes, that billiard ball caused the other billiard ball to fire off into that corner." We can make sense of that. There's a causal relationship between these two balls. Hume goes, "But is there? How do you know?" "Well, you can see it." "What can you actually see?"
- 7.45 What Hume says is, what you actually see is, you see ball A moving. You then see ball A contiguous with, touching, ball B, and then you see ball B fire off into the corner. You don't see ball A cause ball B. You just see them touching, you see them together, and then you see them move in different directions. You don't see any causal fizz between the balls. It may be very weird if, instead of firing off into the corner of the table, the ball that had been struck turned into a hippopotamus. It would be very unusual.
- 7.46 But, says Hume, it would be no offense to logic. It's not like breaking some mathematical law. It's possible. There's nothing that you're observing here that the opposite couldn't be just as possible. There's no magic happening here in the event. All we are justified in forming judgments about are what we can see, and we can't see these causal connections. This is just a psychological habit that we've got, this idea that there is causation.
- 7.47 It's a very radical view in lots of ways and has enormous influence on Anglo-American philosophy throughout the 19th century, and particularly into the 20th century. It's seen as a view, both the regularity view of Hume and also its sibling theory in the laws of nature, the no-laws view. These are just regularities. Those two go together.
- 7.48 That seemed to be much more consistent with Stephen Hawking's quote that I opened with at the very beginning of this series. All we're doing is describing what the scientist sees. We're just the backseat driver, just describing it. We're not bringing in any magic explanations. We're not invoking God or necessitation relations or spooky powers, Aristotelian essences. Everything's just purely descriptive. That's our job.
- 7.49 There's something attractive about that to philosophers who have science envy, who can see that philosophers have been arguing about the same questions for 3,000 years. They look across at their science colleagues who, until recently, were gentleman amateurs. Suddenly, they've transformed the planet beyond recognition, beyond anything that any chapter in human history is achieved. It's not fair. I think there is a certain, in many ways, justified respect of the achievements and successes of science.
- 7.50 But that's brought with it, I think, something like an inferiority complex when it comes to philosophers who are trying to, who do have a legitimate role to play in helping scientists clarify what they mean and explaining the challenges. This may partly explain the popularity of these much leaner and meaner, much more pared-back, philosophically agnostic, you might say, or quietest perspective, on big philosophical questions.

- 7.51** There have been various attempts to make sense of causation in ways that don't invoke what naturalists think of as spooky powers or improbable law-like relations between universals. David Lewis, a very famous American philosopher at Princeton, dominated the 1970s, '80s, and '90s, develops the so-called counterfactual theory of causation, which is very elegant in lots of ways. What it is to say that A causes B is just to say that, if A had not occurred, then B would not have occurred.
- 7.52** We can say that a cause has some role in making that counterfactual statement true; it's somehow essential to it. Take the alleged cause away, does the effect disappear? If there is a change, then we can say that it's earned the label "cause." It's a clever view because it captures our intuition that causes make a difference in the world, that we were explaining something. It also helps us understand the difference between causation and just accidental correlation, something that just accidentally happens to come together.
- 7.53** Something that accidentally happens to come together, you can say that this would not be the case if they had not come together. It wouldn't be the same dependency relationship in the counterfactual if they were just accidentally together, and they were not in any causal relationship. It's quite nice because the idea is that it puts the causal power in the logic of our statements about the world, but it is agnostic about, is the case in the world, but explains why that counterfactual is true. That's the problem.
- 7.54** The final view that we can look at, in many ways—this is to be paired with the laws of nature discussion—is the causal powers view. This is back to this idea that there is, that things in the world belong to a particular category, a natural kind. In virtue of belonging to that natural kind, you've got certain—you're endowed with certain powers. What's happening when we see A cause B is that A is exercising powers that it has on B, and B has certain dispositions to react to those powers when they're exercised by A on B. That's the way to understand it.
- 7.55** You can say that a match has a power to strike, to ignite when it's being struck. The striking is the manifestation of this power, and then the ignition occurs. That's the way to understand it. Salt has the power to dissolve in water, water has the power to dissolve salt, and the manifestation of that power is the dissolution in the water. So, this is an attractive view because, in many ways, it's understanding regularities as something that emerges organically from reality, from nature up, from the kinds of things that things are, all the way up.
- 7.56** Whereas the laws of nature view, as opposed to the powers of natures view, that emerges in the 17th century from Descartes and Newton and others, is much more top-down. It's much more—you've got a divine legislator at the top, puppet master at the top, got all these puppet designs, puppet strings, and then it's just the puppet strings. There's nothing about the way the world is, the distinctive ways that things are, down at the bottom, things that are governed by the laws of natures; their natures, what they

are, is completely irrelevant. It doesn't matter at all. Everything can be explained by the puppet strings.

7.57 This view actually says, no, it matters what kinds of things things are. That's going to be fundamental in the explanation as to why that event, involving that thing, occurred the way it did. That, in some ways, gives us a third relatively new rival and very realist view about what causation is. Causation at that level is the exercising of powers; it's real, it's robust. What gives them that force, the necessity, is because it's part of what it is to be that thing.

7.58 This, by the way, is how a lot of scientists often talk about what they're investigating into substances and particles and organisms. It's as if these molecules and compounds and elements do genuinely have powers of their own, in virtue of, because of being the kinds of things that they are.

7.59 We can see that, again, as with the question of the laws of nature, you've got three really very different camps that certainly can't all be right. Maybe, actually, maybe they're all wrong. But if any of them is right, only one of them is right. The no-laws approach, and then the top-down laws approach, and then the bottom-up powers approach. You've got three analogous, three similar but parallel views about causation to go along with those three camps. We see, again, this fundamental disagreement, and very difficult to see how science can help us resolve that difficulty once again.

7.60 Of course, it raises a lot of the questions that we haven't touched on very much, but we've glanced at in previous lectures—the implication of a powers view or a law's view or a no-law's view on responsibility, on our free will, on how we understand reason, how we understand our minds, mental causation, and so on and so forth. This is something different from just billiard balls knocking into each other.

7.61 So we've looked at laws of nature, we've looked at causation, and we can tie these two together with some reflections on what scientific explanation is. Because, of course, science doesn't just predict, it explains. That's what commends one theory over another to us, the fact that it can explain more; it's got more explanatory power. The question then is, but what counts as an explanation that is scientific?

7.62 Very plausibly, you want to say, "Well, it's an explanation that involves scientific statements, or statements that scientists are best equipped to form accurate judgments about." "Well, what are scientific statements?" "They are statements, when you get down to it, if you think of the whole grammar of science, there are these principles, how to do the verb tables and so on, and how to do the conjugating and the declining.

7.63 Those are the laws of nature, the ones that we've discovered." I mean, they're the most scientifically freighted statements of law: $f = ma$, $e = mc^2$. That's not all of science, but these, as it were, these are the golden tickets to understanding how the world works.

It's these principles, these regularities, these laws. That's what makes a scientific explanation, scientific—appeals to laws, appeals to causal patterns.

- 7.64** One very famous view developed along these lines of what explanation is just says, a good scientific explanation is one that can deduce from the facts a law, or can work out what law is making this the case. This is a very distinguished philosopher of science in the 20th century, a guy called Carl Hempel, in the 1940s. He says that what explanation is, is a deduction from laws. You can say, the reason that X is happening to Y is because of, and I can trace it, deduce it, back to that "nomos", that law. It's the deductive nomological model of Carl Hempel.
- 7.65** He says, if you just give me the laws of nature, you give me the initial conditions, I can tell you that's the most complete scientific explanation you can hope for. Why did this bit of copper conduct electricity? Hempel would say, "What's the law?" The law is that all copper conducts electricity. Initial condition, this is copper. That's all I need to—that's all you should want to ask for as a scientific explanation. That's it. I've given you the law. We've got the initial conditions. That's all you need for it to count as a complete explanation. For Hempel, the laws are fundamental to what explanation is.
- 7.66** And so, it's awkward that there should be such disagreement about what laws are. Remember, we had the no-laws view, just another accidental regularity. There's the top-down laws view and the bottom-up, no-laws but powers view. Here, Hempel is telling us that what really sets scientific explanation apart from all of the others, what makes scientific explanation powerful and serious, is that it relies on this very clear, stable, uncontroversial idea of laws of nature, when in fact, it's an extremely contested and controversial idea; there's no agreement at all on what a law of nature is, or even if there is such a thing as a law of nature.
- 7.67** So, given that uncertainty, how can it be that the gold standard for scientific explanation, what makes scientific explanation the gold standard in explanation, is something, the laws of nature, about which there's no philosophical agreement at all, even up to whether or not they exist? That's a challenge for this leading account of what scientific explanation consists in.
- 7.68** It's also a model that isn't immune to Hume's skepticism. Hume might say, "We don't need a law." We don't need the law that all copper conducts electricity. It's still completely consistent with everything that we know, that we can say "all copper observed in the past conducts electricity." It's the same. It's just another repeating pattern here. It's got nothing to do with other apparent instances.
- 7.69** Hempel's model of explanation does assume that laws make some difference in the world. They've got an oomph on the world, to use a very technical philosophical term. There's something they do. They explain things, and there's a universality to them, and positing them makes the world a lot less mysterious. It would be a very mysterious

world if Hume was right. There's nothing actually connecting different events together, different patterns of events together.

7.70 If it was just nothing there at all, a pure psychological habit, the world would be a very odd place. It would be a constant miracle every time you get into the car that it doesn't turn into Swiss cheese. This is a miracle because, on the Humean view, it might just as well have turned into a big slice of Swiss cheese. Maybe not for very long in the Arizona sun, but we want explanations here. It's not just random regularities that happen to be connected to each other or contiguous with each other.

7.71 So it would seem that explanation, if it is the jewel in the crown of the natural sciences, it has really brought to the world an explanation of how the world works. It does seem as if you're going to have to have a view of laws and causes that is realist enough to do all of that explaining work.

7.72 There are other problems, too, in the sense that there's a question as to how laws could cause things in the world. Isn't the relationship between patterns of events in the world and the law a bit more like the relationship between, say, a storm and a barometer? The barometer measures air pressure. There's always going to be a tight correlation between the barometer and the weather. But do we say the barometer causes the bad weather? No. It's not a cause, but it's correlated with it. It's not an accident. But the barometer doesn't explain the storm. The storm explains the barometer.

7.73 So it's not enough just to have laws. You also got to be able to explain the mechanics of how one thing causes another. This is a debate that continues to rage across all of the top philosophy faculties in the world because, of course, it's tightly connected to—the question of causation is tightly connected, as I mentioned previously, to questions of agency and human agency and human causation, and whether that is a different thing from ordinary causal connections and events that science describes.

7.74 This brings us to the end of looking at these three crucial theoretical building blocks in science: laws of nature, causation, and explanation. We've seen how philosophy helps us to clarify what's clear and what isn't, clarify what's intelligible, and where there are some serious problems and inadequacies. We've looked at what counts as a good theory and so on and so forth. This gives us a sense, again, of how little science simply authenticates itself as a successor to philosophy.

7.75 The more science has progressed, the more philosophical questions it's thrown up. It's not just clearing them up. It's not putting them away. If Hawking was right, we should see a lot more retired or fired philosophy professors, but we've seen the opposite. There's now a whole cottage industry of philosophers just working on the philosophical implications of and the correct interpretations of quantum mechanics and the problem of consciousness, or the problem of life.

- 7.76** How do you get life from non-life? Is the study of the transition from non-life to life, physics, or is it biology? Well, that's a harder question to answer than you might think. Biology is the study of life. "Bios" means life in Greek, and physics is the study of inorganic, or maybe organic chemistry, but inorganic principles. Physics isn't sensitive to the question of life and death. There's no true statement in physics that would have the word life or death, or not feeling terribly well, in it, because that's physics.
- 7.77** But if you're going from physics to biology in a question, namely, what is it? What are the laws governing the transition from something that is inorganic, just matter, pure matter, just to something that has life, has some principle of whatever life is in it. From slime to amoeba—slime, it was nothing, it was dead. Amoeba is something, we're not sure, but it's not dead. How do you describe that? If it's biology, then you're studying something that's inorganic, that's dead, and seems, as it were, intrinsically dead. If you just treat it as physics, then you have to start dealing with questions that physics can't cope with, like animate organic life, death.
- 7.78** Finally, you've got questions around the origin of the universe. So, philosophy has exploded around questions of, the implications of cosmological findings, particularly fine-tuning constants. Big Bang theory has quickened a new strand or recovered a strand of argumentation within the cosmological argument, so God's existence, called the Kalam cosmological argument. We'll be touching on that next time.
- 7.79** So there are all these different ways in which advances in science have driven philosophy to blossom in all sorts of different ways. It's hard to make sense of that, if it really were the case that science was just displacing, was simply displacing philosophy, was answering all of philosophy's questions. It seems, in fact, that we've got the reverse. The more science we do, the more philosophical questions we have. Thank you for listening.

Lecture 8: Unanswered Questions

At a glance — Unanswered questions — the open problems, in the final lecture.

- 8.1** Welcome to the eighth and final lecture in Philosophy of Science. It's been a whirlwind tour in lots of ways, that historical survey from Thales back in the seventh century, all the way up to Newton in the 17th century, and Einstein in the 20th. Just to understand the seedbed of the scientific revolution and how these different intellectual cultures approached the question of how to think about reality, how to think about nature.
- 8.2** What we're going to look at in this final lecture is four areas where there's been huge scientific progress in a way that has thrown up all sorts of new philosophical questions, or asked old philosophical questions in an urgent new way. It's been fascinating, the way in which science has impacted and has also failed to impact what's happening in philosophy.
- 8.3** I mean, two good examples off the top of my head would be, of where a scientific discovery had some impact on a philosophical position, would be Einstein's work on general and special relativity; had a huge impact on philosophers starting to think about the nature of time and whether time was real. So this deep philosophical question that had been around but acquired a new urgency after Einstein's work.
- 8.4** Secondly, the Big Bang theory, and we're going to look at this in a bit more detail as our first of four case studies. There you see the developments in Big Bang cosmology in the '60s and the '70s as having an impact on the old philosophical question of whether the world has existed forever, eternally, or whether it actually started. Intellectual traditions that followed Genesis 1, so all three Abrahamic traditions, said, well, yes, it did begin, or there is some point of origin for the world. It didn't just exist forever, and Aristotle is wrong. We have to accept that the two can't both be right.
- 8.5** So we're going to have a look at four of these different areas where advances in science have thrown up all sorts of interesting philosophical questions, philosophical questions that can help us understand the developments in science a little bit better.
- 8.6** Because it used to be the case that we didn't actually have very much data. I mean, science is relatively young as a set of disciplines that are, in a professional way, classifying and exploring the world. It's really only been 200 years, you could say, that it's even close to being a profession, just an important, highly regarded job that's more than just the gentleman amateurs who like to conduct experiments in the gardens of their vast estates, just for a bit of fun, which was, effectively, what science was until somewhere midway through the 19th century.
- 8.7** The Royal Society in 1660 was just really a group, was like another gentlemen's club in London, a big club in London that gentleman amateurs would attend and discuss important questions relating to the natural sciences. But now we've got almost

overwhelmed by data, overwhelmed by evidence, and the challenge is actually trying to organize it and trying to make sense of it all.

- 8.8** Think of advances in cosmology. I've already mentioned the Big Bang theory, but also multiverse theories, the so-called quantum vacuum. We'll come on to discuss those in more detail. Huge leaps forward in biology, of course, evolutionary biology, and the role that DNA and genes play in Darwinian processes, so much that it's called neo-Darwinism, really, with that supplement of genetic theory to it.
- 8.9** Questions about huge leaps forward in molecular biology and studies around the origins of life and trying to understand what life is. Similarly, in neuroscience, we've seen huge advances in understanding the brain, the most mysterious organ of any in the animal kingdom. It's just that there's been a lot of work on brain imaging and computational models, and lots of debates about what's going on when we're thinking about particular things; are there correlations to different parts of the brain, and so on and so forth.
- 8.10** And of course, in physics, not just cosmology, but physics, quantum physics in particular. Huge leaps forward there, which have really overturned and subverted a lot of our standard assumptions about how the world was structured. Huge leaps forward in particle physics, obviously, and also in the wake of Einstein's relativity theory. All of these are breakthroughs that help to give philosophers confidence that they're really beginning to try to make sense, philosophically, of what is most basic, most basic element of all reality, way beyond the merely atomic level.
- 8.11** What we're going to do in this lecture is to go through each of these four dimensions as case studies, you might say, so that we can understand in quite a concrete way the relationship between science and philosophy, what science's limits and scope might be, what philosophy's limits and scope might be, and how we integrate scientific discoveries with other parts of our philosophy of the world and of reality.
- 8.12** I think with each of these case studies, we're going to see something similar occurring. We're going to see there'll be lots of new data, new laws, new theories, and so on. It can transform the field in all sorts of radical ways, and it can throw up lots of different questions and so on. But as I've already said, the philosophical questions, they don't just remain, some of them fall away, but there are many more that come in their place.
- 8.13** Lots of conceptual issues. How are we interpreting this particular theory? These are questions, many questions, as we saw when we were discussing laws of nature and models of causation in the previous lecture. It's very difficult to arbitrate on scientific grounds between those different theories because the science stays the same.
- 8.14** The first case study, we're going to look at the question of existence, the existence of the world we find ourselves in. Why? This is Leibniz's old question, picked up by Martin Heidegger, the deepest, most fundamental philosophical question: why is there something rather than nothing? Why anything? Why this? It's a legitimate question.

Leibniz says there must be an explanation. Nothing exists without some reason for its existence. For everything that exists, you can say why. It's a legitimate question. Why, when it might not have existed at all?

- 8.15** The traditional answers have been, well, God. God is the reason that anything exists at all, or that what we see that exists at all, and if anything distinct from him exists at all. You might have a Platonist who says there are these necessary beings, necessary universals and forms in Plato's heaven, and they're the architecture of the world. They structure and fix the world, and that's what created it. That's where it comes from. Or it could be the view that stuff happens. This is the materialist view. Stuff happens, that happened, don't ask awkward questions. It's just a brute fact that the entirety of space-time popped out of an infinitesimally small piece of space into the world that we have 13.89, 14 billion years later, and stuff happens.
- 8.16** With the discovery in the '60s of the Hawking-Penrose singularity theorem, these kinds of questions about the universe's origins, space and time, the nature of space and time, become suddenly very urgent. It's an irony here that Hawking was, to the end of his life, I think quite self-consciously, an atheist, and didn't think that any appeal to God at any point would be legitimate. There's an irony that the theory that he puts forward in the 1960s gives new life to a new strand of—I should say, new life to an old strand of natural theology, called the Kalam cosmological argument.
- 8.17** The Kalam cosmological argument goes roughly as premise A: whatever begins to exist has a cause. It's the first premise: whatever begins to exist has a cause. The second premise: the universe began to exist. This is something that's a premise that you couldn't have plugged in before roughly 1965. The universe began to exist. Before that, it was, no, it didn't. Well, we're not sure. What does it even mean to say that it began? What does it mean to say that time began? There was a point before which time began. How does that work? Well, that's what the math says. This is what the theory says.
- 8.18** The argument goes on. This is premise one: whatever begins to exist has a cause. Premise two: the universe began to exist. Therefore, from one and two, from premise one and two: the universe has a cause. Because we've said, whatever begins to exist has a cause. It's pretty plausible, pretty plausible judgment, that. The science tells us that the universe begins to exist, so the universe must have a cause.
- 8.19** Or then you might ask, well, what cause? Because we're outside space-time at this point, so all talk of causation is off. All talk of laws of nature is suspended because you're outside the domain that those laws were describing. That's all that laws of nature are. Laws of nature are the laws of nature, not beyond nature. So whatever the explanation is as to what causes the universe, it can't be subject to the laws of nature. It's somehow outside nature. It's outside time. It's not even in time. Because that's what the science says. This is when time began. Time is stitched into the space, the spatiotemporal continuum.

- 8.20** Now, the argument goes on to claim that the best explanation for what could initiate a cause outside space and outside time, but could, as it were, will it to happen, intend it to happen, and bring it about that it happens, is God, or something very, very close to what the religious traditions, monotheistic traditions, have meant by God. It's not God necessarily, the conclusion to this, or any particular God, but it's probably too thick a slice of God for the atheist's digestive system. That would be the thought.
- 8.21** It doesn't really tell you which God, but it's a pretty powerful argument, using science. This is the interesting feature here. Remember the textbooks would have you believe that, what's the proper ongoing relationship between religion and science? One of conflict, one of warfare, one of displacement. Science is displacing religion, just as Hawking tells us science is displacing philosophy. It's got a very kind of totalitarian impulse, the sciences. It likes to colonize things.
- 8.22** Here we see an example of a development at the cutting edge of cosmology, as it was in the 1960s, and to some extent still is today. Nothing has rivaled the Big Bang theory. Nothing has come even close to offering an alternative. It's been with us 50 years now, make that 60 years. It's striking that it's woken up a whole strand of philosophy of religion to think about the nature of God. In particular, a finding in physics has pushed us to thinking about metaphysics. What is it that's upstream of physics, upstream of space and time, upstream of the only world that physics can describe?
- 8.23** This is a very interesting development. Again, not a development that science itself can settle, because by definition, we'd be asking questions about what's upstream of science, upstream of physics. Some have said, well, other things might have caused the universe. People say there's a quantum vacuum, a field of virtual particles that's governed by quantum field theory, and the universe just popped out of that. Is this a good response?
- 8.24** Well, in my view, it isn't. In my view, what it does, it fails to understand the nature of the argument. A quantum vacuum may be very empty, but it's still something. It's not a nothing. You could still ask for the quantum vacuum, is it something that begins to exist? Presumably, you're going to say, yes, there was once when it was not, or it's not necessary that it exists. It could have been something completely different. We can just say, whatever begins to exist has a cause. The quantum vacuum that brought space and time and the universe into being has a beginning; it began to exist. Therefore, the quantum vacuum, this very, very impressive quantum vacuum, has a cause. What's that?
- 8.25** So you've not really made any progress here. You've pushed the explanatory—you've written another check, but you still haven't cashed it. It's still, what's the ultimate explanation here? You're just saying, this just moves us back. And of course, there's the other worry that quantum mechanics, yes, is a very strange— it subverts and challenges many of our established conventions and conceptions of physics, but it's still physics. It's the subatomic world.

- 8.26** That is to say, it's a world that—the quantum world is a world that is, it's in this world. It's a very spooky, strange world, but it's a world that physicists, quantum mechanics, is something that describes a very, very tiny, fundamental layer of this world, of space and time. There's no more reason to think that quantum laws or quantum patterns or correct quantum theories, unlike laws of ordinary nature, can somehow jump outside of space-time and pronounce what the world is like outside space and time.
- 8.27** There's nothing that enables a quantum physicist to do that any more than an ordinary physicist. Still, as far as we know, these are domains of scientific explanation that apply to this world, even if it is just the subatomic world or the world at the level of the very, very fast, the speed of light, or could there be something more than the speed of light, as in the case of Einstein's relativity theories.
- 8.28** Others have said that you can have a multiverse; and so, the reason that we have the universe that we do, that is strangely suited, at least in one tiny, tiny corner of it, strangely suited to the emergence of carbon-based life and greenery and some mountains and gardens and a broadly temperate climate, that this is so completely improbable, it must be the case that there's this multiverse generator which keeps firing out these different universes, and our lucky number came up eventually.
- 8.29** So many, many different universes with lots of different laws, and we just happened to find ourselves in the life-permitting one. It's a whole cosmological lottery. And is this a good explanation? Is it any better than the quantum vacuum explanation? Well, for the same reason, it just pushes the problem back. Because if you're impressed with the universe and you're curious about where the universe came from, being told there's this multiverse machine that creates trillions of universes like this, I suspect that kind of person is not going to just be less curious as to what this extraordinary machine is.
- 8.30** The obvious next question is, what is it? How did it emerge? Who could possibly have put something as incredible as that together? I thought cracking how the universe came into existence was tough. That was going to be difficult, but now there's this whole new layer. And that will still be a legitimate question. I mean, an obviously legitimate question: what is this machine? How does it do it? How is it programmed? Where does it get its causal powers from? Does it will anything? Is it just a machine, or is it a kind of? So many questions.
- 8.31** And there's also a feeling with this response, the multiverse hypothesis, that this is an inverse version of the gambler's fallacy. The man who goes to play vingt-et-un roulette at the casino in Monaco every night for 20 nights. And he says, on the 21st night, "I might have lost every single time. I have to win tonight. I have to win tonight. It's just got to be my lucky night because none of the other nights have been lucky. And it is so likely that I'm going to win tonight."
- 8.32** Now, what are the chances of him winning on night 21? Identical to the chances of him winning on night one, one in 21. But the mind has a habit of spreading itself out upon

the world, as Hume says, and you think, fate now is going to favor me, obviously. I think part of the problem with this cosmological lottery that the multiverse theory proponent pushes is to say that if you've got a trillion throws of the dice by this multiverse maker, then it had to be the case that our lucky number came up.

- 8.33** What are the chances? Well, they're much, much weaker because the machine's been going forever. And so, the chance is—now, does that affect the philosophical puzzle? Namely, why something rather than nothing? Why us? Why this rather than anything else? No, there's still not a one-in-21 chance, but it's still extraordinary that this one throw of the machine's dice should result in this. It still requires explanation. So that's the first case study, questions about cosmology and the origins of the universe.
- 8.34** And the next one, the next case study I wanted to look at, is moving from physics to biology. We've touched on this briefly already, this distinction between life and non-life. This will help us think about how we make sense of biology in the context of the various philosophical questions that arise as a result. In many ways, this is a very old philosophical question.
- 8.35** What is it that distinguishes living from non-living? Is it because living things have a soul? This is Aristotle's view in his *De Anima*, his *On the Soul*. Aristotle's view is that all living things have a soul of some kind, are ensouled to some degree. So you've got plants, you've got the plant kingdom. Plants have souls, says Aristotle. They have a nutritive soul, so they at least have in a way that a rock doesn't. A rock hasn't got the inclination to feed itself. A plant has an inclination to feed itself. It has life and soul of a very primitive kind, says Aristotle. That's what makes it different from an inorganic dead thing like a rock.
- 8.36** Then, at the second layer, there's what all the animal kingdom and the plant kingdom have—sorry, what all the animal kingdom has, but the plant kingdom does not have— a sensory soul, a soul that can have senses and perceive things in the world. The capacities to sense is another kind of ensoulment that plants don't have, but that all animals do have. Sensory functions or capacities of some degree, of some kind, even if it's very limited. And on top of that, the nutritive inclination that he calls the nutritive soul, which plants have as well.
- 8.37** Then at the very top, says Aristotle, you've got this third principle, which only one tiny corner of the animal kingdom has, namely human beings, and that is the rational soul. It's a capacity to reason and to understand and to engage in arguments and speech and to understand meaning. Aristotle says, human beings alone are driven by this desire to understand, can stand in wonder at philosophy, at the big questions. But again, to go back to the basic point here, that for Aristotle, ensoulment, or having some principle of life or energy of some sort, it's that that distinguishes life from non-life. This is his view.
- 8.38** What has science discovered about life? What does it tell us? Well, in biochemistry, life is really chemistry. It's compounds and elements interacting with each other in various

complex ways. Cells are effectively molecular machines. DNA stores information. We don't need to posit some vital force. We don't need to posit a soul. Who believes in souls, anyway? It's not very scientific. We want to avoid those explanations, if at all possible. That's biochemistry's answer.

- 8.39** Then you might have a larger argument from biology, from evolution, an explanation about how this life evolved from simpler forms and make some suggestions as to how natural selection operates in these different transitions and so on and so forth. Then there's this idea of this creation of artificial life in a lab. Does this just show all life is, is chemistry? If you can do it in a chemistry lab, then there's no designer needed, no God needed. No, it's just anybody with a lab who's ended up in the right species can create life.
- 8.40** This opens up lots of deep philosophical puzzles. How does life begin? How do you get from physics to chemistry to biology? How do you study those transitions in a way that doesn't push you onto one particular side of the fence? Because you're trying to explain, in order to be able to explain the transition, it can't just be from non-life to life. It can't just be physics, and it can't just be biology, as I've mentioned before. There's something in between. Something needs to bridge the gap.
- 8.41** And this is not something that scientists feel particularly comfortable answering. I mean, it's a challenge to avoid a dialogue of the deaf between the physicists and the chemists and the biologists on these kinds of questions. You started to see emerge all across, in many universities all across the West, research institutes on the origins of life or the beginnings of life, which try to combine experts from the different fields and try to forge a collective conversation.
- 8.42** But there's still a lot of dispute as to how you would hold such a conversation together. What's the overall conceptual framework? It's not obvious at all. Is there a sharp boundary between life and non-life? I think plausibly there is. If you think there's a gradual continuum, then it seems as if life is a bit of a vague concept, much vaguer than we believe it to be, and so on and so forth.
- 8.43** So we've looked at origins of the universe as our first case study, and secondly, this deep problem of the origins of life as our second case study, the way in which advances in science has stimulated us to think philosophically about the nature of reality.
- 8.44** The third case study, I thought we'd turn to from questions in cosmology to start with, and then biology next, and chemistry to some extent, is neuroscience, or we might say the problem that consciousness raises in relationship to neuroscience. That's, of course, one of the oldest questions and debates in philosophy. What is the mind and what is the brain? Are they two distinct things? If the mind is something that's real, it's immaterial, and therefore, how can something that's immaterial have any causal influence over the material?

- 8.45** It's like trying to explain something by saying, "The fairies did it." Something that's not material is doing things in the world. This is spooky. How is this possible, if that's what the mind is? How do they interact? Again, you've got a problem of transitions. It can't be the laws of physics. If there's such a thing as the mind that is immaterial, that is not the brain, and you've got the brain here, well, there's some interaction or patterns between the brain states and the mind states.
- 8.46** Well, what are the laws that govern that? If that's a scientific statement, a claim, what are the laws that govern those? They can't be laws of physics. Laws of physics describe what might be happening in the brain, neuroscientific laws, as it were, and at base, laws of physics. But if the mind is something that's not material, not a physical object, then the laws of physics don't apply to it. So the laws of physics can't tell us about, can't describe the traffic between brain states and mind states, if there's such a thing as mind states that are distinct, different, not material.
- 8.47** Now, we know maybe the neuroscientists can tell us that, when you're experiencing particular feelings or even thinking particular thoughts, then different parts of your brain light up. There is a connection or a correlation between feeling certain things, thinking about certain things, being in certain states of mind, and the results of brain imaging at that time. Therefore, all the mind is, say the neuroscientists, is the firings of these different parts of the brain.
- 8.48** Now, in many ways, you might say, what does the science really add here? Did I really need to know that there was some relationship between my brain and how I think, how I feel? The first time a caveman, Fred Flintstone, smacks his friend over the head with a club, I think the relationship between psychological states of pain and the physical world, how my brain is doing in the world, yes, do have very deep connections. Yeah, I felt that. That was painful. We know that there's a correlation between our bodies and our psychological states.
- 8.49** The question is, what is that connection? Even if you can describe perfectly what's happening on one half side of that connection, which the neuroscientists claim to be able to do, that doesn't help us explain what's going on in this completely distinct category of being, namely consciousness, subjectivity, mind. So physics can tell us about physics, but it can't tell us about, as it were, our mental physics, our mental lives, if our mental lives are not physical domains.
- 8.50** Now, this is the so-called hard problem. This is a hard problem if you're just committed to physics being the be-all and end-all when it comes to explanation. Why does it feel like something to be you? Why does it feel like something to be you? It does feel like, there is something that it is like to be you, and you have knowledge of that in a way that nobody else ever could or ever will. It's subjective. It's not like just objective knowledge, descriptions of the world. It's something that uniquely and pervasively involves you and only you.

- 8.51** That seems as if it's something that a full explanation, an explanation that was attempting to be comprehensive but only expressed in the language of physics, would never, in fact, be an adequate explanation, wouldn't properly capture any explanation. You could say, almost by definition, because scientific explanations are public. They're public explanations. That is to say, in principle, an explanation is something you could give to another person when confronted with the same evidence, and they'd say, "Oh, OK, right, this is true. I can understand this."
- 8.52** Now, could you ever make somebody else know, to the degree that you know, what it is like to be you? No, because if you could, they would be you. But they're not you. You're you. Now, you could be a brilliant diarist. Maybe you could describe in extraordinary detail, or maybe you were a film director, and you could do an incredible film of what it's like to be you. To some extent, maybe you could convey what it's like to be you, to some extent. You could quicken feelings that others might have had, that there'd be moments of recognition.
- 8.53** But that's not the same thing. It's not the same thing. You could have a complete printout of everything that's going on in your life. Giving it to another person so that they could read it and understand it would not turn them into you, wouldn't furnish them with memories that you have because you were there and so on and so forth. So it seems as if there are limits to neuroscience, limits to what the omniscient neuroscientist, the neuroscientist who knows all truths that will ever be known in neuroscience, is still not going to know what it's like to be you.
- 8.54** It's not a limitation of science. It's just saying science studies science, and the mind is not a scientific object on many views. And so, by the way, this is why the phrase "mental health" is quite an interesting phrase, because the assumption with the phrase "mental health," as opposed to psychological distress or some other equivalent phrase, what's interesting about that phrase is that health is primarily a word that we apply to physical things, physical states, material states.
- 8.55** Mental health, well, if you think that the mind just is the brain, then yeah, OK, that's fine. That's just saying, is your brain healthy? If a neuroscientist could improve it, but it's a physical way of thinking and talking. If, on the other hand, the mind is something that's different from, connected to, but distinct from the body, from the brain, but is not a material thing, is not something that science can study, then it turns out the word health is just being used as a metaphor. It's not being used in a literal way.
- 8.56** Saying, well, just as we say, "I'm feeling OK physically," you could say, "I'm feeling OK mentally," and those are similar. Well, maybe they are, maybe they're not, but there are some key differences, it seems to me. For example, what do you often do with a doctor? You go, the doctor says—the doctor will give you a clean bill of health. "Look, you're fine. I checked you over; nothing to worry about. You're fine." If you don't think so, you can go and get a second opinion, and you would think the second, third opinion would say, "Yeah, clean bill of health."

- 8.57** What would a clean bill of mental health be? You go to a therapist, you say, check through. "I know you're struggling with this, but that's fine. Yeah, clean. You're totally fine. Resting state, completely happy all the time. You're healthy. Your mind is healthy." Well, first of all, could any human being get a clean bill of mental health? That is to say, could any human being say they're completely calm, totally balanced, not ruffled by anything, permanently happy? It may be, but this would be an outlier, an anomaly. What on Earth is wrong with this person?
- 8.58** The point is, there isn't the same objective criteria that a doctor could apply to work out if you've got a broken arm or poor vision or, heaven forbid, cancer. These are objective assessments that can be made with evidence, and you can get a second opinion and so on and so forth. When it comes to the mind or your feelings or your cycle or your headspace, as it were, what's the science of that? If you went and got a second opinion from another therapist, I think you would probably come out to have a completely different story as to what explains this. It wouldn't be the convergence that you want to see in the sciences.
- 8.59** It seems as if there is whatever else we're trying to do when we're trying to relieve ourselves, soothe ourselves, comfort ourselves psychologically, out of states of psychological distress. It's not the same thing as putting a plaster cast on or having radiotherapy or something. It's a completely different kind of thing. And so that neuroscience must, therefore, for all of its discoveries, we've got to be careful that science envy and science worship doesn't allow—it doesn't mean that the authority that neuroscience now invests itself with should determine whether or not we think we are we or you are you.
- 8.60** We shouldn't lose too much sleep over that. We're seeing extraordinary developments in neuroscience, but neuroscience is just the science of the brain, and the science of the brain can't tell us if there's something more than a brain, because if there was something more than a brain, there'd be, let's say, a mind or a soul. But what does brain science study? The brain. It doesn't study the soul, doesn't study the mind. So it's agnostic. It has to be agnostic about whether or not there's something more than the brain. That's just what it is. It may be that there is, that maybe there isn't, but nothing in brain science is going to tell us whether there's something more than the brain.
- 8.61** So, this has been a very important and significant application or point of intersection between philosophy and science. Finally, I just wanted to, we've touched on this before, but I wanted to just come back to it briefly as our fourth case study, and that's the extraordinary leaps forward in quantum mechanics. Just to go into enough detail to give you a sense of the strangeness of quantum mechanics, I know we've talked about it now and again in previous lectures, but why is quantum mechanics such a challenge, and why does it throw up so many interesting philosophical questions?
- 8.62** Remember our old friend Newton in the Principia Mathematica, 1687, comes up with this incredibly powerful explanatory model for motion, for mass in motion, where every

object has a particular set of properties, mass and momentum and volume and extension and so on and so forth. It's pretty deterministic. Everything happens where it's supposed to happen, in proximity to, with other objects, other masses. And quantum mechanics just gets rid of all of that, pretty much.

- 8.63** First of all, it says there aren't—no particles have definite properties, in the way that Newton assumed. No particle has a particular mass or a particular extension. What you have, it seems, for the maths to work in quantum mechanics, you have a particle that is in not any one position, but in a superposition. And so this is a strange idea that an electron in a superposition, or a particle in a superposition, could be positively charged, it could be negatively charged. It is both positively charged and negatively charged.
- 8.64** One particle, in other words, we have to believe, is in multiple states simultaneously. This is the puzzle. If we can't wrap our heads around it, then there's all sorts of technology that is completely mysterious. The reason that quantum mechanics, the questions are so urgent in quantum mechanics, is not that this is some fancy new theory in physics that may or may not be true. That's string theory, which probably isn't true. They've had 25 years, and goodness knows how many millions of dollars to make some progress, and they just haven't, really.
- 8.65** But with quantum mechanics, what it makes possible in the modern world, from computing to transportation, to any number, almost any major technological breakthrough, depends upon quantum mechanics being true, being a successful guide to the way the world is at the subatomic level. It's unbelievably reliable when we factor in. If it was just a theory, if it was just a very speculative idea, then most of the modern world wouldn't work, wouldn't function. But it's got real-world predictive power and, as it were, contact with reality.
- 8.66** We have to make sense. And yet, in order to make sense of it, we have to believe several impossible things before breakfast, including that a particle can be in state A and state not-A at one and the same time. It can be both what it is and what it's not at the same time. How do we make sense of this? And to make things even worse, the only way to resolve it being both A and not-A, or both being positively charged and negatively charged, is for little old us to come along and measure it.
- 8.67** And as soon as we measure it, boom, it becomes one or the other. It goes from superposition to position, just from us looking at it from the other side of the galaxy. How is this possible? How is it that things in the subatomic world could depend upon little old me measuring, just clocking, not doing anything, just measuring? How is it that reality depends upon the observer, when we thought that it was the other way around? It was the world to be observed, and we are, as Kepler says, "thinking God's thoughts after him." It's all there. It's given, and we've got to work it all out.
- 8.68** Whereas this seems that we—this is telepathy, as I think I've mentioned before. It's the mind can measure something and make it be what it wasn't before. The measurement

problem, the so-called measurement problem, arises because quantum mechanics seems to have two completely contradictory sets of rules.

- 8.69** Rules number one: when unobserved, particles will exist in superpositions of multiple states simultaneously, deterministically obeying certain laws that are captured in Schrödinger's equation. Set of rules number two: when observed, particles will exist in a particular position, and what that position will be can only be probabilistically determined. It's not deterministic. Sometimes it'll be that, sometimes it won't be. It's the point of measurement when the superposition collapses into one thing or the other, but you can never determine what that's going to be. It's indeterminate.
- 8.70** Now, obviously, the biggest problem with that, with those two, the reason for the conflict between those two rules is, as I stated them, rules number one begins with the word "unobserved." The first word that I used to describe second set of rules number two is "observed." When nobody's looking, this is how the world behaves. But if you look, it's going to be this whole different set of rules applies. This is the biggest conceptual headache at the heart of quantum mechanics, that somehow we are involved in the way the world, at the level of not doing stuff, reasoning about, not causing things, just measuring, reasoning about the world, observing and reasoning.
- 8.71** You could see how if this theory just came along as something that somebody suggested, you'd laugh at it; nobody would be talking about it today. But as I said, it is so indispensable to so many crucial leaps forward in technology that we're stuck with it. We've just got to try and make sense of it. Or if we don't, just sit with it, works in practice, maybe we shouldn't be too worried about how it works in theory. Maybe the problem of the modern world is that things work great in practice, and we stop everything and say, "Ah, but have we done a survey of how it works in theory? It's got to go through the theory test first."
- 8.72** So, this is part of the strangeness of the world of quantum mechanics, and talked about the indeterministic probability, that you don't know once you measure what it's going to be. Questions also to do with entanglement. This principle in quantum mechanics, at least a way in which we can make sense of quantum—the only way in which we can make sense of quantum mechanics is to say that it's possible for two particles to mirror the other's behavior.
- 8.73** It seems as if, what quantum mechanics tells us, is that if you measure, say, particle A's spin on one side of the galaxy, and it's up, then you can say that particle B's, there are scenarios where you can say that particle B will be down, will be negatively charged, no matter how far away the particles are from each other. They could be many, many millions of light-years away from each other. But to make the maths work, this so-called entanglement of two particles across inconceivably large distances is a phenomenon of what the quantum— seems to be a phenomenon, an aspect of the quantum world we have to make sense of.

- 8.74** This is one of the reasons Einstein couldn't stand quantum mechanics. He said this is "spooky action at a distance," but it's like magic. There's a kind of magic. It's very hard to make sense of it. This has thrown up all sorts of different theories within the sciences. There's the so-called Copenhagen theory, which broadly says, we always told you that it was maps all the way down. We always told you there isn't any—you can't take your sunglasses off and see what the world is like from a non-human perspective. That's never going to happen.
- 8.75** The world is basically dependent upon how we interpret it, how we see it. This paradox, the measurement problem, is just not a problem. It shows that measurement creates reality. That's what we always said, the map guys. It's just us. All that science is, is just coming up with different maps. The reality is what we make of it through our maps or through our tinted sunglasses or whatever it might be. That's why we finally—quantum mechanics has helped us realize that we were right all along. There isn't a world out there. The world is what you invent, what you construct, not what you find, not what you discover. That's the Copenhagen model.
- 8.76** There's a number of other models, too, that are very strange and require—it can be very difficult to keep a straight face when some of these theories are being elaborated. But the point is that, even though quantum mechanics has revolutionized the modern world, it's shown that our classical picture of the science we used to have is basically discredited. It forces all these questions about time and the nature of space and ultimate reality and matter.
- 8.77** So, extraordinary revolutions and extraordinary practical leaps forward, but the philosophical questions intensify. The world is getting more mysterious, the more science we do on it, not less mysterious. It's becoming clear that the number of questions that science simply cannot answer is proliferating for every success science has within its own field.
- 8.78** And this has been, really, one final way to respond to the challenge that the great Stephen Hawking gave us at the very beginning of the series, namely, this claim that scientists would force the philosophers out of business. Turns out that the more scientists there are, the more scientific discoveries there are, the more philosophers are needed, and the more philosophy needs to be done. Thank you for listening to this series.
- 8.79** I have a million questions, but I only have time for one. But do you believe that through all the theories that we've touched upon through these eight lectures, one takes a primary role or a basis for everything else that connects and links, for example, neuroscience and the mind? To me, sitting here and listening to everything that I've listened to, I feel like that would be almost the primary basis that everything else leads off of or branches off of. What do you think about one of them taking on a primary role or the basics where all the other ones branch off of?

- 8.80** No, it's a very good question. The first thing I would say is that one's got to be careful about casting suspicion on claims, just because those claims have emerged at the end of a process where you can appeal to all sorts of external factors as to why that belief came to be held by that person.
- 8.81** So, you could say to the Muslim who is born in Saudi Arabia, he grows up believing in Islam and studying it, and so on, you say to him, "Well, you only believe that Islam is true because you were born in Saudi Arabia. If you'd been born somewhere else, you'd have believed something different." Now, is that a good argument? Well, no, it's not a good argument because it doesn't go to the heart of the truth claim. You can't discredit a belief because you can point to some contingent circumstances in which the belief arose.
- 8.82** One good example might be the Babylonians, Babylonians or the Mayans, unbelievably good at maths, coming up with true mathematical truths and mechanisms for yielding, generating mathematical truths. Then you could say, you only think that the abacus works in the way that it does because you were obsessed with the stars and you had this weird theology and this weird religion that meant that you had to get the calendar exactly right, and that meant that you developed an intellectual culture that was very good at maths. So we're not going to believe any of this math stuff, because you picked this up by accident, because of the particular culture that you found yourself in.
- 8.83** I would say there are truths that are true no matter what. In certain cultures, you might be pushed towards those truths and grasping them better; in other cultures, maybe not. But it doesn't affect the truthiness of the truths, how you came to hold that true belief. It doesn't affect it.
- 8.84** When your question about whether neuroscience should be the queen of the sciences or the queen of philosophy of science, yes and no. It's funny, as soon as you talked about neuroscience, and then you started to talk about the self and one's own decisions and consciousness and all that sort of thing, and the claim is that, as soon as you start using the I-word, you're already beyond neuroscience. Neuroscience can see your brain. It can't see the world from the way you are. That's something that's totally different, that neuroscience will never be able to settle. So there's that worry about a limitation.
- 8.85** But in a deeper sense, your instinct, I think, is a very powerful one, namely that, if you look across all of the whole history of Western and Eastern philosophy, one dominant, or I wouldn't say dominant, but one leading view is that the only way to make sense of this old world is to make sure that you've got some mind upstream of it. If it's just matter, just stuff, and then the mind is just little old us, we could be here today, gone tomorrow. It'd be strange if all the laws and all the principles depended on us being around to be true.

- 8.86** On the other hand, all the laws and things are complicated. They're the things that need to be thought about. It's not just rocks that can be doing this or comprehending this. So, many traditions have said there must be a mind upstream. There must be a consciousness, will, desire, intelligence, rationality, upstream of this whole messy material process that is nevertheless unbelievably ordered.
- 8.87** So your intuition there, that mind is somehow primary, I think is the right one. I think it's probably less plausible to say that it's the human mind that is the source of it all, just because we seem to be such fragile reeds, as Pascal, but we're thinking reeds. How are we thinking reeds? We're fragile reeds, but thinking reeds. Thank you.
- 8.88** I was listening to a podcast where a prominent evolutionist was being interviewed, and he was asked about intelligent design. The issue he had with it was, basically, who created God? If there is a God, then the question becomes: who created God? To me, the thought went up, is that even the right question to ask? I was just curious if you had any philosophical thoughts on how to frame that discussion.
- 8.89** Yes. One of the most obvious quick response to that is to say that what's being explained in cosmological arguments or teleological arguments like that, the reason God is invoked as a plausible explanation in those philosophical arguments, is precisely because he's a plausible candidate for the buck to stop there. You're trying to come up with some explanation or some solution that is going to help you say, "I'm going to turn my spade here. I don't need to keep digging. I've got the answer now."
- 8.90** The whole point of saying that the reason why there is anything at all, why there's life at all, why there are carbon-emitting processes at all, you've got to be appealing to something that is a good candidate. So, if you're talking about the universe, it can't be something that's in space. It can't be something that's in time. Because that's what the universe, that's the universe. You can't appeal to something in the universe to explain how the universe happened. Because the cause is part of what you're explaining, and nothing can cause itself. Nothing can cause itself.
- 8.91** If nothing can bring itself into being, in order to bring yourself into being, it means you're not in being. But if you are in being, you can't be causing yourself to come into existence. This is a kind of bootstraps philosophy. It's got to be something beyond it. It's got to be something outside space, outside time. It's got to have all these causal powers, and you can start to develop an argumentative trajectory along those points.
- 8.92** But the key thing is that, in the classical tradition of theism, Abrahamic monotheism, God exists no matter what. What it is to be God is to exist no matter what, not to depend upon anything for your existence, not to rely on any parts for your own existence, just exist no matter what. It's as incoherent as saying, "What made God?", as saying, "What made the number two? What made it?" Number two is not made. It just exists. It's not made; nothing brought it into being, because it didn't have a beginning in

time. Nothing can extinguish it. Not even the heat death of the universe can make it cease to be the case that $2 + 2 = 4$.

8.93 And so, you haven't understood the quest, you haven't understood the attraction of the theistic posit, if you're saying, "What made God?" Because it's the same thing as saying, "Who made the thing that couldn't fail to exist? Who made the thing that couldn't fail to exist?" Nobody. Because something that can't fail to exist doesn't need to be made. That's what it is. It's what it does. And this is the standard view of God, that his existence is identical to his essence. "Esse"—"Esistere" and "esse" are the same, uniquely in God. There's no difference.

8.94 What God is is the same as the fact that God is. What God is is the same as the fact that God is. Two are completely fundamental. For everything else, you can say, why that? Why this? Why is it like that? Why is it like that? It's a plausible place for the buck to stop.